Machine Learning Capstone Project: Used Cars Price Prediction Low-Code Milestone Report

MIT Applied AI and Data Science Program

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Executive Summary

The Cars4U used-car startup company intends to enter the used-car market in India. This particular used-car market is characterized by rapid growth, price volatility, and notable information asymmetry. The principal mission of this company is to develop a machine-learning-driven pricing engine that brings fairness, clarity, and scalability to every used car in Cars4U's inventory. Given the previously stated characteristics of the Indian used car market, the model must generate accurate, reliable, and scalable pricing for each used car, while also avoiding bias, price instability, and information discrepancies. A robust predictive model can significantly enhance Cars4U's market position by reducing negotiation disputes. This results in increased trust between buyers and sellers and enables automatic, data-driven pricing for the company.

The exploratory analysis of the initial 7,253 used cars from the provided dataset revealed a rich, highly diverse dataset. However, the initial dataset had multiple challenges, including missing values and skewed distributions across the independent variables (features). Essential numerical independent variables, including engine displacement, power, and new prices for used cars, exhibited right-skewness due to the presence of luxury vehicles. Essential categorical independent variables, such as Fuel Type and Owner Type, aligned with clear Indian used-car market characteristics, as diesel cars and first-owner vehicles comprised most of the dataset. There were over 6,247 missing values for the New Price variable, and similar notable gaps in Power, Engine, and Seats. Careful preprocessing and imputation strategies were required. Regardless of these issues, the dataset still provided powerful signals for modeling after it was cleaned and properly engineered.

The modeling procedure involved multiple regression models. These included linear models (e.g., Multiple Linear Regression), regularized linear models (e.g., Ridge Regression and Lasso Regression), K-Nearest Neighbors (KNN) Regression, and tree-based models (e.g., Decision Tree Regression and Random Forest Regression). The linear models were interpretable but could not capture the nonlinear interactions exhibited in used-car pricing. The tree-based models best captured the nonlinear interactions more efficiently. The Random Forest Regression model had the preeminent performance in R-squared, Root Mean Squared Error (RMSE), Mean

Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). This was due to the model's ability to handle outliers, nonlinear relationships, and mixed independent variable types, as demonstrated by its ensemble nature. The Random Forest Regression model's performance made it the most dependable contender for deployment. Thus, this aligns with Cars4U's goals of improving accuracy, potency, and generalization across various used car brands, manufacturing years, and conditions.

The chosen Random Forest Regression model serves as the final solution design for Cars4U's production pricing engine. The model's stability and interpretability are underscored by the importance of independent variables and interpretability tools, which help Cars4U use used-car price valuations for customers and domestic stakeholders. The model aids strategic actions such as dynamic pricing, inventory refinement, and fraud detection. To clarify, the model, as a practical automated pricing engine, allows Cars4U to achieve price standardization, lower pricing ambiguity, aid both buyers and sellers, and establish a competitive advantage in the Indian used car market.

In general, Cars4U intends to implement data science as a key business capability approach. By using continuous data fortification, recurring model retraining, and customer-facing workflows, the pricing engine's Random Forest Regression model can become a strategic resource that compounds value over time.

Business Problem Overview and Solution Approach

Business Problem Overview

Cars4U faces business challenges entrenched in the Indian used-car market. These challenges include an inconsistent, unclear, and eminently variable pricing atmosphere. This atmosphere entails numerous buyers and sellers operating with incomplete information and notable regional disparities. Nevertheless, the entire Indian used car market depends on authentic, dependable, and transparent pricing to combat bias, inconsistent costs, geographic diversity, and information asymmetries. These standardization deficiencies found in the nature of the Indian used car market cripples Cars4U's capability to contend against other used car platforms. To overcome these obstacles, Cars4U must adopt a machine-learning pricing engine that can predict unbiased market values based on used-car attributes such as brand, model, year, and

condition. Therefore, the machine-learning pricing engine would empower Cars4U to provide clear valuations, reduce negotiation conflicts, improve user confidence, and establish an adaptable, data-driven pricing strategy that strengthens long-term competitive advantage.

Analysis Explanation in Achieving Business Objective

To accomplish Cars4U's business objective, it needs to convert an incomplete, intuition-driven pricing environment into a methodical, data-driven valuation framework that can operate reliably. A machine-learning pricing engine model allows Cars4U to navigate India's diversified used-car market, where brand reputation, geographic demand, past ownership history, and vehicle condition interact in nonlinear ways that humans cannot easily evaluate. For that reason, the model must be trained on these nonlinear relationships in the historical dataset to effectively reduce the bias and inconsistency that currently inhibit customer trust and impede transactions. This analytical foundation further allows Cars4U to standardize used car prices across Indian cities, ensuring that similar used cars receive comparable price valuations despite local negotiation dynamics. Consequently, Cars4U can be recognized as a transparent and dependable marketplace in an industry where pricing is opaque.

The pricing engine machine learning model best supports Cars4U's functioning and strategic goals by allowing accurate predictions, inventory planning, and margin enhancement. By utilizing reliable price predictions from the machine learning model, Cars4U can identify underestimated used cars for acquisition, identify overpriced listings that could stagnate, and predict how depreciation patterns may diverge across brands and segments. These insights can assist Cars4U in reducing holding costs, increasing turnover rates, and allocating marketing resources more efficiently. The machine learning model's explanatory power, particularly regarding the importance of certain independent variables (features) and price-driver analyses, can better reinforce valuations for both buyers and sellers. Hence, the model's explanatory power can reduce negotiation friction and overall customer disapprobation. As time passes, Cars4U can capitalize on this data-driven machine learning model to gain a competitive advantage over other platforms, while also refining its pricing acumen as the company continues to grow and new market patterns emerge.

Solution Approach or Methodology

The project follows a data-driven, multi-faceted methodology, using Python-based statistical modeling and computations conducted by a Data Science team to address the business problem and objective, including critical issues, key insights, operational findings, strategies, and recommendations from Cars4U's provided dataset. The approach emphasizes Python-powered statistical analyses to deliver comprehensive data insights that support strategic interventions and recommendations. This approach focuses on using machine-learning workflows to reveal key pricing drivers, address data-quality issues, and produce actionable insights for Cars4U's decision-making process. Since the provided dataset ranges from a multitude of used car vehicles across brands, Indian cities, fuel types, and ownership type histories, the methodology centers on powerful preprocessing steps, including data cleaning, handling missing values, extracting brand and model information, feature engineering independent variables such as Car Age and Depreciation Amount, and encoding categorical variables for modeling. The solution configuration incorporates descriptive analytics, correlation analysis, and exploratory visualizations to uncover market patterns and steer model selection. A Multiple Linear Regression model, regularized linear regression models (e.g., Ridge Regression and Lasso Regression), a K-nearest neighbors (KNN) Regression model, and nonlinear tree-based models (e.g., Decision Tree Regression and Random Forest Regression) are trained and evaluated using a train-test split (70 to 30) to ensure generalization and to classify the model that best highlights the complex, nonlinear structure for the used car prices dataset.

Ancillary procedures best strengthen the model's comprehensibility and operational preparedness for Cars4U's business climate. Outlier analysis is implemented to ensure that each model can distinguish genuine luxury-segment extremes from erroneous entries, while also learning meaningful variation rather than noise. Variable importance techniques also aid in identifying the most powerful price predictors, allowing Cars4U to defend price valuations to customers and refine procurement strategies. Each model's performance is assessed using metrics such as R-squared, Adjusted R-squared, RMSE, MAE, and MAPE to balance accuracy with consistency across diverse used car vehicles. The final model, also known as the final methodology, is firmly grounded in Random Forest Regression, as it predicts fair market values and provides a scalable, logical foundation for Cars4U's pricing automation, inventory optimization, and long-term competitive strategy.

Problem Definition

Context

The problem this project aims to solve is important because the entire used car market in India relies on accurate, reliable, and extensible pricing, yet is characterized by bias, unpredictable costs, regional diversity, and information inequities. Historically, in 2018-2019, while new car sales totaled 3.6 million units, approximately 4 million second-hand cars were purchased and sold (a 0.4 million difference). Based on this historical fact, there appears to be a decline in new-car sales in India, suggesting greater demand for pre-owned cars. Given the size and shift of the used-car market in India, it is now considered much larger than the new-car market, necessitating data-driven pricing; otherwise, it becomes inept and unethical. Notable advantages for Cars4U in solving this uncertain used car pricing and supply problem include prominent market positioning driven by pricing intelligence, superior trust from both buyers and sellers, quicker sales transactions, user procurement and retention, scalable and automated operations, versatile revenue opportunities, improved fraud detection and quality control, and a data advantage that compounds over time. It is clear that many Indian customers are more interested in buying used cars, and a tech start-up featured in this project, Cars4U, is seeking a foothold in the used car market. These customers are specifically interested in purchasing used cars from Cars4U at reliable prices, while sellers on Cars4U are simultaneously trying to sell used cars to generate revenue. Cars4U thus requires assistance from a Data Science team to identify the most critical factors (e.g., mileage, brand, model, year) that influence the actual value of used cars in their inventory. A pricing model that can predict fair values for used cars can empower Cars4U sellers to set competitive prices for their used-car inventory, resulting in quicker sales and preventing exploitation by mediators. In the same manner, if Cars4U can effectively utilize a pricing model, it can outperform its competitors by attracting more sellers and buyers, reducing negotiation conflicts, leveraging assets, and building a transparent, favorable reputation. As such, a machine-learning-based pricing model can help Cars4U achieve price standardization, reduce pricing uncertainty, assist both buyers and sellers, and empower the company without relying on traditional methods such as manual appraisals.

A valuable additional viewpoint on Cars4U's business problem is the growing operational and financial risks posed by unstable pricing across a rapidly scaling inventory. As Cars4U grows, manual or intuition-driven appraisals can become steadily more unmanageable, leading to mispriced used cars that remain unsold for extended periods or are sold rapidly at significant losses. This results in direct revenue outflows, inefficient inventory turnover, and difficulty predicting supply needs. If Cars4U lacks a standardized, data-driven pricing configuration, it will be unable to adjust margins, plan appropriation, or maintain pricing consistency across Indian cities, where rival competitors may also employ algorithmic pricing models. As such, the business problem also involves correcting market opacity for customers, safeguarding Cars4U's internal economics, enabling predictable growth, and ensuring that pricing decision-making scales effectively as it expands.

Objectives

As stated before, the objective of this project is to develop a machine-learning pricing model for the Cars4U tech start-up that accurately predicts used-car prices and helps the business devise cost-effective differential pricing strategies. The main goal of Cars4U is to develop a trustworthy, data-driven machine-learning pricing model that accurately estimates the market value of any used car in its inventory. By developing such a machine learning model, the Cars4U company meets other goals such as allowing for clarity in a tumultuous market, entrusting sellers with fair cost estimates, raising customer trust through dependable pricing, boosting a competitive advantage in the Indian used car marketplace, enhancing platform competence, curtailing possible negotiation abrasions, and implementing extensible, automatic cost assessments.

Another objective that advances Cars4U's long-term strategy is the need to convert pricing from a volatile operational task into a forecasting intelligence capability that explains countless business functions beyond valuation. A well-constructed machine learning model should estimate accurate prices and can also aid Cars4U in extrapolating demand across Indian cities, anticipating inventory gaps, determining high-value used-car segments, and enhancing procurement decision-making before used cars appear on the Cars4U platform. By evolving used-car pricing into a forward-looking, systematic asset, Cars4U can enhance margin operations management, lower holding costs, and enable data-driven decision-making. Examples of these data-driven

decisions include which used cars to acquire, promote, or retire from Cars4U's inventory. These broader objective positions the machine learning pricing engine as a fundamental element of Cars4U's comprehensive growth engineering, rather than merely a strictly independent valuation instrument.

Key Questions

The curated key questions related to the problem that must be addressed for the Cars4U tech start-up to succeed are listed as follows. Foremost, key questions regarding insights into the used car market and business include: How large is the opportunity cost for Cars4U in the Indian used car market? What are the challenges and difficulties that Cars4U must face when setting and managing the prices of its used cars? How can Cars4U provide accurate used-car prices using a machine learning model to enhance trust in the platform? What are some competitive advantage factors that Cars4U avails by utilizing a pricing engine in the form of a machine learning model? In addition, key questions or important insights related to the provided Cars4U dataset include: Which features most strongly influence used-car prices? How do the depreciated values operate across brands and segments in Cars4U's used car inventory? Does a particular patterning of overpriced or underpriced cars exist in the Cars4U's provided dataset? How orderly and trustworthy is the available data? Furthermore, key questions regarding the modeling and performance of the chosen machine learning model include: Which model best balances accuracy, interpretability, and scalability? What is a decent error margin for real-world prices for used cars in India? How does the chosen machine learning model perform across various used car segmentations? Can the chosen machine learning model generalize to new, unseen used car model prices? Moreover, key questions regarding the used-car products and user experience include: How will Cars4U display predicted used-car prices to its customers and users? How will the machine learning model be integrated as a valuation instrument into the Cars4U platform? What forms of explanations will the machine learning model provide to Cars4U to build trustworthiness with both its Indian buyers and sellers? Lastly, a key question regarding Cars4U's strategic and operational insights is: How can Cars4U use pricing data to improve revenue? How will Cars4U regularly improve the chosen machine learning model? How can Cars4U build a durable, sustainable, and competitive advantage over its competitors?

Problem Formulation

As a data scientist, I would frame this as a supervised regression problem: Cars4U aims to address pricing unpredictability in the Indian used car market by predicting a car's fair market value from various features. Using historical data on used cars and their selling prices from Cars4U's provided dataset, a data scientist is trying to learn a function that predicts a car's fair market price based on its features. In particular, this function should generalize well to unseen car data and remain effective across various brands, models, years, and conditions. Data science best fits into the spectrum of solving the pricing problem of used cars through the following strategies: involving high-dimensional and nonlinear relationships by combining interacting used car features, assessing considerable variability and noise by uncovering patterns, generating used car prices in scalable and automatic means, and necessitating the need for transparency and trust through feature importance, price ranges, and confidence intervals. The nature of the data science problem can, hence, be described as a supervised learning regression task where a data scientist is trying to calculate the target numerical continuous variable, the price of a used car in Indian rupees (INR), by determining which input features or predictors from Cars4U's provided dataset, best determine a predicted price or price range for a used car in its used car inventory among the different performances of robust regression machine learning models. Among these input features or predictors include the categorical variables such as brand name and model name of the used car, the location which the used car is being sold or is available for purchase, fuel type of the used car, and type of transmission used by the car, numerical variables such as manufacturing year of the car, the total kilometers driven in the car by the previous owner or owners, the standard mileage offered by the car company in Kilometers Per Liter (KMPL) or Kilometers per kilogram (KM/KG), and displacement volume of the used car's engine in Cubic Centimeters (CC), and derived features such as age, and depreciation value.

Data Exploration

Data Description

In this section, the structure of the data from Cars4U's provided dataset is presented. The dataset includes 7253 unique used cars (each represented by a row) with 14 distinct features (e.g., independent variables).

Categorical Variables

The collected dataset includes 6 categorical features. The first is the “Serial Number” of a used car. The second is the “Name” of a used car, which includes the brand and model. The third is the “Location” (i.e., cities in India) where a used car is being sold or available for purchase. The fourth is “Fuel_Type,” the type of fuel (e.g., Petrol, Diesel, Electric, Compressed Natural Gas–CNG, and Liquefied Petroleum Gas–LPG) used by the car. The fifth is “Transmission,” which is the type of transmission (e.g., Automatic and Manual) used by the car. The sixth is “Owner,” which indicates the number of previous owners of a used car (e.g., First–one previous owner, Second–two previous owners, Third–three previous owners, and Fourth–four previous owners).

Numerical Variables

The collected dataset includes 8 distinct numerical features. The first is a discrete numerical feature, “Year,” which represents the car's manufacturing year. The second is “Kilometers_driven,” which is the total number of kilometers the previous owner(s) drove in the car. The third is “Mileage,” which is the standard mileage offered by the car company in Kilometers per liter (KMPL) or Kilometers per Kilogram (KM/KG). The fourth is “Engine,” which is the engine's displacement volume in Cubic Centimeters (cc). The fifth is “Power,” which is the engine's maximum power in Brake Horsepower (BHP). The sixth is another discrete numerical feature, “Seats,” which is the number of seats in the used car. The seventh is “New_Price,” which is the price of a new car of the same used-car model, in 100,000 Indian Rupees (INR). The eighth is “Price,” which is the price of a used car in 100,000 Indian Rupees (INR). In this dataset, Price is the primary target feature (i.e., the dependent variable).

Understanding the Structure of the Data from the Dataset

Table 1. Sample preview of the First 5 Rows in the dataset

	S. No.	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_price	Price
0	0	Maruti Wagon R LXI CNG	Mumbai	2010	72000	CNG	Manual	First	26.60	998.0	58.16	5.0	NaN	1.75

	S. No.	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_price	Price
1	1	Hyundai Creta 1.6 CRDi SX Option	Pune	2015	41000	Diesel	Manual	First	19.67	1582.0	126.20	5.0	NaN	12.50
2	2	Honda Jazz V	Chennai	2011	46000	Petrol	Manual	First	18.20	1199.0	88.70	5.0	8.61	4.50
3	3	Maruti Ertiga VDI	Chennai	2012	87000	Diesel	Manual	First	20.77	1248.0	88.76	7.0	NaN	6.00
4	4	Audi A4 New 2.0 TDI Multitronic	Coimbatore	2013	40670	Diesel	Automatic	Second	15.20	1968.0	140.80	5.0	NaN	17.74

As noted earlier in the description of both categorical and numerical variables, the table shows each variable featured in separate columns from left to right as: S.No (e.g., Serial Number), Name (e.g., name of the used car including brand and model name), Location (e.g., the location at which the used car is being sold or is available for purchase in separate Indian cities), Year (e.g., manufacturing year of the car), Kilometers_driven (e.g., The total kilometers driven in the car by the previous owner(s)), Fuel_Type (e.g., the type of fuel used by the used car in Petrol, Diesel, Electric, Compressed Natural Gas, and Liquefied Petroleum Gas), Transmission (e.g., the type of transmission used by the car (e.g., Automatic or Manual), Owner (e.g., the type of ownership (e.g., First–one previous owner, Second–two previous owners, Third–three previous owners, and Fourth–four previous owners) of a used car, Mileage (e.g., the standard mileage offered by the car company in Kilometers per liter (KMPL) or Kilometers per Kilogram (KM/KG), Engine (e.g., the displacement volume of the engine in Cubic Centimeters (CC), Power (e.g., the maximum power of the engine in

Horsepower (BHP), Seats (e.g., the number of seats in the used car), New_Price (e.g., the price of a new car of the same model in 100,000 Indian Rupees (INR)) and Price (e.g,. the price of a used car in 100,000 Indian Rupees (INR)).

Table 2. Sample preview of the Last 5 Rows in the dataset

	S. No.	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_price	Price
7248	7248	Volkswagen Vento Diesel Trendline	Hyderabad	2011	89411	Diesel	Manual	First	20.54	1598.0	103.6	5.0	NaN	NaN
7249	7249	Volkswagen Polo GT TSI	Mumbai	2015	59000	Petrol	Automatic	First	17.21	1197.0	103.6	5.0	NaN	NaN
7250	7250	Nissan Micra Diesel XV	Kolkata	2012	28000	Diesel	Manual	First	23.08	1461.0	63.1	5.0	NaN	NaN
7251	7251	Volkswagen Polo GT TSI	Pune	2013	52262	Petrol	Automatic	Third	17.20	1197.0	103.6	5.0	NaN	NaN
7252	7252	Mercedes-Benz E-Class 2009-2013 E 220 CDI Avan...	Kochi	2014	72443	Diesel	Automatic	First	10.00	2148.0	170.0	5.0	NaN	NaN

Similarly, the last five rows in the dataset follow the same tabular structure as the first five rows, beginning with Serial Number and ending with Price.

Shape of the Dataset

Code Output 1. Shape of the Dataset

(7253, 14)

Observations on the Shape of the Dataset

The dataset has 7253 rows and 14 columns. The collected dataset provides a comprehensive, medium-sized set of observations (rows) for each column (independent variables, including the target dependent variable). 80/20 split

Data Types found in the Dataset

Table 3. Column Variables with associated Data Types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7253 entries, 0 to 7252
Data columns (total 14 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   S.No.              7253 non-null   int64
 1   Name               7253 non-null   object
 2   Location           7253 non-null   object
 3   Year               7253 non-null   int64
 4   Kilometers_Driven  7253 non-null   int64
 5   Fuel_Type          7253 non-null   object
 6   Transmission       7253 non-null   object
 7   Owner_Type         7253 non-null   object
 8   Mileage            7251 non-null   float64
 9   Engine             7207 non-null   float64
 10  Power              7078 non-null   float64
 11  Seats              7200 non-null   float64
 12  New_price          1006 non-null   float64
 13  Price              6019 non-null   float64
dtypes: float64(6), int64(3), object(5)
memory usage: 793.4+ KB
```

Observations on Dataset Types

The dataset provided has 14 columns (variables). It is important to indicate that the S.No. column is an integer type. The Name, Location, Fuel_Type, Transmission, and

Owner_Type variables are all of type object. The Mileage, Engine, Power, Seats, New_price, and Price variables are all of type float—the S.No. The column is initially dropped from further analyses since it serves as an index for data entries and would not provide any predictive power.

Missing Values in the Dataset

Table 4. Number of Missing Values in the Dataset

	0
S.No.	0
Name	0
Location	0
Year	0
Kilometers_Driven	0
Fuel_Type	0
Transmission	0
Owner_Type	0
Mileage	2
Engine	46
Power	175
Seats	53
New_price	6247
Price	1234

dtype: int64

Observations on Missing Values in the Dataset

Missing values are present in the Mileage, Engine, Power, Seats, New_Price, and Price variables in the dataset. There are 2 missing values for the Mileage variable. There are 46 missing values for the Engine variable. There are 175 missing values for the Power variable. There are 53 missing values for the Seats variable. There are 6247 missing values for the New_price variable. There are also 1234 missing values for the target Price variable. These findings suggest that the dataset was not properly collected, possibly due to input errors or insufficient sampling. In other words, the

analysis in this report is not fully polished and may not yield reliable insights and conclusions that support accurate, justified recommendations.

Duplicate Values in the Dataset

Code Output 2. Number of Duplicate values in the Dataset

np.int64(0)

Observations on Duplicate Values in the Dataset

No duplicate entries were found across all columns in the dataset. This finding suggests that the dataset was collected accurately, without duplicate entries.

Exploratory Data Analysis (EDA)

Summary Descriptive Statistics for Numerical Variables

Table 5. Summary Descriptive Statistics for all Numerical Variables

	count	mean	std	min	25%	50%	75%	max
Year	7253.0	2013.37	3.25	1996.00	2011.00	2014.00	2016.00	2019.00
Kilometers_Driven	7253.0	58699.06	84427.72	171.00	34000.00	53416.00	73000.00	6500000.00
Mileage	7251.0	18.14	4.56	0.00	15.17	18.16	21.10	33.54
Engine	7207.0	1616.57	595.29	72.00	1198.00	1493.00	1968.00	5998.00
Power	7078.0	112.77	53.49	34.20	75.00	94.00	138.10	616.00
Seats	7200.0	5.28	0.81	2.00	5.00	5.00	5.00	10.00
New_price	1006.0	22.78	27.76	3.91	7.88	11.57	26.04	375.00
Price	6019.0	9.48	11.19	0.44	3.50	5.64	9.95	160.00

Descriptive Statistics for Year (numerical discrete variable)

The minimum Year of a used car is 1996. The average Year of a used car is approximately 2013. The maximum Year of a used car is 2019. The table indicates that the most used cars are from 2011 through 2016. There are also a few very old used cars from 1996 through 2000.

Descriptive Statistics for Kilometers_Driven (numerical continuous variable measured in KM)

The minimum total kilometers driven of a used car by the previous owner is about 171 kilometers. The average total kilometers driven by the previous owner of a used car is 58,700. The maximum total kilometers driven by the previous owner of a used car is 6,500,000 kilometers. The maximum amount is most likely a data-entry error, since it is unrealistic for a used car to have over 6,500,000 kilometers on the odometer. Furthermore, 75 percent of used cars have been driven by previous owners for fewer than 73,000 kilometers, indicating a tail with few outliers.

Descriptive Statistics for Mileage (numerical continuous variable measured in KMPL or KM/KG)

The minimum standard mileage offered by the car company for a used car is 0. The minimum amount may be inaccurate, as it is unrealistic for a used car to have 0 miles. This is most likely due to a data entry error or to missing data. The average standard mileage for a used car from the car company is 18.14. The maximum standard mileage offered by the car company for a used car is 33.54. The distribution of standard mileage is roughly symmetric. The upper range of the standard mileage offered by the car company for a used car is credible for small diesel cars.

Descriptive Statistics for Engine (numerical continuous variable measured in CC)

The minimum displacement volume of the engine of a used car is 72 CC. The average displacement volume of the engine of a used car is about 1616 CC. The maximum displacement volume of the engine of a used car is 5998 CC. There is evidence of a strong right skew, probably due to the use of luxury or performance cars, with ranges from 3,000 to 6,000 CC.

Descriptive Statistics for Power (numerical continuous variable measured in BHP)

The minimum engine power of a used car is 34.20 BHP. The average engine power of a used car is 112.77 BHP. The maximum Power of the engine of a used car is 616 BHP. There is evidence of a right skew due to high-performance cars with 400 to 600 BHP.

Descriptive Statistics for Seats (numerical discrete variable)

The minimum number of seats in a used car is 2. The average number of seats in a used car is 5. The maximum number of seats in a used car is 10. There is evidence of a high concentration of 5-seat used cars, with very low variance.

Descriptive Statistics for New_price (numerical continuous variable measured in INR 100,000)

The minimum new price for a car of the same model is 100,000 INR. The average new price of a new car of the same model is 22.78 INR 100,000. The maximum new price for a new car of the same model is 100,000 INR. There is evidence of a strong right skew due to luxury used cars.

Descriptive Statistics for Price (numerical target continuous variable in INR 100,000)

The minimum Price of a used car is 0.44 INR 100,000. The average Price of a used car is 9.48 INR 100,000. The maximum price of a used car is 100,000 INR. There is evidence of a strong right skew due to luxury used cars. The minimum value is credible for old hatchback used cars.

Summary Descriptive Statistics for Categorical Variables

Table 6. Summary Descriptive Statistics for all Categorical Variables

	count	unique	top	freq
Name	7253	2041	Mahindra XUV500 W8 2WD	55
Location	7253	11	Mumbai	949
Fuel_Type	7253	5	Diesel	3852
Transmission	7253	2	Manual	5204
Owner_Type	7253	4	First	5952

Descriptive Statistics for Name

There are 2,041 unique car brands and model names across all used cars. Among the most frequently used car brands and names is the Mahindra XUV500 W8 2WD. There is also evidence of a long-tailed distribution. It is preferable to extract the brand and model found in each used car for further analysis.

Descriptive Statistics for Location

There are 11 unique locations, each a city in India. Among the cities, Mumbai is the most frequent location, with over 949 occurrences in the dataset, suggesting that it is a major market hub. This could also imply that prices might be higher at this location due to demand.

Descriptive Statistics for Fuel_Type

There are a total of 5 unique Fuel Types across each used car. Among fuel types, Diesel is the most frequent, with over 3,852 occurrences. This observation resonates with Indian market patterns, where diesel cars are usually held at higher resale values.

Descriptive Statistics for Transmission

There are 2 unique Transmissions across all used cars. Of the two transmission types, Manual is the most frequent, with over 5,204 occurrences. This observation aligns with the fact that Automatic used cars are less common and usually cost more.

Descriptive Statistics for Owner_Type

There are a total of 4 unique Owner_Type categories across each used car. Of the four unique Owner Types, the most frequent is First, with over 5,952 occurrences. This observation is common since first-owner used cars maintain their price value, and are easier to sell.

Table 7. Unique Number of observations in each categorical variable

```
For column: Name
['Maruti Wagon R LXI CNG' 'Hyundai Creta 1.6 CRDi SX Option'
 'Honda Jazz V' ... 'Ford EcoSport 1.5 Petrol Ambiente'
 'Jeep Compass 1.4 Sport' 'Hyundai Elite i20 Magna Plus']
Number of unique values: 2041
Name
```

```
Mahindra XUV500 W8 2WD                    55
Maruti Swift VDI                          49
Maruti Swift Dzire VDI                    42
Honda City 1.5 S MT                       39
Maruti Swift VDI BSIV                     37
                                          ..
Mercedes-Benz E-Class E240 V6 AT           1
Nissan Terrano XE 85 PS                    1
Mahindra TUV 300 2015-2019 T8 AMT          1
Hyundai i20 2015-2017 1.4 CRDi Sportz      1
Hyundai Creta 1.6 VTVT Base                1
Name: count, Length: 2041, dtype: int64
--------------------------------------------------
For column: Location
['Mumbai' 'Pune' 'Chennai' 'Coimbatore' 'Hyderabad' 'Jaipur'
'Kochi'
 'Kolkata' 'Delhi' 'Bangalore' 'Ahmedabad']
Number of unique values: 11
Location
Mumbai        949
Hyderabad     876
Coimbatore    772
Kochi         772
Pune          765
Delhi         660
Kolkata       654
Chennai       591
Jaipur        499
Bangalore     440
Ahmedabad     275
Name: count, dtype: int64
--------------------------------------------------
For column: Fuel_Type
['CNG' 'Diesel' 'Petrol' 'LPG' 'Electric']
Number of unique values: 5
Fuel_Type
Diesel      3852
Petrol      3325
CNG           62
LPG           12
Electric       2
Name: count, dtype: int64
--------------------------------------------------
For column: Transmission
['Manual' 'Automatic']
Number of unique values: 2
Transmission
```

```
Manual       5204
Automatic    2049
Name: count, dtype: int64
--------------------------------------------------
For column: Owner_Type
['First' 'Second' 'Fourth & Above' 'Third']
Number of unique values: 4
Owner_Type
First             5952
Second            1152
Third              137
Fourth & Above      12
Name: count, dtype: int64
--------------------------------------------------
```

Number of unique observations for Name

As noted earlier, there are over 2,041 unique names for each used car in the dataset. Among the top 5 most frequently used car names are the Mahindra XUV500 W8 2WD, Maruti Swift VDI, Maruti Swift Dzire VDI, Honda City 1.5 S MT, and Maruti Swift VDI BSIV, with frequencies of 55, 49, 42, 39, and 37, respectively.

Number of unique observations for Location

As noted earlier, there are 11 unique cities in India where used cars are sold. Aside from Mumbai, the other locations are Hyderabad, Coimbatore, Kochi, Pune, Delhi, Kolkata, Chennai, Jaipur, Bangalore, Ahmedabad, where each location has a total of 949, 876, 772, 772, 765, 660, 654, 591, 499, 440, and 275 frequencies, respectively.

Number of unique observations for Fuel_Type

As noted earlier, there are 5 unique Fuel Type categories for each used car. Aside from Diesel, the other Fuel Types are Petrol, CNG, LPG, and Electric, with frequencies of 3,852, 3,325, 62, 12, and 2, respectively.

Number of unique observations for Transmission

As noted earlier, there are 2 unique Transmission categories for each used car. Aside from Manual, the other Transmission type is Automatic. Manual has 5,204 frequencies, and Automatic has 2,049.

Number of unique observations for Owner_Type

As noted earlier, there are 5 unique Owner Type categories for each used car. Aside from the First category, the other Owner Types are Second, Third, and Fourth & Above, with frequencies of 5,952, 1,152, 137, and 12, respectively.

Checking for Kilometers_Driven extreme values

Table 8. Kilometers_Driven in Descending Order

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_price	Price
2328	BMW X5 xDrive 30d M Sport	Chennai	2017	6500000	Diesel	Automatic	First	15.97	2993.0	258.00	5.0	NaN	65.00
340	Skoda Octavia Ambition Plus 2.0 TDI AT	Kolkata	2013	775000	Diesel	Automatic	First	19.30	1968.0	141.00	5.0	NaN	7.50
1860	Volkswagen Vento Diesel Highline	Chennai	2013	720000	Diesel	Manual	First	20.54	1598.0	103.60	5.0	NaN	5.90
358	Hyundai i10 Magna 1.2	Chennai	2009	620000	Petrol	Manual	First	20.36	1197.0	78.90	5.0	NaN	2.70
2823	Volkswagen Jetta 2013-2015 2.0L TDI Highline AT	Chennai	2015	480000	Diesel	Automatic	First	16.96	1968.0	138.03	5.0	NaN	13.00

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_price	Price
3092	Honda City i VTEC SV	Kolkata	2015	480000	Petrol	Manual	First	17.40	1497.0	117.30	5.0	NaN	5.00
4491	Hyundai i20 Magna Optional 1.2	Bangalore	2013	445000	Petrol	Manual	First	18.50	1197.0	82.90	5.0	NaN	4.45
6921	Maruti Swift Dzire Tour LDI	Jaipur	2012	350000	Diesel	Manual	First	23.40	1248.0	74.00	5.0	NaN	NaN
3649	Tata Indigo LS	Jaipur	2008	300000	Diesel	Manual	First	17.00	1405.0	70.00	5.0	NaN	1.00
1528	Toyota Innova 2.5 G (Diesel) 8 Seater BS IV	Hyderabad	2005	299322	Diesel	Manual	First	12.80	2494.0	102.00	8.0	NaN	4.00

Observations and Implications on Kilometers_Driven in Descending Order Table

According to the Kilometers_Driven table in descending order, there is evidence of an extreme outlier in the top entry: the BMW X5 xDrive 30d M Sport (manufactured as recently as 2017) used car has been driven over 6,500,000 kilometers by a previous owner. This extreme value is not plausible and can be considered a data entry error, so that this row can be removed from the dataset.

Checking for Mileage extreme values

Table 9. Mileage in Ascending Order

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_price	Price
7226	Hyundai	Ahmedabad	2014	41000	Petrol	Manual	First	0.0	1086.0	62.0	5.0	NaN	NaN

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_price	Price
	Santro Xing GL												
5647	Toyota Qualis Fleet A3	Mumbai	2001	227000	Diesel	Manual	Fourth & Above	0.0	2446.0	NaN	8.0	NaN	2.20
707	Mercedes-Benz M-Class ML 350 4Matic	Pune	2014	120000	Diesel	Automatic	First	0.0	2987.0	165.0	5.0	NaN	30.00
6697	Hyundai Santro Xing XL	Jaipur	2007	85000	Petrol	Manual	Second	0.0	1086.0	NaN	5.0	NaN	NaN
6093	Hyundai Santro Xing XL	Bangalore	2007	47000	Petrol	Manual	Second	0.0	1086.0	NaN	5.0	NaN	NaN
2597	Hyundai Santro Xing XP	Pune	2007	70000	Petrol	Manual	First	0.0	1086.0	NaN	5.0	NaN	1.12
2267	Toyota Qualis RS E2	Pune	2004	215750	Diesel	Manual	Second	0.0	2446.0	NaN	10.0	NaN	3.50

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_price	Price
1460	Land Rover Range Rover Sport 2005 2012 Sport	Coimbatore	2008	69078	Petrol	Manual	First	0.0	NaN	NaN	NaN	NaN	40.88
3061	Hyundai Santro GS	Ahmedabad	2005	58000	Petrol	Manual	Second	0.0	999.0	NaN	5.0	NaN	1.51
6090	Hyundai Santro Xing GL	Ahmedabad	2013	63831	Petrol	Manual	First	0.0	1086.0	62.0	5.0	NaN	NaN

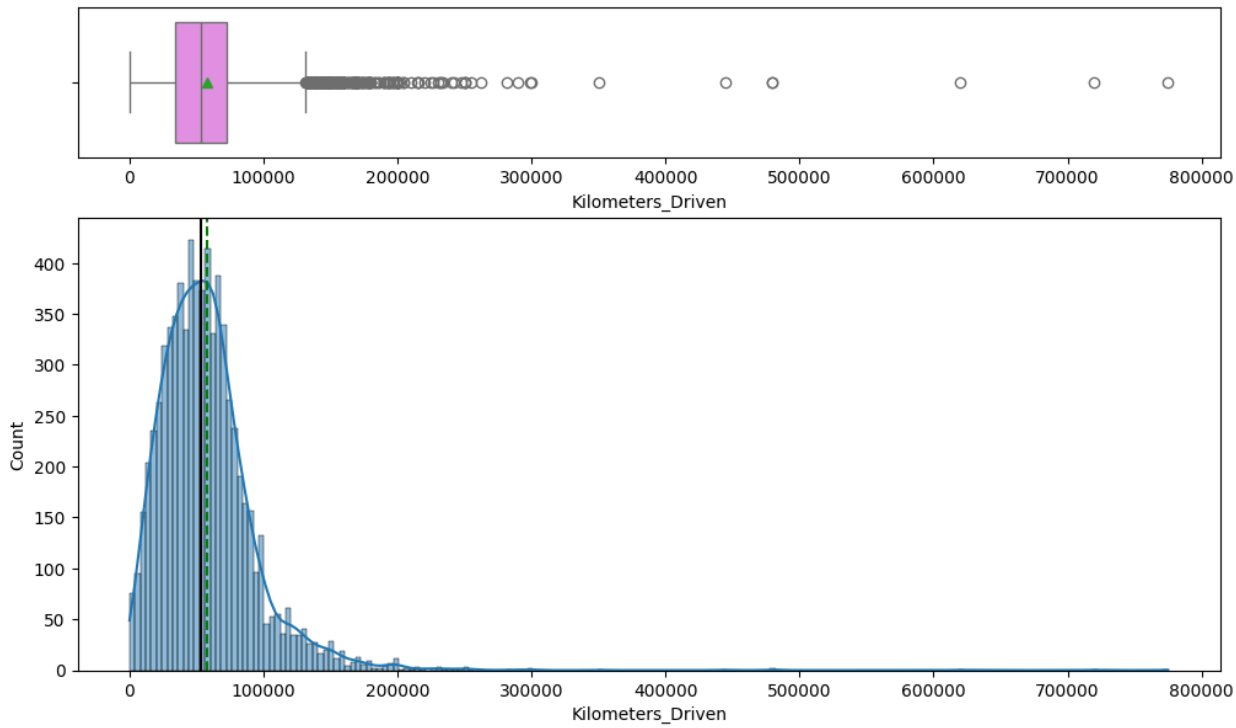
Observations and Implications on Mileage in Ascending Order Table

According to the Mileage in ascending order table, there is evidence of a series of Mileage values of “0.0,” which indicates an implausible fact where the mileage of cars cannot be 0. Due to these values, they will be treated as missing and imputed later in the analyses.

Univariate Analysis

A univariate analysis was conducted to examine the nature, distribution, and implications of each variable in the dataset. The median is the middle value in a dataset, splitting the data into two halves. A combined histogram and boxplot function was used to graph the remaining numerical variables: Year, Kilometers_Driven, Mileage, Engine, Power, Seats, New_Price, and Price. A bar plot function was used to graph the four categorical variables (excluding the Name variable until it is properly feature engineered): Location, Fuel_Type, Transmission, and Owner_Type.

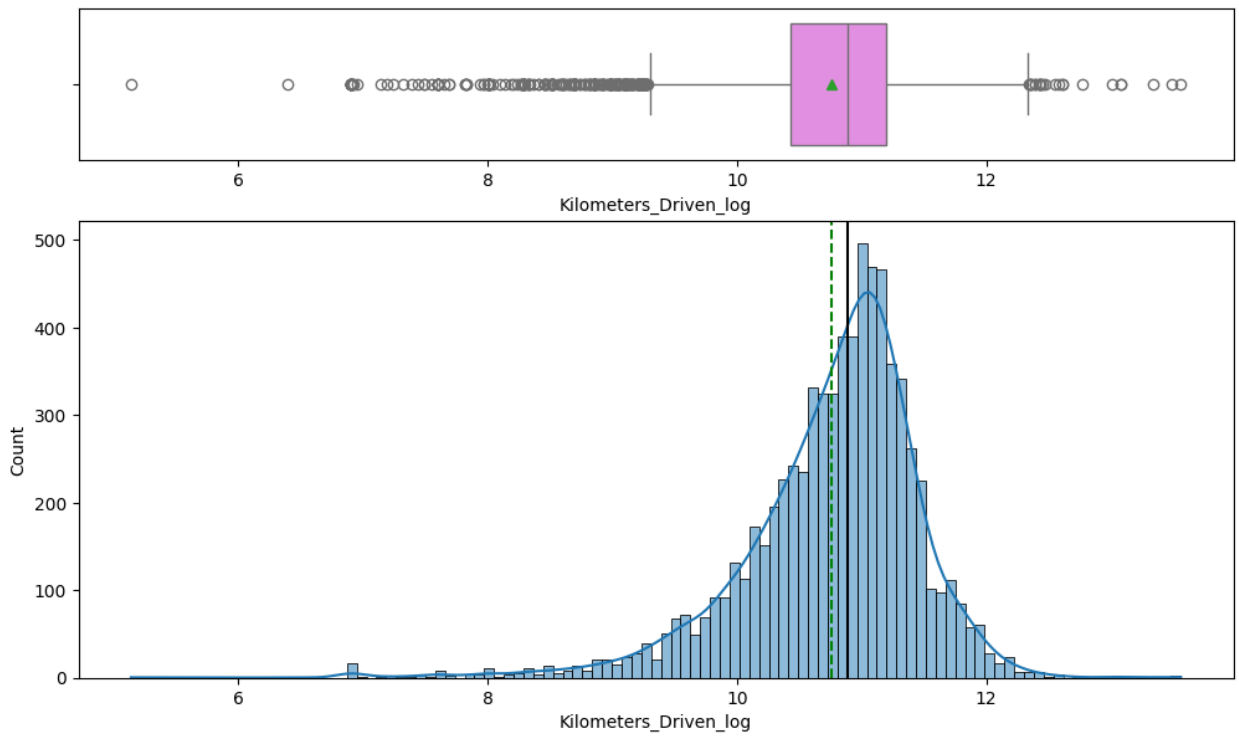
Graph 1. Kilometers_Driven Boxplot and Histogram



Observations and Implications on Kilometers_Driven Boxplot and Histogram

The histogram and boxplot of the Kilometers_Driven numerical variable are heavily right-skewed. The majority of used cars driven by previous owners have relatively low to moderate mileage, below 100,000 kilometers, as shown in the boxplot. There are many used cars with extremely high mileage, above 200,000 kilometers, and some even have driven close to 700,000 to 800,000 kilometers. This suggests that these outlier values represent used cars that are commercialized vehicles, fleet cars, long-distance highway usage, and Diesel SUVs with longer engine longevity. The median (denoted by a solid black line in the histogram) is at 53,404 Kilometers Driven, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is at 57,810.85 Kilometers Driven. The histogram indicates that the most commonly used cars have fewer than 100,000 kilometers on the odometer. This is common, as most customers prefer to purchase used cars with fewer than 100,000 kilometers on the odometer. The used cars in the right tail of the distribution indicate that these cars have significantly low prices.

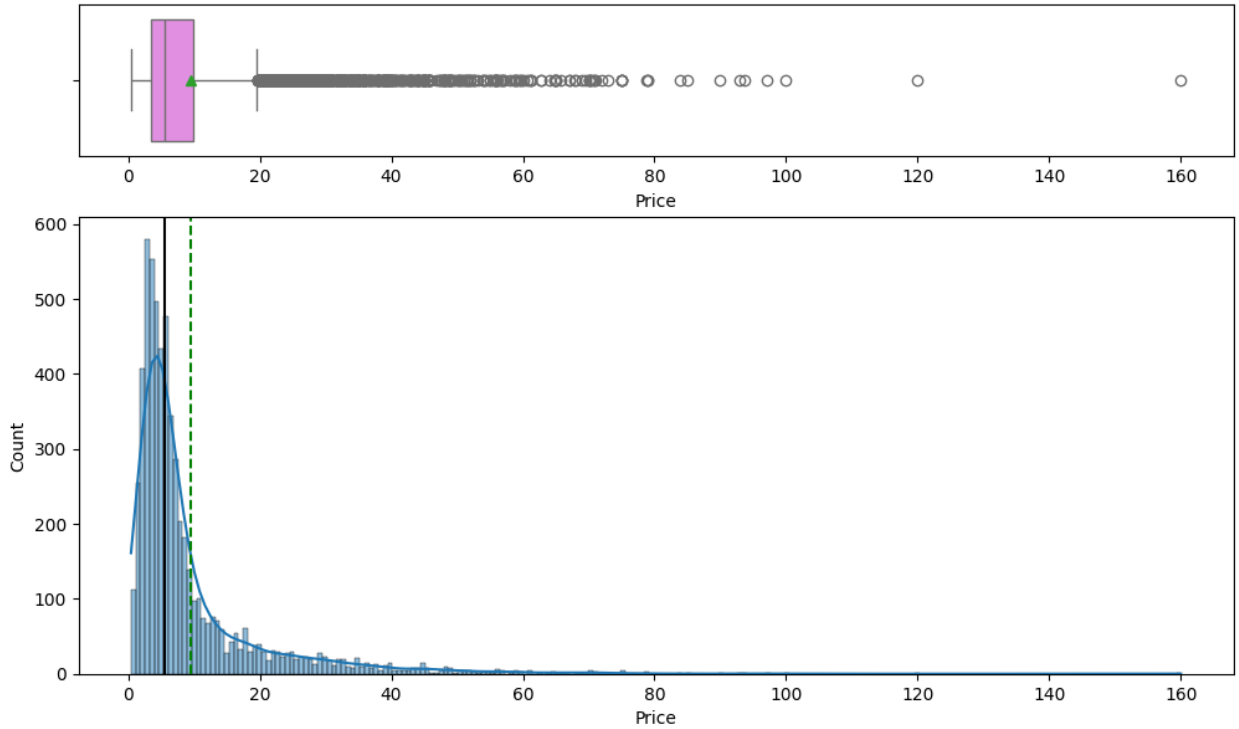
Graph 2. Kilometers_Driven Log transformed Histogram and Boxplot



Observations and Implications on log-transformed Kilometers_Driven_log Boxplot and Histogram

Given the importance of this independent variable (i.e., feature), it must undergo a log transformation to reduce or eliminate its skewness. Other reasons for this log transformation include that Kilometers_Driven spans several orders of magnitude and has a non-linear, diminishing-returns relationship with the target price variable. Also, applying a log transformation reduces leverage from extremely high-mileage cars, improves the linearity of the relationship with the target price variable, and enhances interpretability. Now, the median (denoted by a solid black line in the histogram) is at 10.89 Kilometers Driven (in log-transformed units), and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is at 10.76 Kilometers Driven (in log-transformed units). Regarding outliers, there are now far fewer than in the original non-log-transformed version. The histogram shows that the log-transformed variable is now roughly bell-shaped, after compressing large values and spreading out small values.

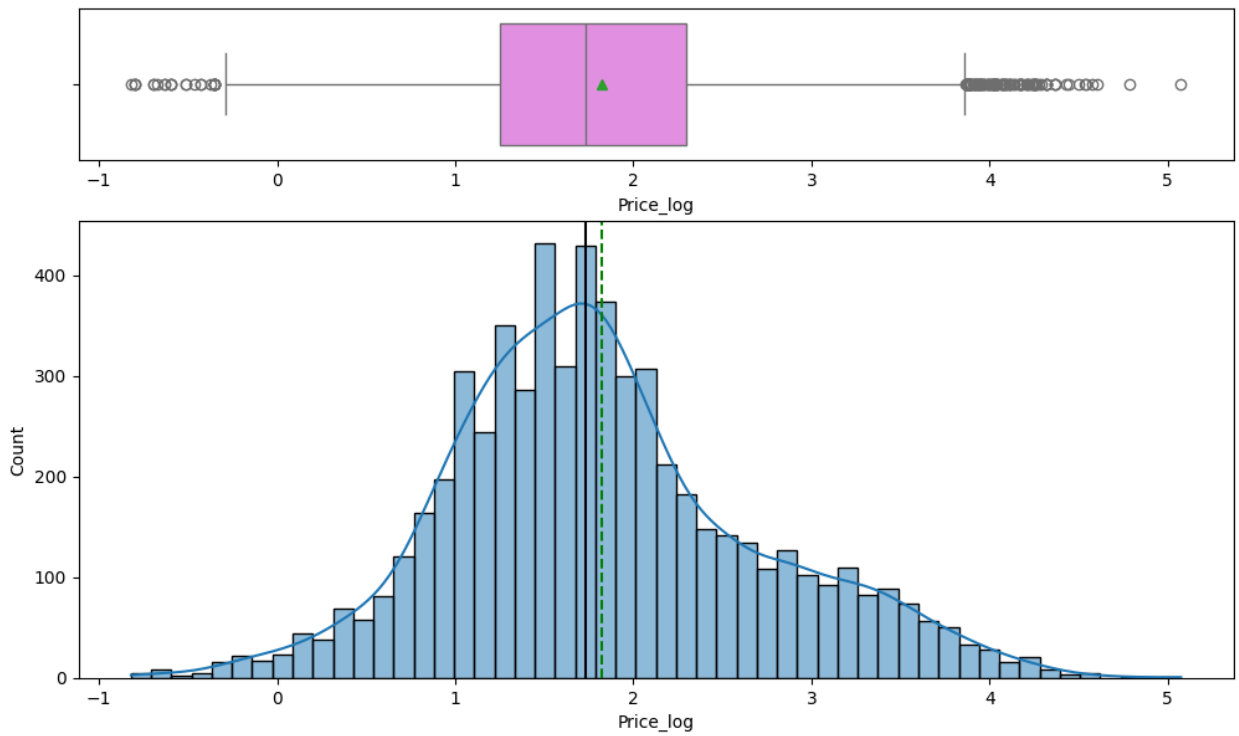
Graph 3. Price Boxplot and Histogram



Observations and Implications on Price Boxplot and Histogram

The boxplot and histogram of the Price target variable are heavily right-skewed and contain many outliers. Most used cars are priced lower, ranging from 0.44 to 5.64 INR 100,000, while a few are very expensive, reaching up to 160.00 INR 100,000. The median (denoted by a solid black line in the histogram) is at 9.48 INR 100,000, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 9.48 INR 100,000. The high-priced used cars valued as outliers include luxury-brand models (e.g., BMW, Mercedes, and Audi), newer used cars, high-trim SUVs, and premium diesel models. The histogram suggests that Cars4U's used car inventory primarily consists of budget and mid-range models. The mean is substantially higher than the median, suggesting that luxury and premium used-car models drive up the average price.

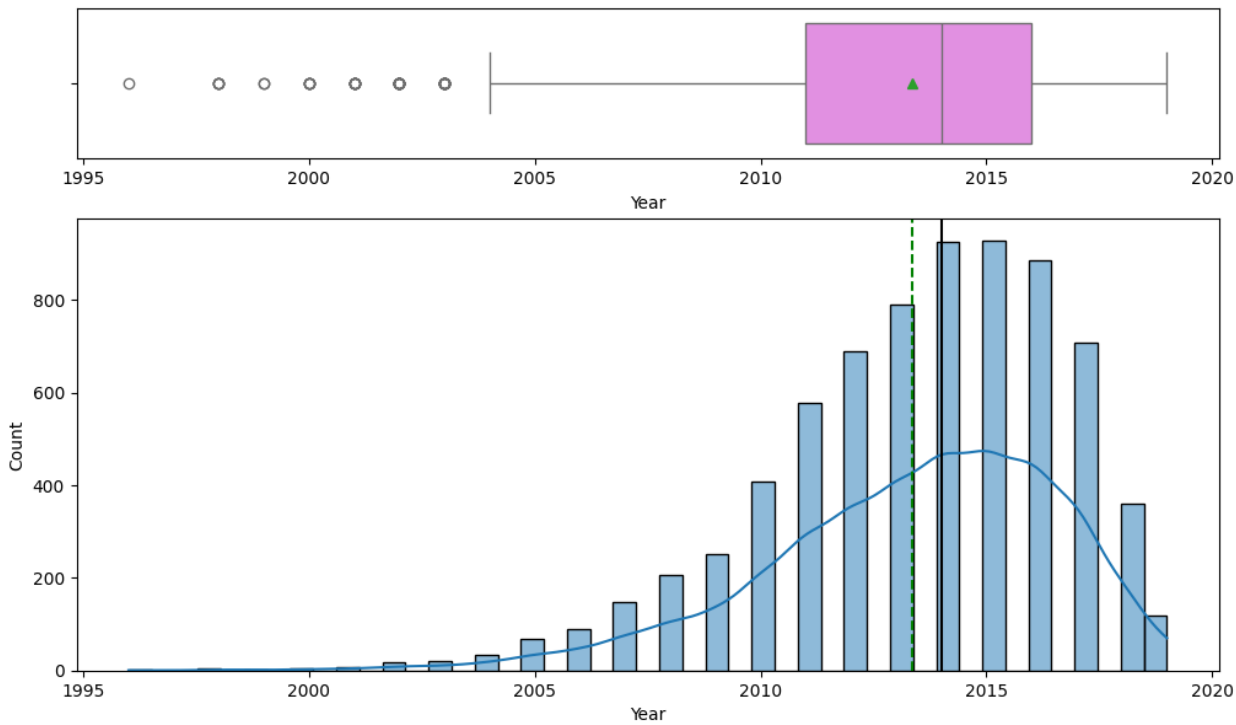
Graph 4. Price Log transformed Boxplot and Histogram



Observations and Implications on Price Log, transformed Boxplot, and Histogram

Because this variable is the target, it must undergo a log transformation to reduce skewness and improve model stability. Also, the distribution of the log-transformed Price variable becomes more symmetric, with reduced skewness and far fewer outliers. The median (denoted by a solid black line in the histogram) is 1.73 INR 100,000 (in log-transformed units), and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 1.82 INR 100,000 (in log-transformed units). The histogram now shows an approximately normal distribution, and the boxplot confirms that the mean and median are roughly equal. By applying the log transformation, there is now no variance and improved linearity of relationships with other independent variables (or predictors), indicating that the log-transformed Price variable is more suitable for regression modeling.

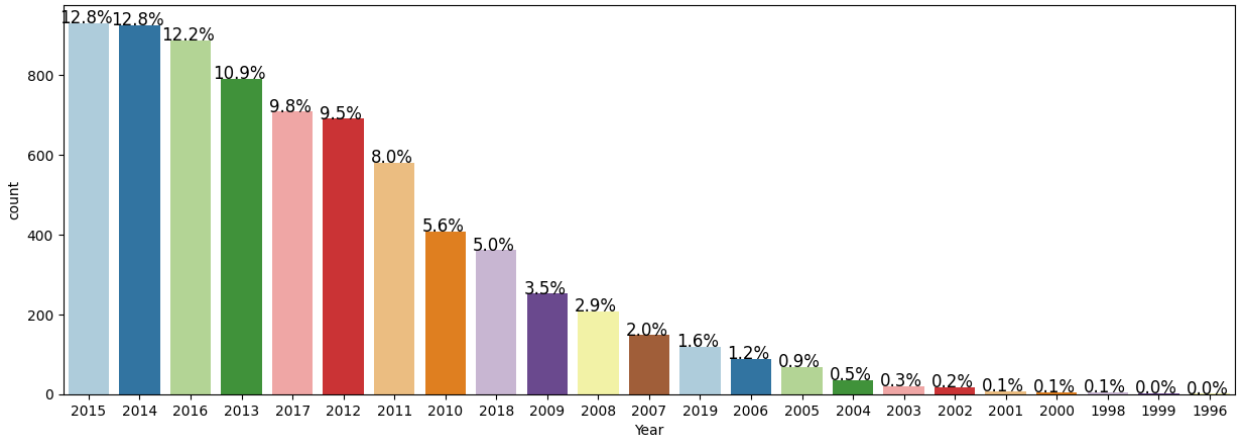
Graph 5. Year Boxplot and Histogram



Observations and Implications on Year Boxplot and Histogram

The boxplot and histogram for the Year numerical discrete variable show that the most used cars were manufactured between 2010 and 2017. This indicates that most used cars are between 8 and 14 years old, which implies mid-aged used cars. Outliers consist of a few older used cars from the late 1990s to early 2000s. These are rare, older vehicles that could indicate discontinued models or low-value listings. There is also a slightly left-skewed shape. The median (denoted by a solid black line in the histogram) is 2014, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 2013. The older used-car outliers, slight left-skewness, and tight Interquartile range (a measure of statistical dispersion representing the spread of the middle 50 percent of the dataset for a variable) all support a feature-engineered Car_Age numerical variable.

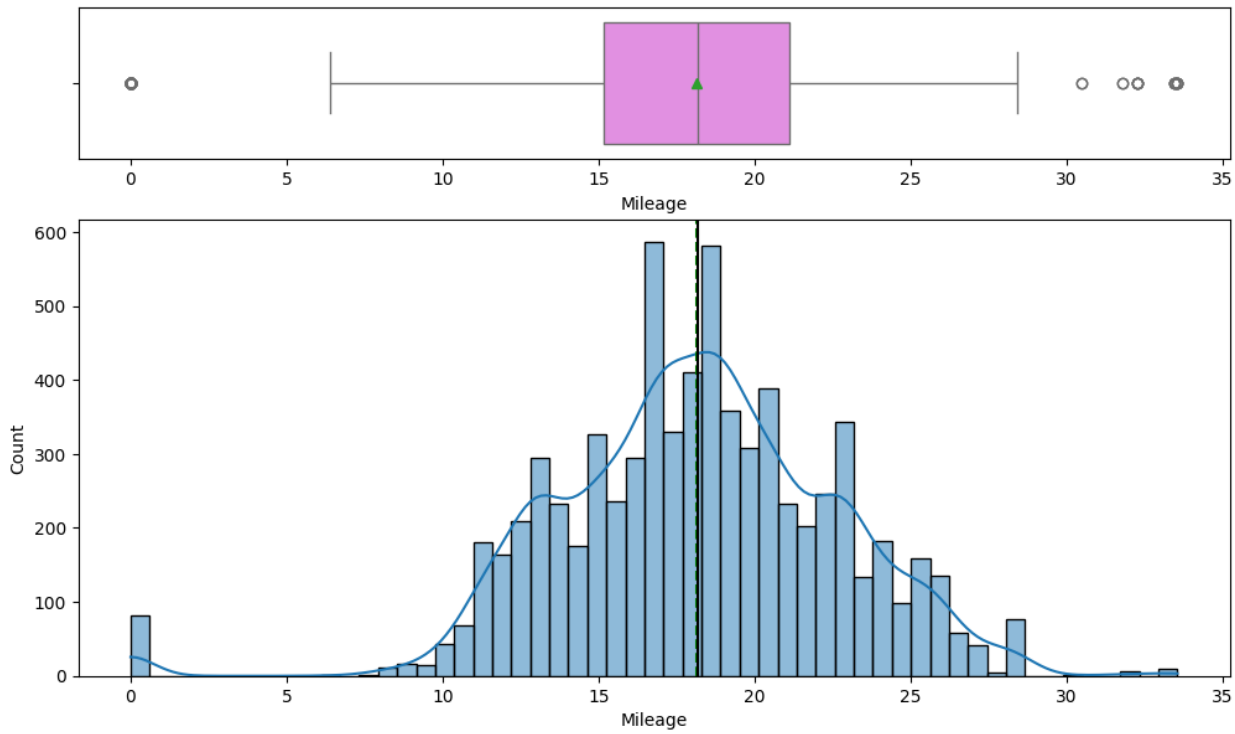
Graph 6. Year Barplot



Observations and Implications on Year Barplot

Because the Year independent variable is a discrete numerical variable, an alternative bar plot is provided to better visualize the frequencies of used cars by manufacturing year; here, it is much clearer that the distribution is right-skewed, with most used cars manufactured between 2010 and 2017. There is also significant dominance among used cars manufactured between 2013 and 2016, accounting for about 11 to 13 percent of all observations. In clearer terms, there is a strong concentration in the modern Indian used-car market rather than in older antique used cars. There are also significant declines in counts before 2010, and years before 2005 have lower inventories. This can be interpreted as older cars having little influence on the Indian used-car market, since newer used cars are more prevalent in most resale markets than older ones. Again, this bar plot further confirms that the Year variable follows a non-uniform distribution, with most used cars from the last decade of production, and further corroborates the need for a Car_Age engineered feature.

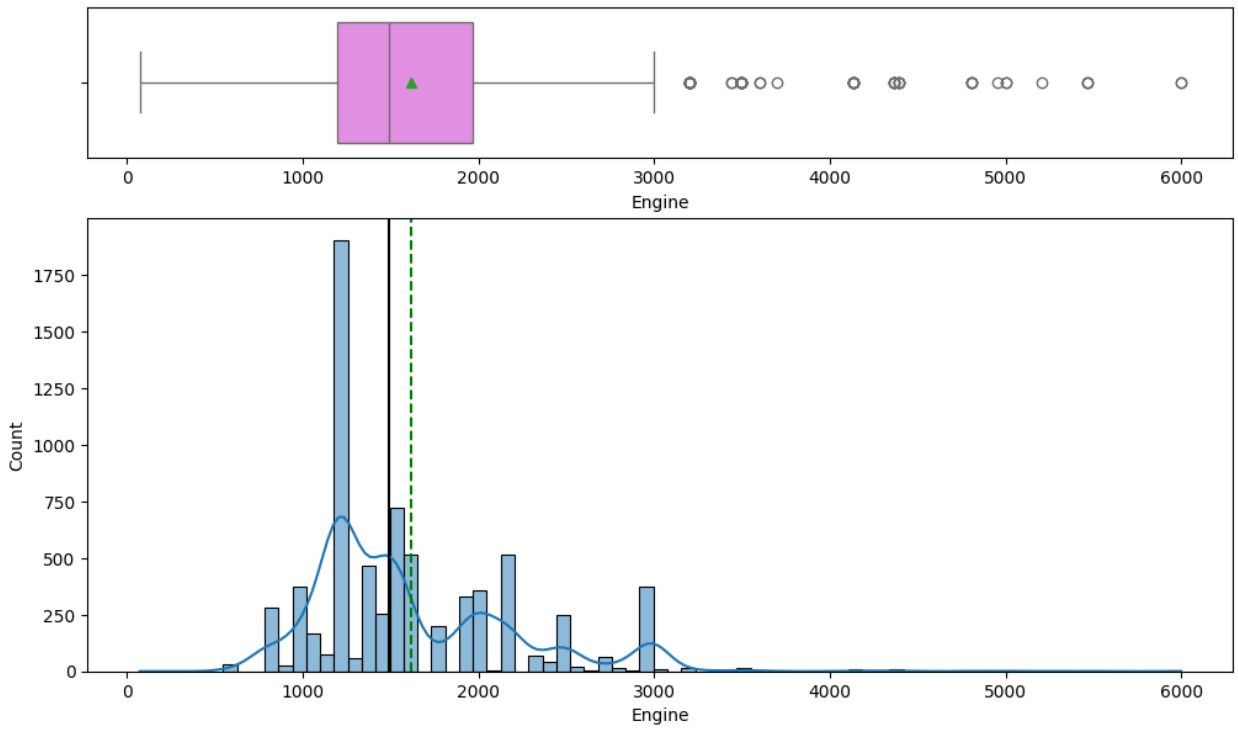
Graph 7. Mileage Boxplot and Histogram



Observations and Implications on Mileage Boxplot and Histogram

The boxplot and histogram for the Mileage continuous numerical variable show that the most commonly used cars have mileage between 15 and 22 KMPL (KM/KG). The distribution is roughly unimodal. There is a long right tail, indicating very high-end outliers. These outliers are a few extremely high-mileage used cars, which could be CNG or hybrid used cars with higher efficiency, or cars with mileage listed in KM/KG rather than KMPL. The median (denoted by a solid black line in the histogram) is 18.16 KM/KG or KMPL, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 18.14 KM/KG or KMPL. The median and mean both fall within the expected range for Indian used cars, suggesting a dominance of petrol and diesel cars. However, the core distribution for the mileage variable is stable and concentrated around realistic values.

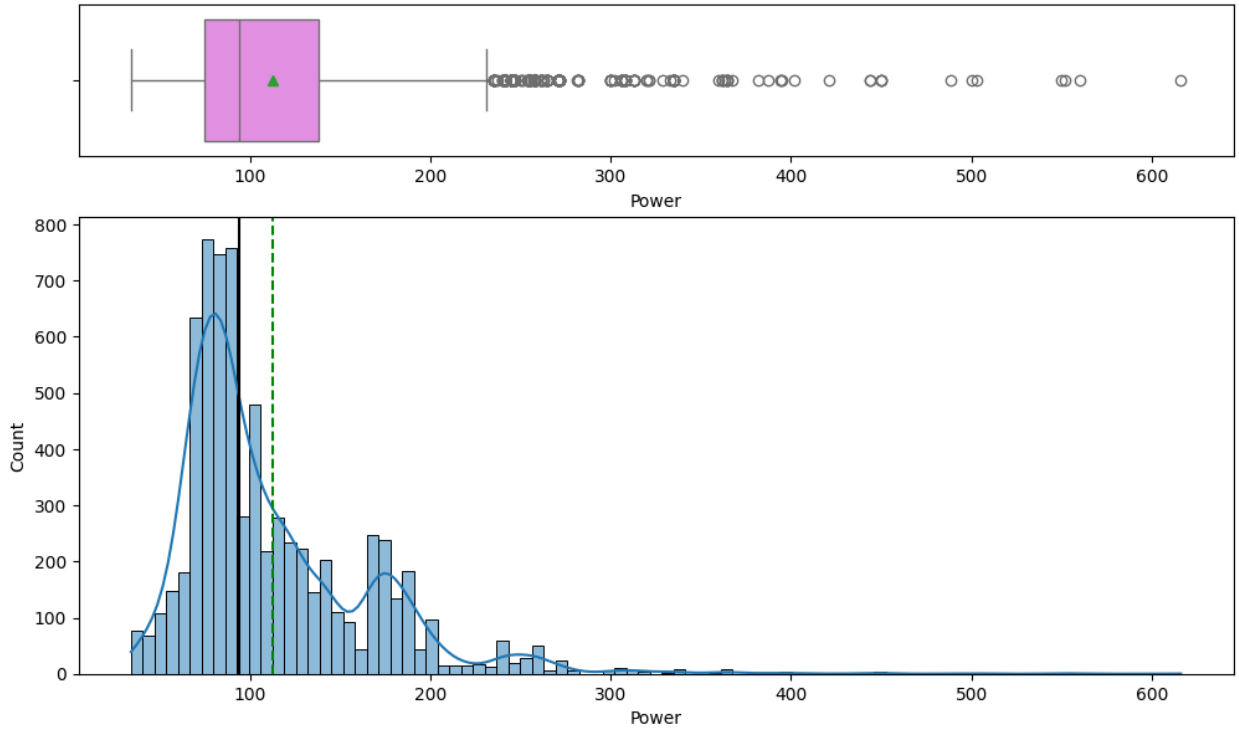
Graph 8. Engine Boxplot and Histogram



Observations and Implications on Engine Boxplot and Histogram

The boxplot and histogram of the Engine numerical variable indicate a right-skewed distribution. The core distribution of the Engine numerical variable is stable, centered around plausible values, and unimodal. Most used cars have engine sizes in the 1,000-1,600 CC range. There is a long right tail with higher engine capacities ranging from 2,000 to 6,000 CC, indicating that these used cars are likely typical Indian passenger cars. The majority of mass-market used cars have small- to mid-size engines, whereas used cars such as SUVs, performance cars, or luxury vehicles are outliers. The median (denoted by a solid black line in the histogram) is 1,493 CC, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 1,616.38 CC. These central tendency measures align with the Indian used car market, where hatchbacks and compact sedans dominate, while used cars with mid-size engines (1,500 through 2,000 CC) are less common, and large engines are uncommon.

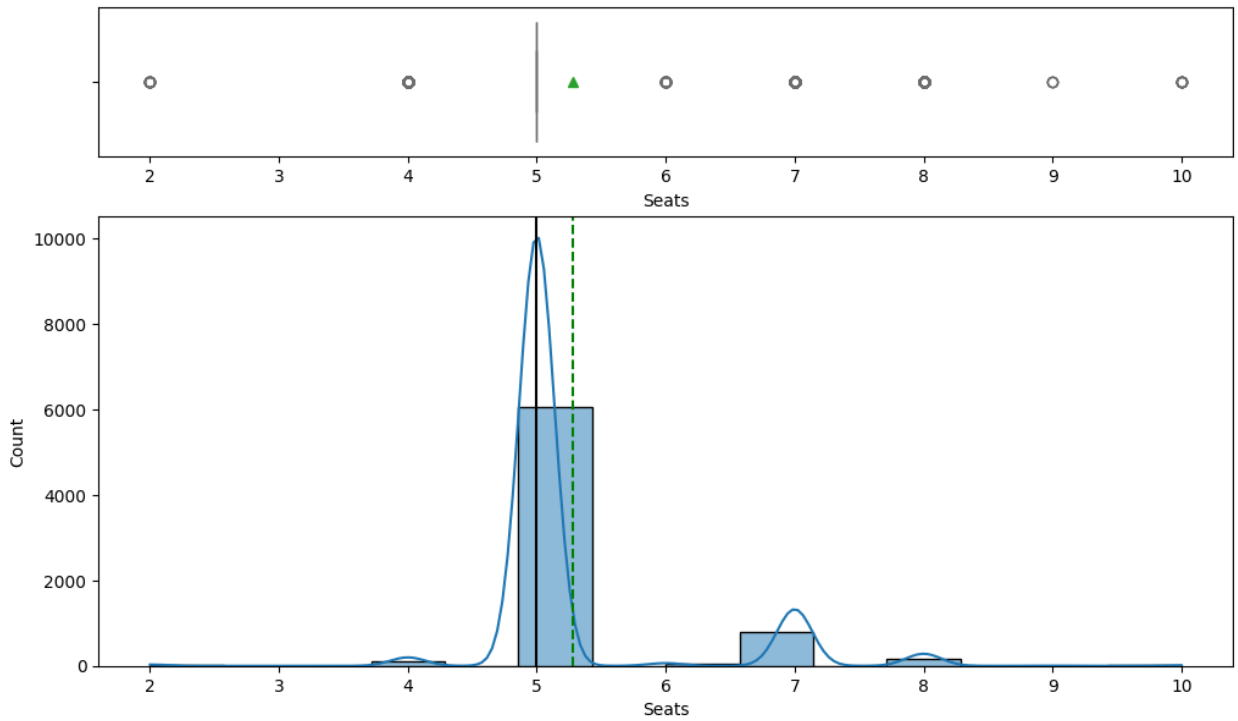
Graph 9. Power Boxplot and Histogram



Observations and Implications on Power Boxplot and Histogram

The boxplot and histogram for the Power numerical variable indicate a heavily right-skewed distribution. There is a large concentration of used cars with power falling between 70 and 110 BHP. There is also a long right tail of used cars with power values from 150 to 600 BHP. The right-skewed distribution is due to most Indian cars being small hatchbacks and sedans with moderate power. Again, there is evidence of a few high-performance or luxury used cars with higher power. The median (denoted by a solid black line in the histogram) is 94 BHP, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 112.77 BHP. These central tendency measures align with the Indian used-car market, since used cars usually have 1.2L petrol-powered engines producing 80 to 90 BHP and 1.5L diesel-powered engines producing 100 to 115 BHP, rather than larger, more powerful engines.

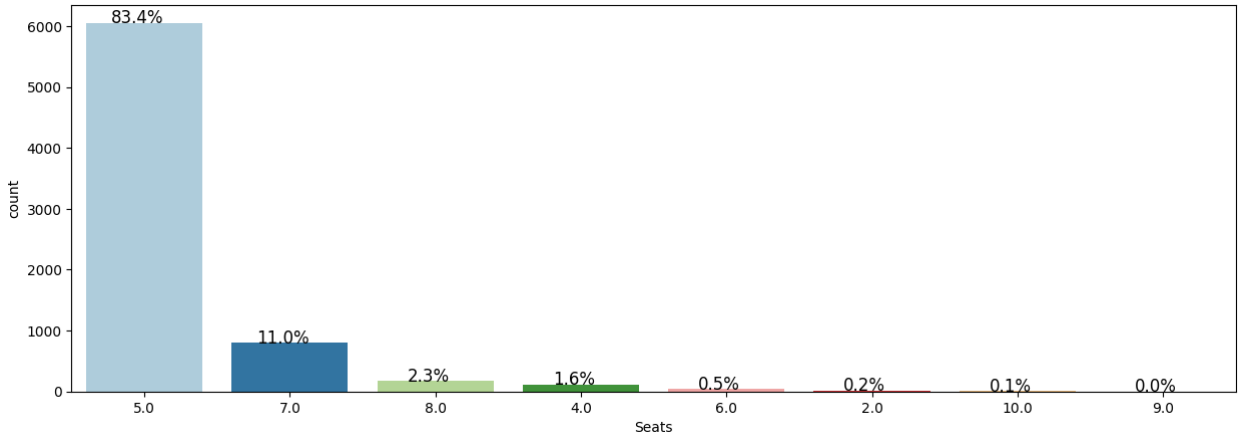
Graph 10. Seats Boxplot (Strip plot) and Histogram



Observations and Implications on Seats Boxplot (Strip plot) and Histogram

The seats boxplot (strip plot) and histogram indicate that it is a heavily discrete numerical variable, with used cars having 5 seats being the dominant category and those with 8 or more seats being less common. The histogram shows a large spike at 5 seats, and the boxplot (strip plot) indicates a dense cluster at 5 seats. The distribution is also right-skewed, with most used cars having 3 to 5 seats, and a small portion having more than 5 seats. The median (denoted by a solid black line in the histogram) is 5 seats, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 5.28 seats or about 5 seats. These central tendency measurements align with the Indian used-car market, in which most hatchbacks and sedans have 5 seats, SUVs or MPVs are less common and have 7 seats, and used cars with 4 or 6 seats are the least common.

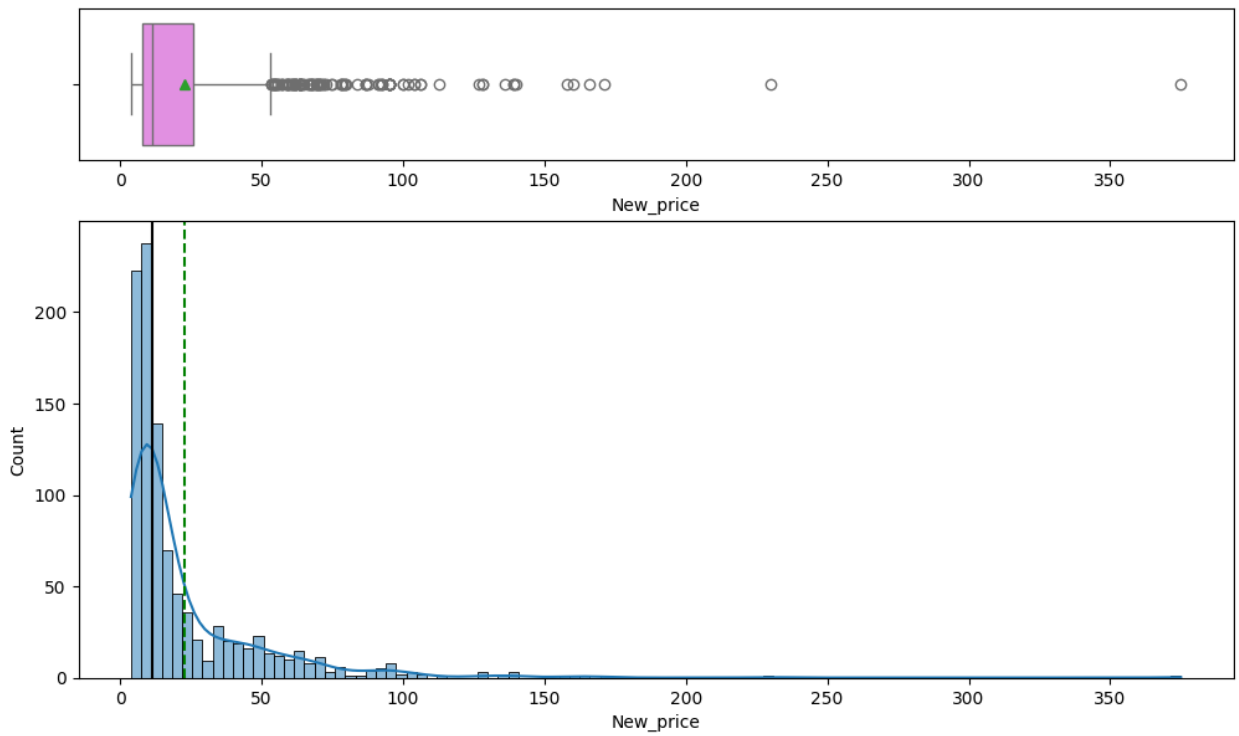
Graph 11. Seats Barplot



Observations and Implications on Seats Barplot

Because the Seats variable is a discrete numerical variable, an alternative bar plot is provided to better visualize the varying seat counts across the used cars in the dataset. Here, it is much clearer that 5-seat used cars dominate the dataset at about 83.4 percent. This implies that 5 seats is a standardized baseline that could potentially affect most modeling decision-making. Afterward, 7-, 8-, and 4-seat used cars are also significant, at 11 percent, 2.3 percent, and 1.6 percent, respectively. This implies that these used cars are typically SUVs and MPVs. Due to there being infrequent categories of used cars with 2, 6, 9, and 10, these rare seat values will later be combined into one category named “Rare_Seat_Values” to stabilize linear models and avoid sparse-category issues.

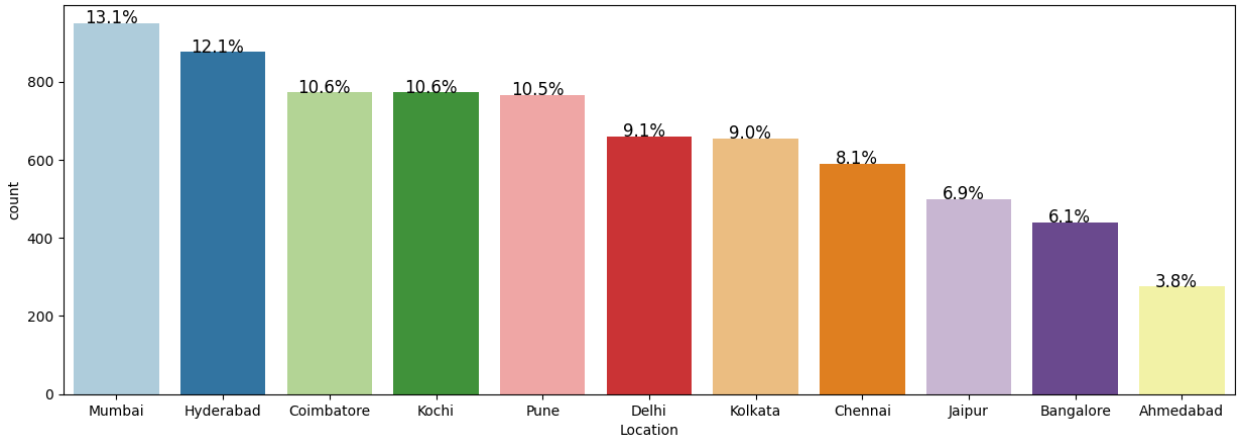
Graph 12. New_Price Boxplot and Histogram



Observations and Implications on New_price Boxplot and Histogram

The New price boxplot and histogram are heavily right-skewed and unimodal. The median (denoted by a solid black line in the histogram) is 11.57 INR 100,000, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 22.78 INR 100,000. There is a large majority of used cars with new price values below 50 INR 100,000, and a long right tail extending from 100 to 300 INR 100,000. The majority of used cars are low-budget or mid-range. There is further evidence that a small number of used cars are luxury or premium-value, and are considered outliers. Again, these are brands of cars, including BMW, Mercedes, Audi, Jaguar, High-end SUVs (e.g., Land Cruiser, GLS, and Q7), and luxury sports cars (e.g., Porsche and Mustang). Even though the New price numerical variable is highly skewed, a log transformation is unnecessary because it is neither directly predicted nor used in the regression equation. Most importantly, a log transformation cannot be applied to this variable because a feature-engineered numerical Depreciation Amount variable will be created, which necessitates keeping New Price in its raw currency units.

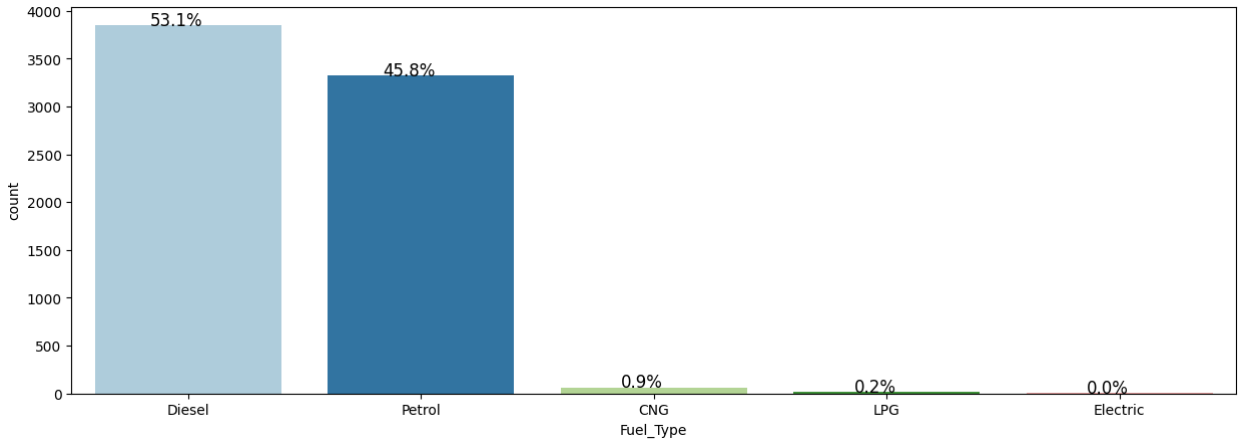
Graph 13. Location Barplot



Observations and Implications on Location Barplot

The location bar plot shows 11 major Indian cities, with no single city dominating across all of them. In other words, there is no geographic bias, and there is enough data per city for meaningful comparisons. Mumbai accounts for the majority of used cars, at 13.1 percent. This aligns with real-world facts, as Mumbai is known to have a massive used-car market with higher purchasing frequency, shorter car replacement cycles, and strong demand for both low- and high-cost used cars. The cities of Hyderabad, Coimbatore, Kochi, and Pune instead form a strong middle cluster, where each city contributes around 10 to 12 percent. These cities also have highly active used car markets. Cities such as Delhi, Kolkata, Chennai, Jaipur, Bangalore, and Ahmedabad have smaller shares, ranging from 3.8% to 9%. Overall, South Indian cities, including Chennai, Hyderabad, and Kochi, are well represented in the dataset.

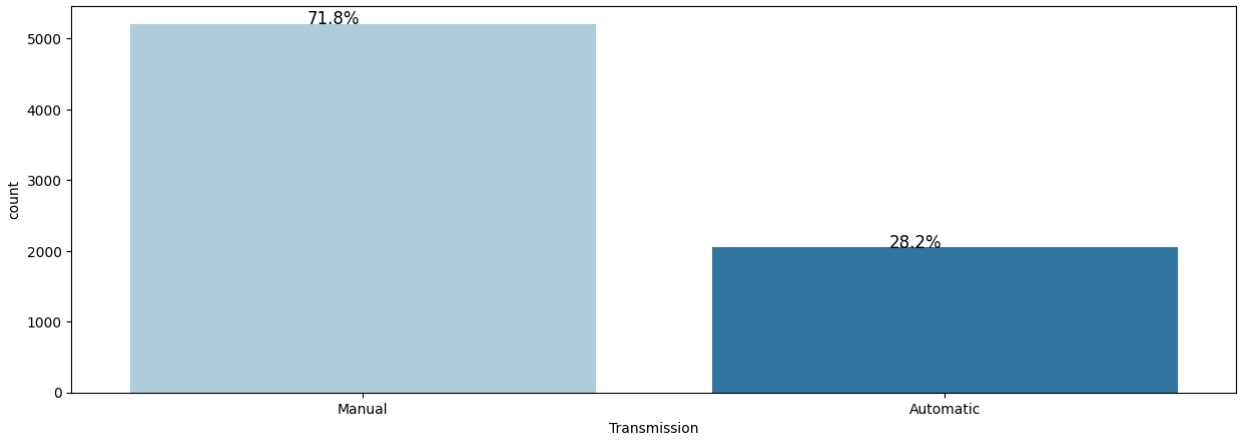
Graph 14. Fuel_Type Barplot



Observations and Implications on Fuel Type Barplot

The Fuel Type bar plot shows 5 distinct Fuel Types among the used cars in the Cars4U inventory—the majority run on diesel or petrol. About 53.1 percent of all used cars run on diesel, and about 45.8 percent run on petrol. This dominance of diesel and petrol fuel types is expected in the actual Indian used-car market, since diesel cars were prominent in the 2010-2016 era due to fuel price fluctuations and torque advantages. Along the same lines, used cars that run on petrol fuel are usually small hatchbacks and compact sedans. As for used cars that run on CNG or LPG, or are electric, they are very uncommon. Because these categories occur much less frequently, these three fuel types will later be combined into an "Other_Fuel-Type" category to help stabilize linear models and avoid sparse-category issues.

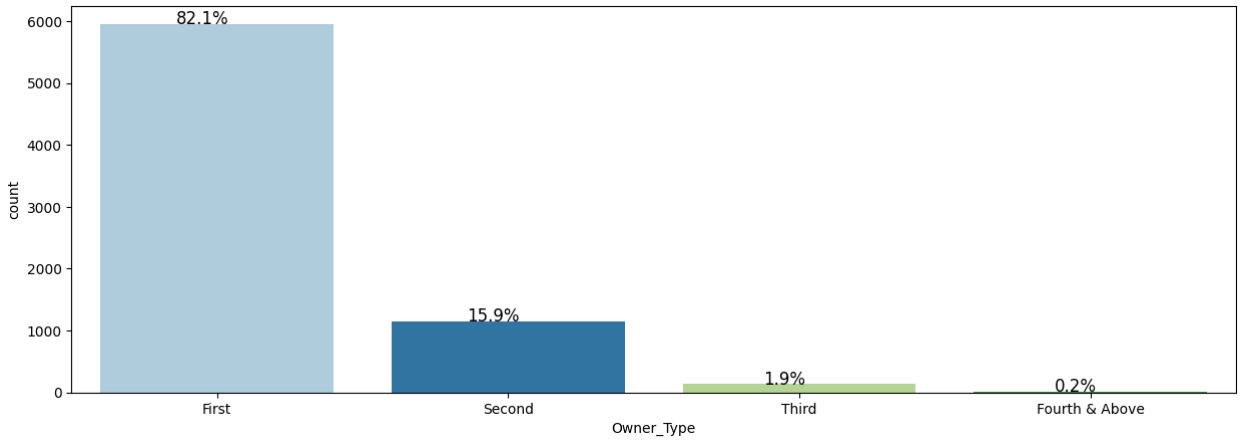
Graph 15. Transmission Barplot



Observations and Implications on Transmission Barplot

The Transmission bar plot shows that the majority of used cars in the Cars4U inventory are manual, at approximately 75 percent. This aligns with the realities of the Indian car market, since manual used cars are cheaper to purchase, offer better mileage, and are dominated by hatchbacks and compact sedans. In contrast, used cars are instead uncommon or more costly until more recent years. As such, used cars are represented in the dataset close to 30 percent. This aligns with the fact that the share of automatic used cars is higher than in older Indian car market averages and has increased significantly since 2015. Also, automatic used cars tend to have higher resale prices compared to their manual counterparts.

Graph 16. Owner Type Barplot

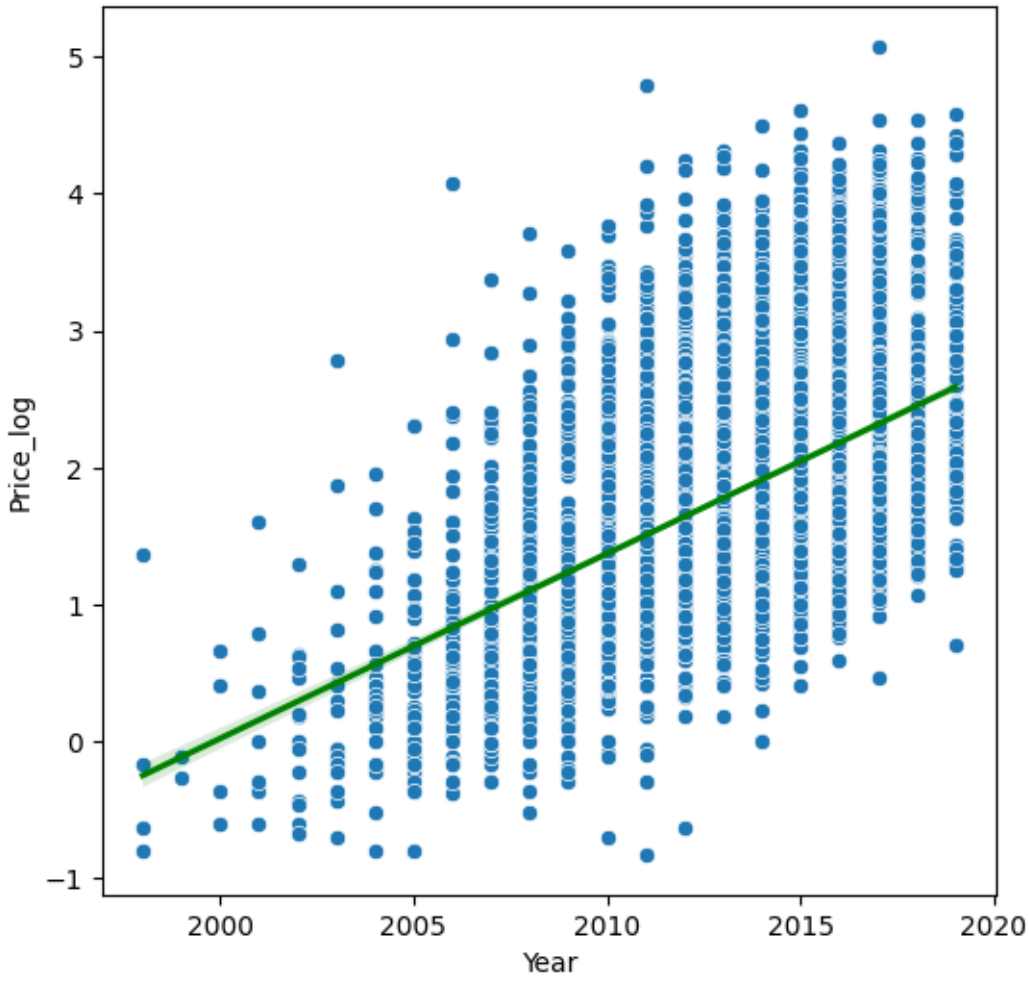


Observations and Implications on Owner Type Barplot

The Owner Type bar plot indicates a clear dominance of first-owner used cars, which aligns with realistic customer behavior. As 82.1 percent of used cars are classified as first-owner type, this means they are more robust, easier to sell, command higher resale prices, are recognized for better maintenance, and are associated with cleaner service histories. Second-owner used cars still account for a significant minority, about 15.9 percent. These used cars are typically still less costly than first-owner used cars and often have higher mileage. Since third-owner and fourth-owner and above used cars are less common, these categories will later be combined into an "Owner_3plus" category to help stabilize the model and avoid sparse-category issues.

Bivariate Analysis

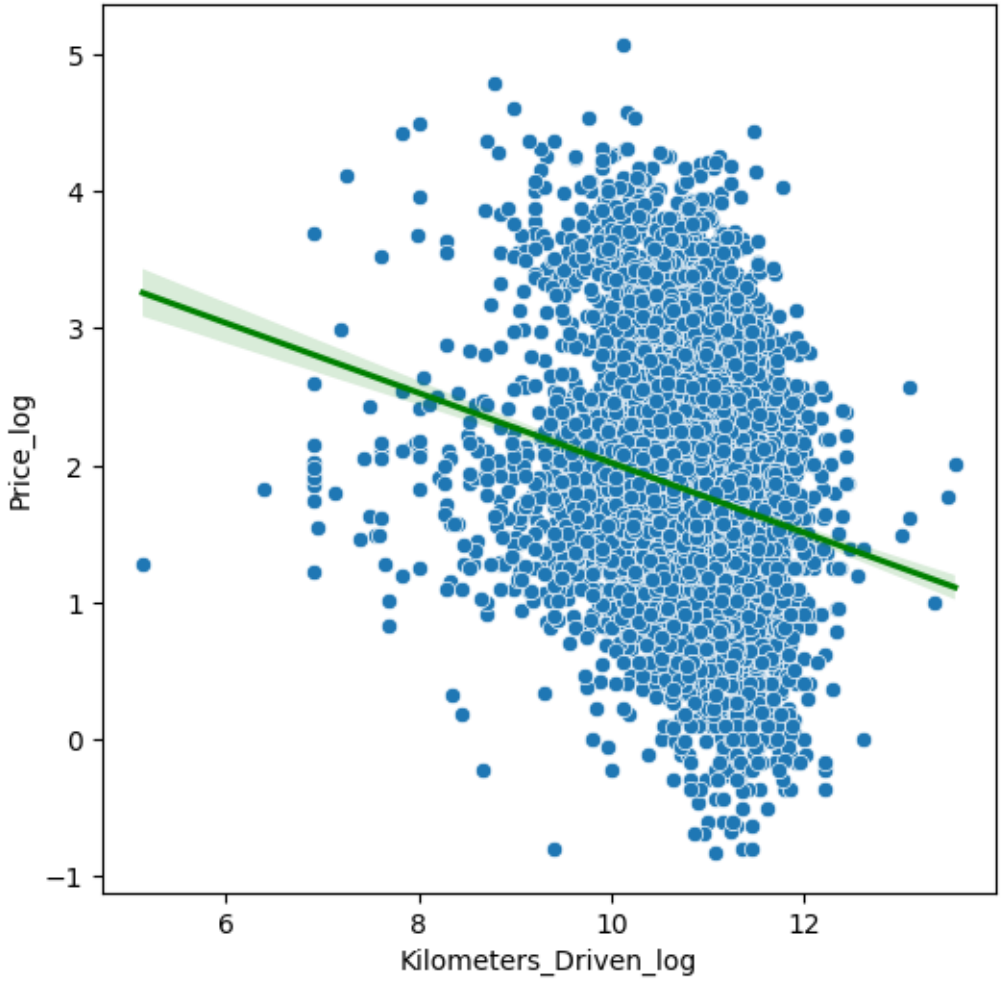
Graph 17. Year vs. Price_log Scatterplot



Observations and Implications on Year vs. Price_log Scatterplot

The Year vs. Price_log scatterplot shows a clear positive correlation between the manufacturing year and the log-transformed price of a used car. The regression line represents a percentage change in price per additional year, rather than a fixed amount of INR 100,000. There is notable clustering of used cars from 2014 to 2019 and greater variability from 1998 to 2008. In clearer terms, the variances increase as the used cars become older. The relationship is robust, but not absolutely linear. Overall, Price_log increases with the manufacturing year, suggesting that newer used cars have higher INR 100,000 price values even after a log transformation. The relationship is clearly linear, even though older used cars have greater variability; this is most likely due to dissimilar conditions and past usage histories. This validates that Year or Car_Age (an engineered feature) can be a significant predictor in modeling price.

Graph 18. Kilometers_Driven_log vs. Price_log Scatterplot

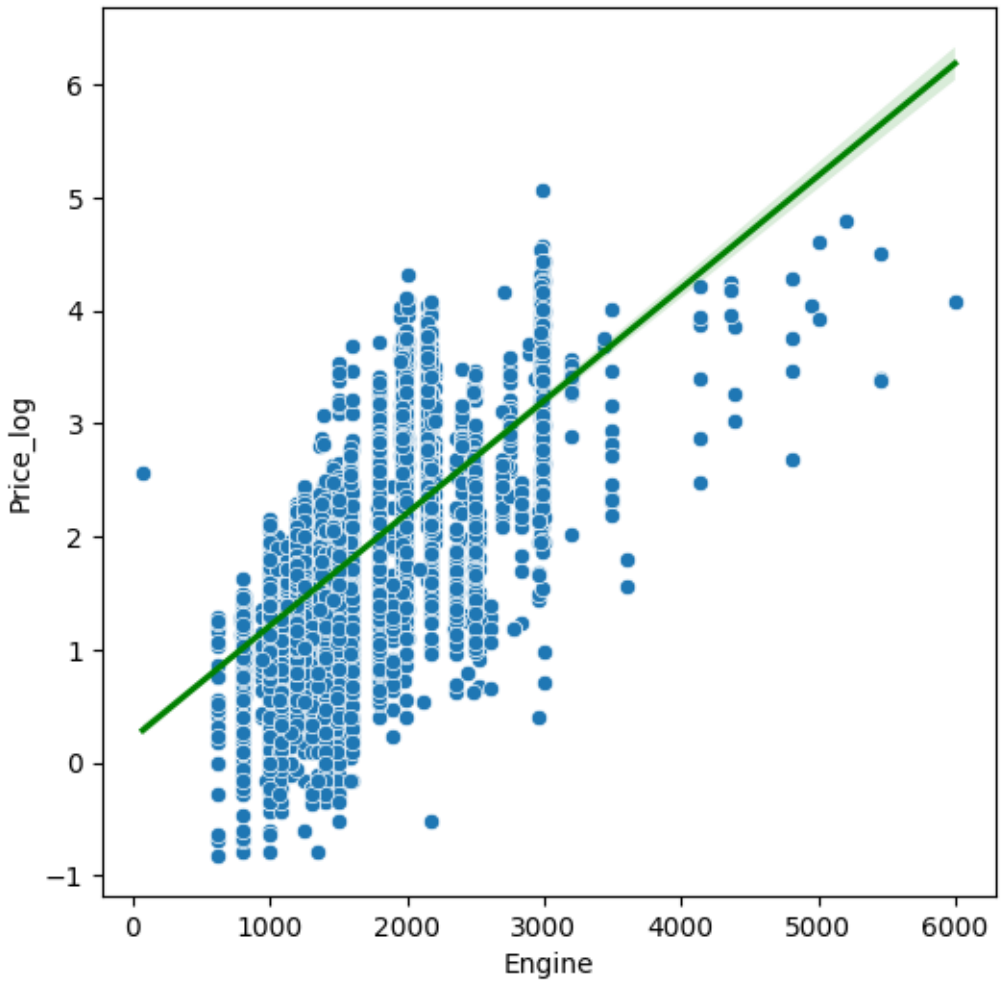


Observations and Implications on Kilometers_Driven_log vs. Price_log Scatterplot

The Kilometers_Driven_log vs. Price_log scatterplot shows a clear negative correlation between the log-transformed Kilometers_Driven variable and the log-transformed Price of a used car. As the number of kilometers driven increases, the price of a used car decreases. The log transformation of the two variables reveals a linear trend. The regression line fits closely to the points, which suggests that the total number of kilometers driven by a previous owner is a significant independent variable of a used's price. There is also evidence of vertical spread at each kilometers-driven level, implying key differences in brand, age, condition, and engine features, but the overall pattern remains stable. Overall, the visualization confirms that higher usage of a used car is associated with lower used-car values, and that the log transformation

effectively preserves variance, making the relationship suitable for subsequent linear modeling.

Graph 19. Engine vs. Price_log Scatterplot

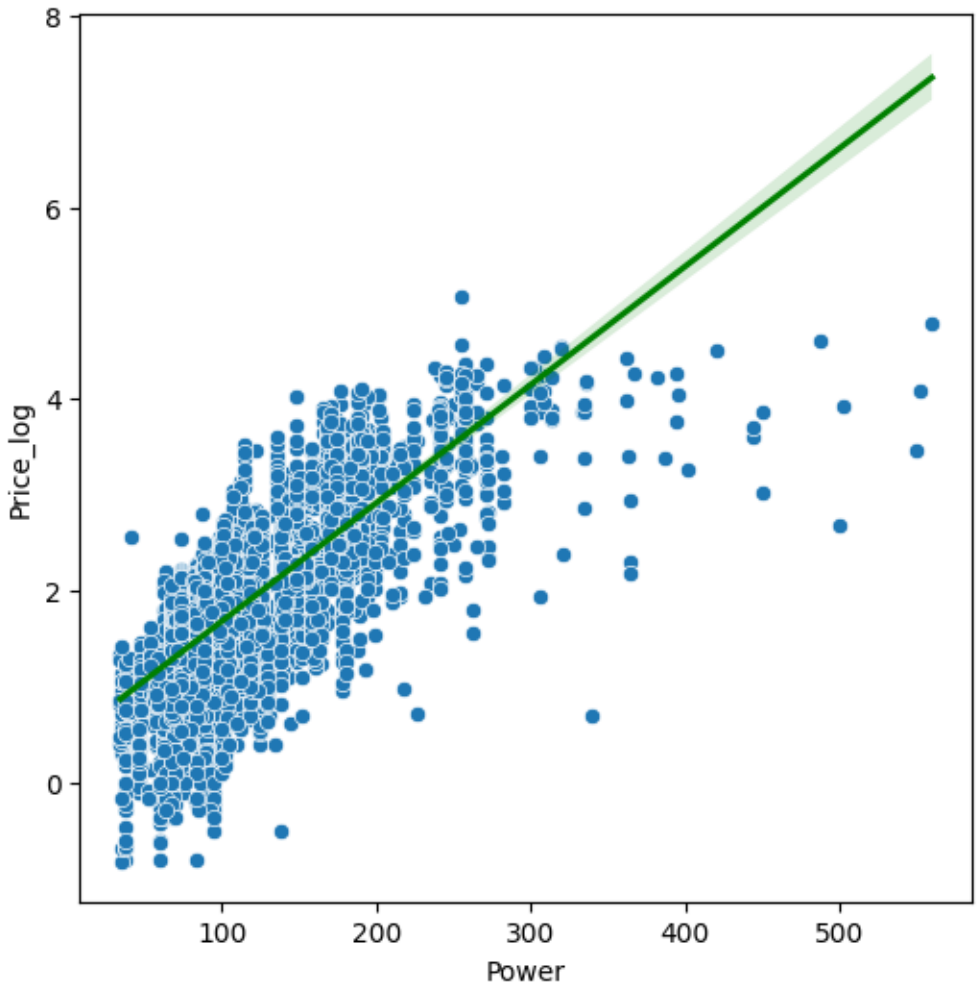


Observations and Implications on Engine vs. Price_log Scatterplot

The Engine vs Price_log scatterplot indicates a clear positive relationship between engine size and the log-transformed price of used cars. Used cars with larger engine displacements tend to be more expensive. The regression line indicates an upward trend, suggesting that engine displacement can be a significant predictor of the target price value even after applying a log transformation. Even though the overall pattern is positively linear, there is evidence of a vertical spread that widens for used vehicles with larger engine sizes, which follows market dispersion. Specifically, premium used-car brands, performance trims, and luxury segments cluster in this

region, affecting price variability. Although there is evidence of dispersion, engine size remains a significant predictor of used-car price. More importantly, higher-capacity used cars with more powerful engines (e.g., 1800 CC, 2000 CC, and 2400 CC) correspond to higher log-price values. Thus, the relationship is smooth and expected across the range of values along the regression line.

Graph 20. Power vs. Price_log Scatterplot

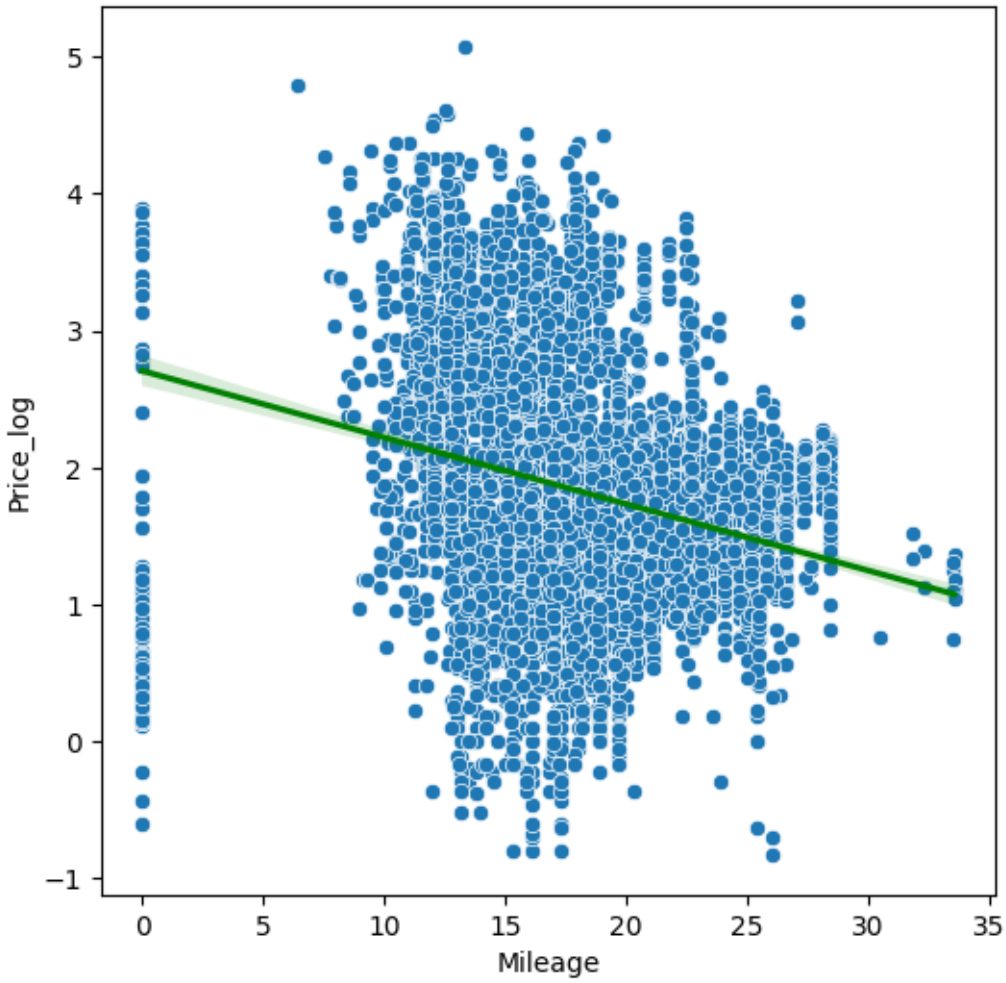


Observations and Implications on Power vs. Price_log Scatterplot

The Power vs Price_log scatterplot indicates a strong positive relationship between used car power and log-transformed price. As the horsepower of a used car increases, the log price also increases. Hence, the regression line clearly captures this pattern. Even though the overall relationship is linear, there is evidence of vertical variability that widens at higher power levels, which realistically reflects the diversity

of higher-performance and luxury used-car models that command different price premiums even at comparable power outputs. Otherwise, these used cars have more powerful engines and command higher log prices. Even though this vertical variability exists, the basic relationship holds: power is a significant and consistent variable that effectively determines the price of a used car.

Graph 21. Power vs. Price_log Scatterplot

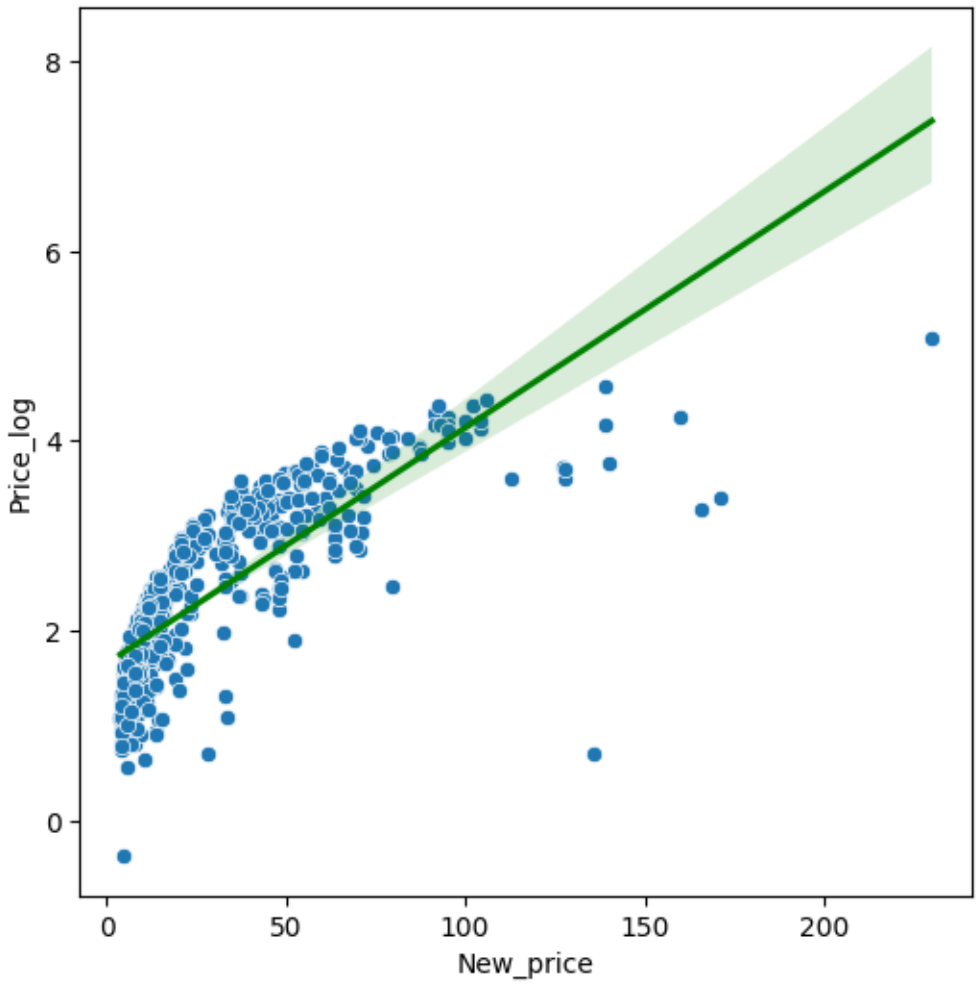


Observations and Implications on Mileage vs. Price_log Scatterplot

The Mileage vs. Price_log scatterplot indicates a clear negative relationship between mileage and the log-transformed price variable. As used cars accrue more mileage, their value declines in a predictable, consistent pattern. In particular, higher-mileage cars are typically cheaper, with a steeper decrease in log price at lower mileage and a flatter downturn at higher mileage. The regression line clearly

represents this downward trend. The log transformation helps stabilize the variance, suggesting a slight linear depreciation effect. Even though there is notable variability in vertical spread at each mileage level, this variability reflects the differing characteristics of brand, age, condition, and engine across used cars. Thus, the scatterplot confirms that mileage can also effectively predict the price of a used car, as it clearly indicates long-term vehicle wear and lower prices.

Graph 22. New_price vs Price_log Scatterplot



Observations and Implications on New_price vs. Price_log Scatterplot

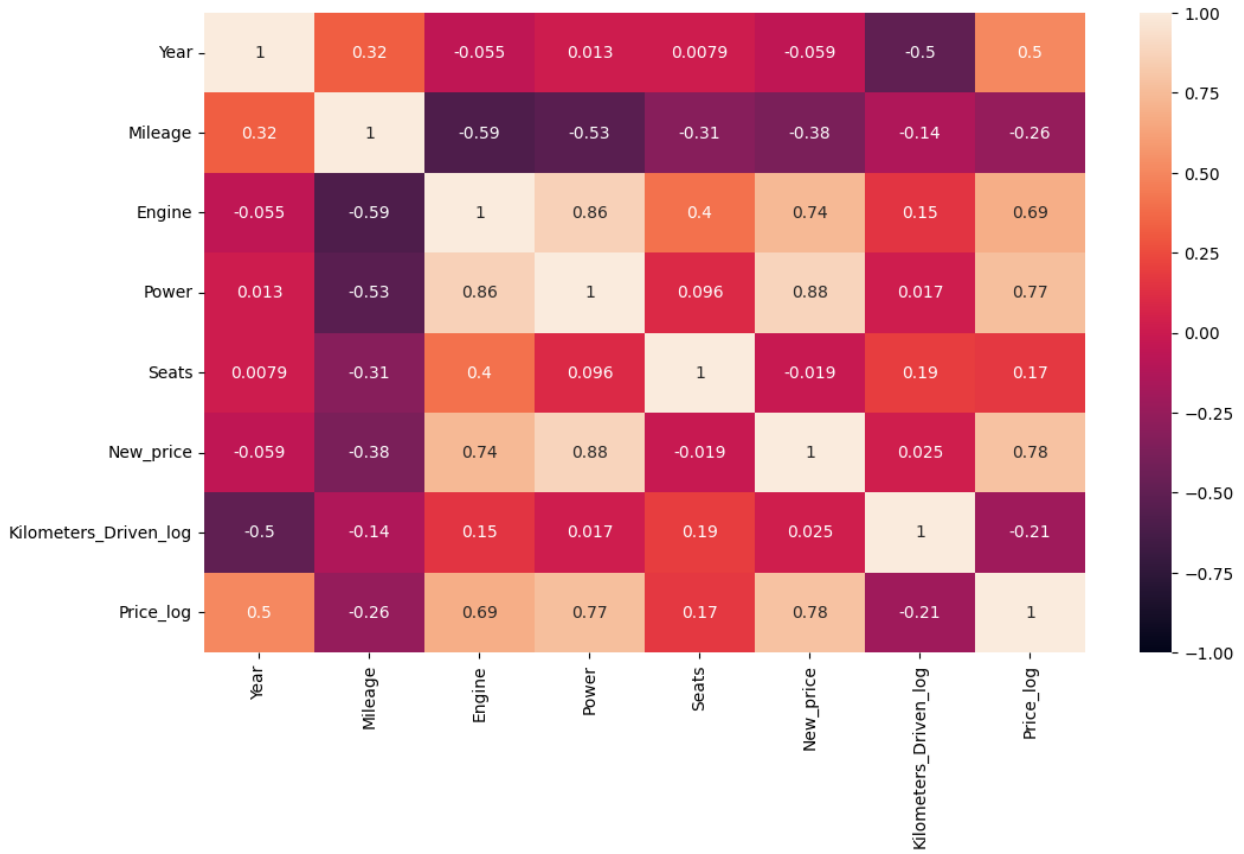
The New_price vs. Price_log scatterplot indicates a strong, smooth positive relationship between a used car's original new car price value and its current log-transformed price value. While the new price increases, the price log values also increase. The regression line also fits the data points well. In other words, used cars

that were initially costly tend to have higher resale values. This also aligns with the fact that these specific used cars, which were once expensive, still maintain their higher used-car prices, even after depreciation and market variability. There is further evidence that the vertical spread widens at higher new price values, indicating the presence of premium and luxury used cars in the collected dataset. Overall, New_price can be considered one of the most potent predictors of used-car prices in the dataset, and the Price_log transformation helps better visualize this relationship.

Correlation among numerical variables

Correlation is defined (using a correlation coefficient) as how two numerical variables move together: either in a direct, positive relationship, an inverse, negative relationship, or no correlation. As in, a positive correlation suggests that both variables increase together, a negative correlation indicates that one variable increases while the other decreases, and no correlation indicates that neither variable increases or decreases. Correlation is measured using a Pearson correlation coefficient. However, a correlation does not imply causation; it only shows how variables move together.

Graph 23. Heatmap Visualization for Correlations among Numerical Variables



Observations and Implications on the Heatmap Visualization for Correlations among Initial Numerical Variables

The Heatmap visualization of correlations among initial numerical variables highlights a clear structure in how used-car attributes affect used-car prices. The target Price_log variable has a strong positive correlation with New_price, with a Pearson correlation coefficient of 0.78. The target Price_log variable also has a strong positive correlation with Power, with a Pearson correlation coefficient of 0.77. Also, the target Price_log variable has a strong positive correlation with Engine, with a Pearson correlation coefficient of 0.69. These strong positive correlations align with the fact that used cars, once established as having higher prices or performance, still command higher resale prices. There is also a positive correlation between Year and Price_log, which follows the expected depreciation pattern, with newer cars usually more valuable. As for significant negative correlation relationships, Kilometers_Driven_log and Mileage show balanced inverse relationships with Price_log. In fact, Kilometers_Driven_log and Price_log have a Pearson correlation coefficient of -0.21,

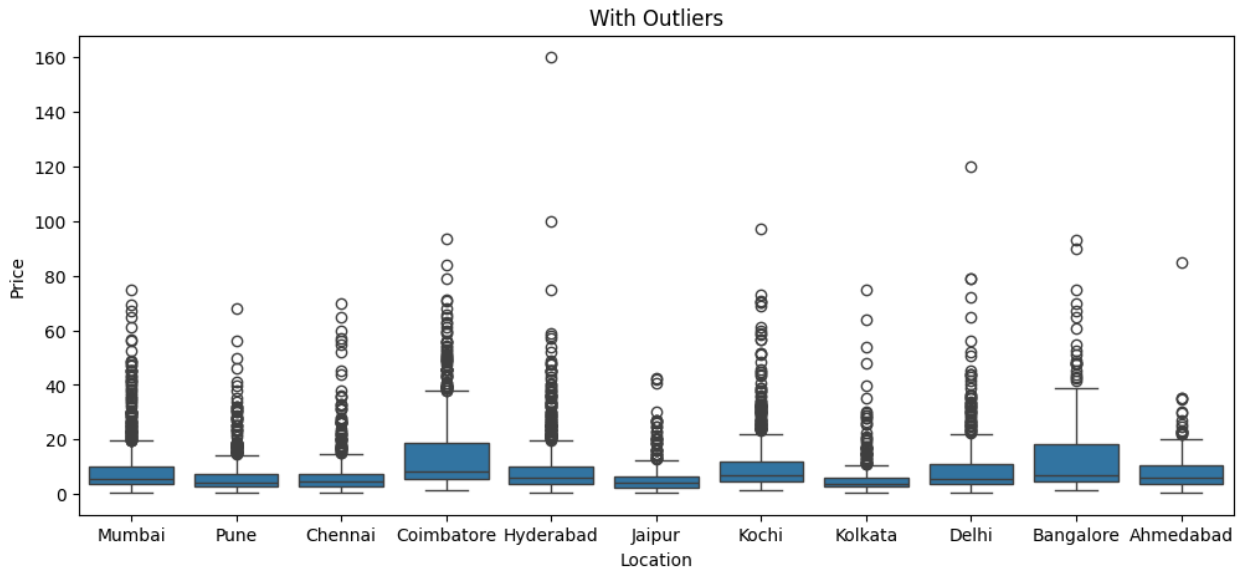
and Mileage and Price_log have a Pearson correlation coefficient of -0.25. These negative correlations indicate that usage and wear reduce the price of a used car.

Aside from correlations with the Price_log target variable, there are other correlations to note. There is a strong positive correlation between Engine and Power, with a Pearson correlation coefficient of 0.86. In fact, this relationship holds because larger engines produce more power. There is a strong positive correlation between Power and New_price, with a Pearson correlation coefficient of 0.88. In fact, this relationship holds because more powerful used cars usually cost more when new. There is a strong positive correlation between Engine and New_price, with a Pearson correlation coefficient of 0.74. In fact, this relationship holds, since larger engines also mean higher new-car values. There are negative correlations between Mileage and Engine, Mileage and Power, and Mileage and New_price, with Pearson correlation coefficients of -0.59, -0.53, and -0.38, respectively. In fact, these negative correlations are valid, since cheaper, lower-spec used cars usually accrue more mileage, and there is a distinct socioeconomic pattern in which lower-priced used cars typically wear out more thoroughly. Additionally, there are positive correlations between Seats and Engine, and between Seats and Price_log, with Pearson correlation coefficients of 0.40 and 0.17, respectively. In fact, these positive correlations are valid, as they still capture the type of used car, typically corresponding to its classification. As such, the seat count can indicate whether a used car is a sedan, SUV, or minivan.

Analyze categorical variables, including Seats (a numerical discrete categorical variable), with the target Price variable.

Categorical variables such as location, fuel type, transmission, owner type, and the discrete numerical seats variable are analyzed in conjunction with the target Price variable, both with and without outliers. The rationale for this analysis is that it captures structural market characteristics, reveals price segmentation, detects non-linear effects, pilots proper encoding techniques, and provides focal points for both high-value and low-value groups within each categorical variable, including seats.

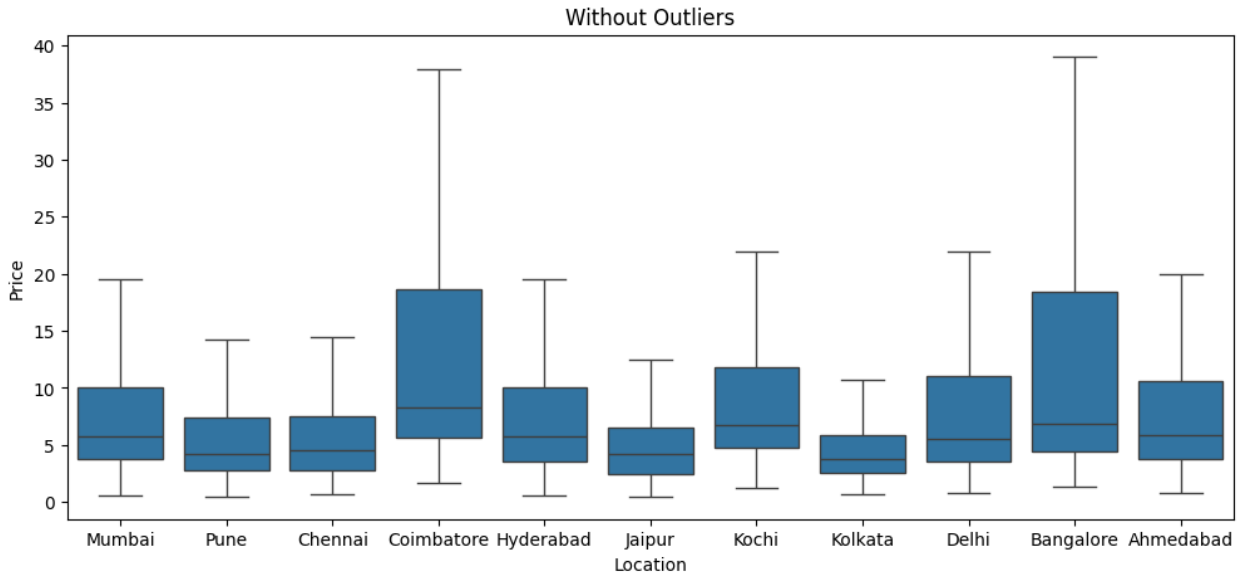
Graph 24. Location vs. Price Boxplot (With Outliers)



Observations and Implications on the Location vs. Price Boxplot (With Outliers)

The Location vs. Price boxplot with outliers indicates very high outliers and significant price variation across Indian cities. There is evidence that several markets show widespread, numerous high-value outliers. Coimbatore and Bangalore have the broadest interquartile ranges and the largest clusters of extremely high-priced points, suggesting amalgamated markets with both budget and premium segments. On the other hand, cities including Jaipur, Kochi, and Ahmedabad indicate firmer distributions, indicating more homogeneous pricing and far fewer luxury-tier vehicles. Likewise, cities such as Mumbai, Delhi, Bangalore, and Coimbatore show numerous outliers, indicating that luxury used-car brands, high-spec trims, and newer used vehicles are more common in these cities and fall above common market ranges. These outliers emphasize that location relates to other predictors (e.g., brand, power, and new price) and can shape price variability. The boxplot indicates that location affects price levels, with some cities exhibiting higher and more volatile price ranges than others.

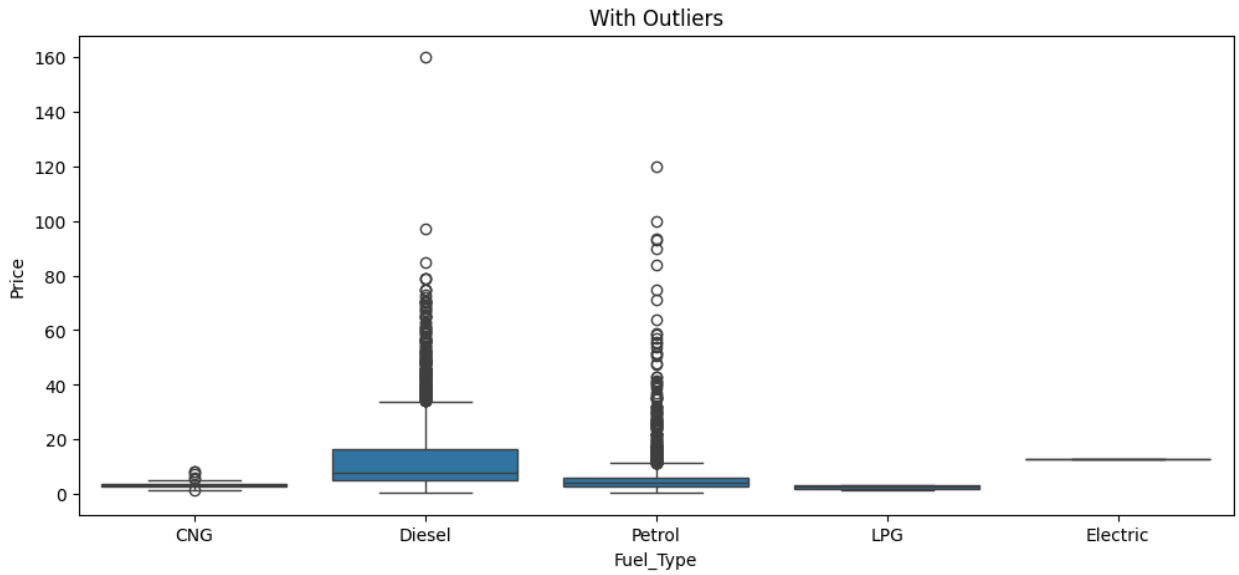
Graph 25. Location vs. Price Boxplot (Without Outliers)



Observations and Implications on the Location vs. Price Boxplot (Without Outliers)

The Location vs. Price boxplot, excluding outliers, indicates the true central pricing structure of used cars across different Indian cities. Cities such as Coimbatore and Bangalore still have the most variable price ranges, suggesting more diversified mid-market segments once extreme values are removed. On the other hand, cities such as Pune, Chennai, and Kolkata show tighter boxplots, indicating more consistent pricing and fewer mid-range variations. Cities such as Mumbai, Delhi, and Chennai show clustering of used cars with higher price ranges. Cities such as Ahmedabad, Jaipur, and Kolkata show tighter, lower-priced distributions. Metro cities, including Mumbai, Delhi, Bangalore, and Chennai, have higher medians, which suggest ample economic distinctiveness and prime vehicle existence. The following business interpretations can be made: (1) Bangalore and Coimbatore have markets with diversified customer segmentation, a wide range of vehicle conditions, or brands with more complex pricing changes. (2) Mumbai and Delhi have high-end outliers that are notable with higher purchasing capacity and increased availability of luxury used car brands.

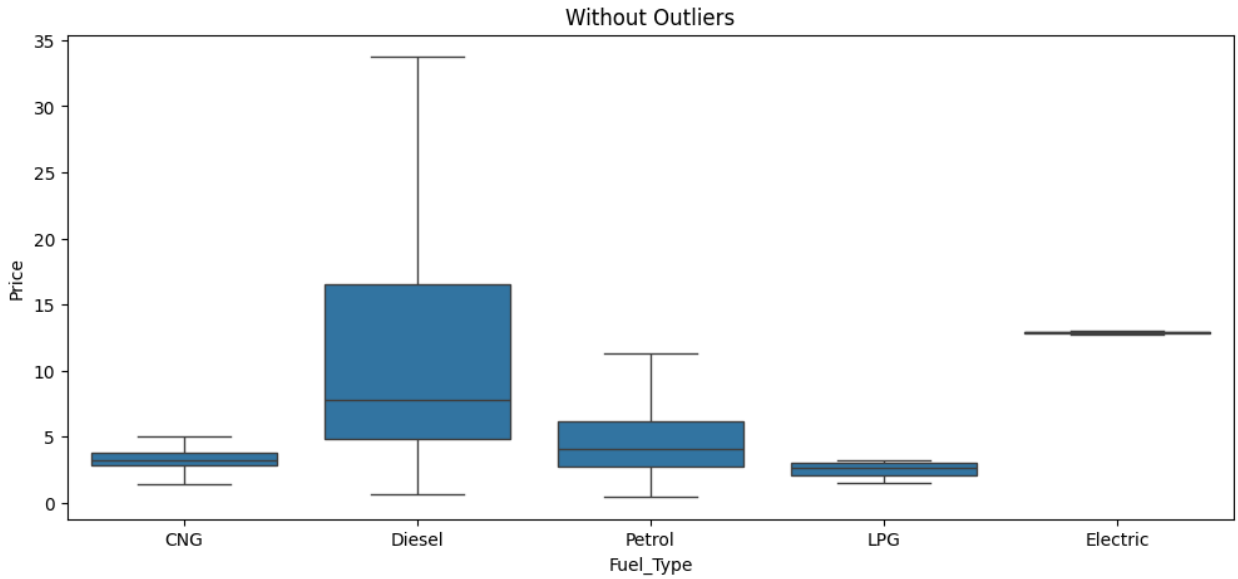
Graph 26. Fuel_Type vs. Price Boxplot (With Outliers)



Observations and Implications on the Fuel_Type vs. Price Boxplot (With Outliers)

The Fuel_Type vs. Price boxplot with outliers indicates that diesel- and petrol-fuel-type used cars comprise the majority of the upper price ranges in the dataset, where variability is greater. This indicates broad market diversity, from low-priced used cars to higher-premium used cars. CNG- and LPG-fuel-type used cars fall squarely within the low-cost segment, with notably lower medians and tighter distributions. Electric fuel-type cars also consist of much lower frequencies in the dataset. Altogether, CNG, LPG, and Electric fuel types used cars in the overall inventory are more compact and predictable. The outliers indicate that the traditional Diesel and Petrol fuel type categories feature premium and luxury used car models that form the upper tail of the price distribution.

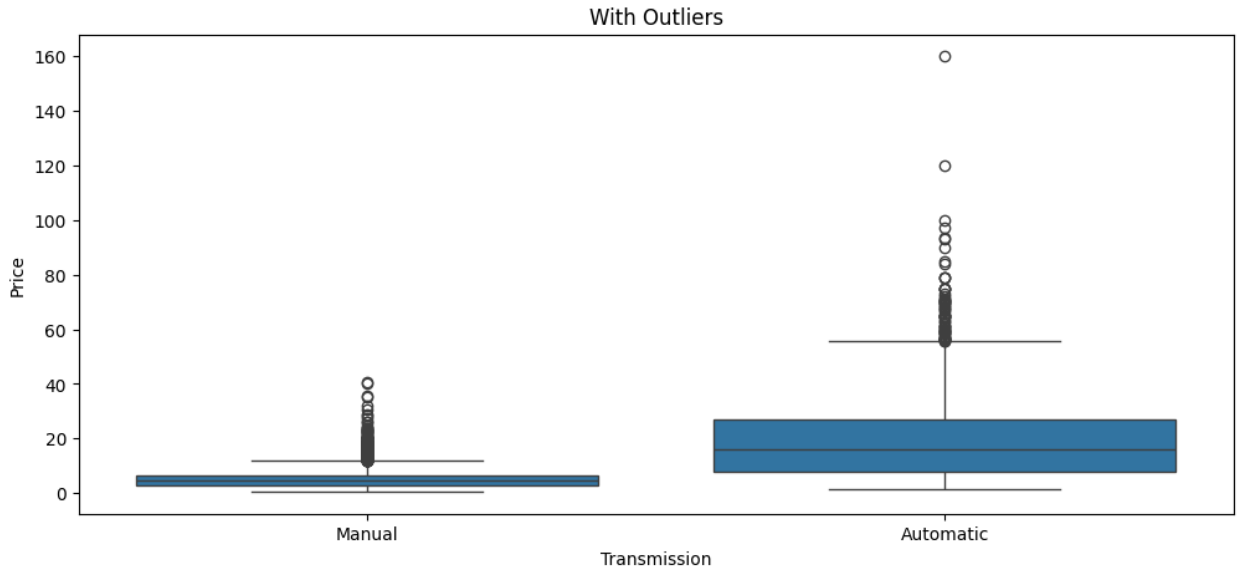
Graph 27. Fuel_Type vs. Price Boxplot (Without Outliers)



Observations and Implications on the Fuel_Type vs. Price Boxplot (Without Outliers)

The Fuel_Type vs. Price boxplot without outliers indicates that Diesel fuel-type used cars are clearly the highest-priced, with the tallest box and widest spread. This suggests a mix of mid-range and higher-spec diesel. Without outliers, it is now clearer that Petrol, CNG, and LPG used-car models are classified as having compact distributions, indicating more consistent, uniform pricing for used cars running on these fuel types. As for Electric used cars, there is an indication that this segment is a younger, regulated market with only a few similarly priced options.

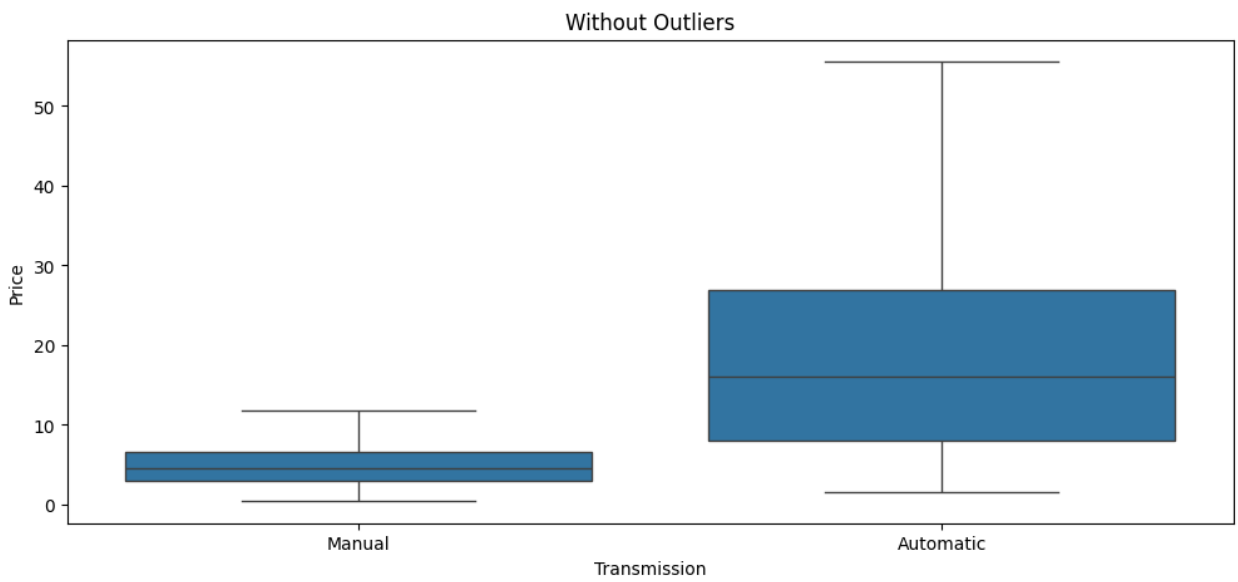
Graph 28. Transmission vs. Price Boxplot (With Outliers)



Observations and Implications on the Transmission vs. Price Boxplot (With Outliers)

The Transmission vs. Price boxplot with outliers indicates that automatic used cars are consistently higher-priced than manual used cars. Manual used cars are priced much less than automatic used cars. There are also significantly more outliers in automatic used cars than in manual used cars. As such, the majority of used cars have automatic transmissions and are associated with premium luxury models.

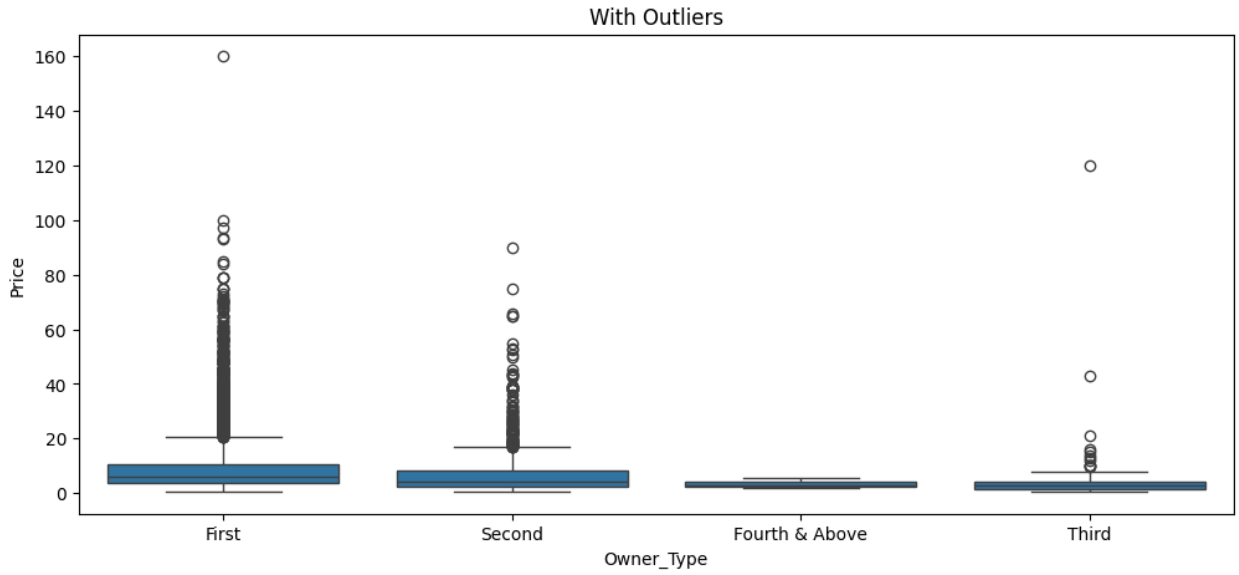
Graph 29. Transmission vs. Price Boxplot (Without Outliers)



Observations and Implications on the Transmission vs. Price Boxplot (Without Outliers)

The Transmission vs. Price boxplot without outliers more clearly indicates that automatic cars have a higher median and much greater variability. Manual cars generally show greater concentration in lower price brackets and have more budget-oriented segmentation. The difference in price spread between automatic and manual used cars suggests that automatic used cars span a broad mid-range market, while manual used cars are grouped more tightly around lower price points.

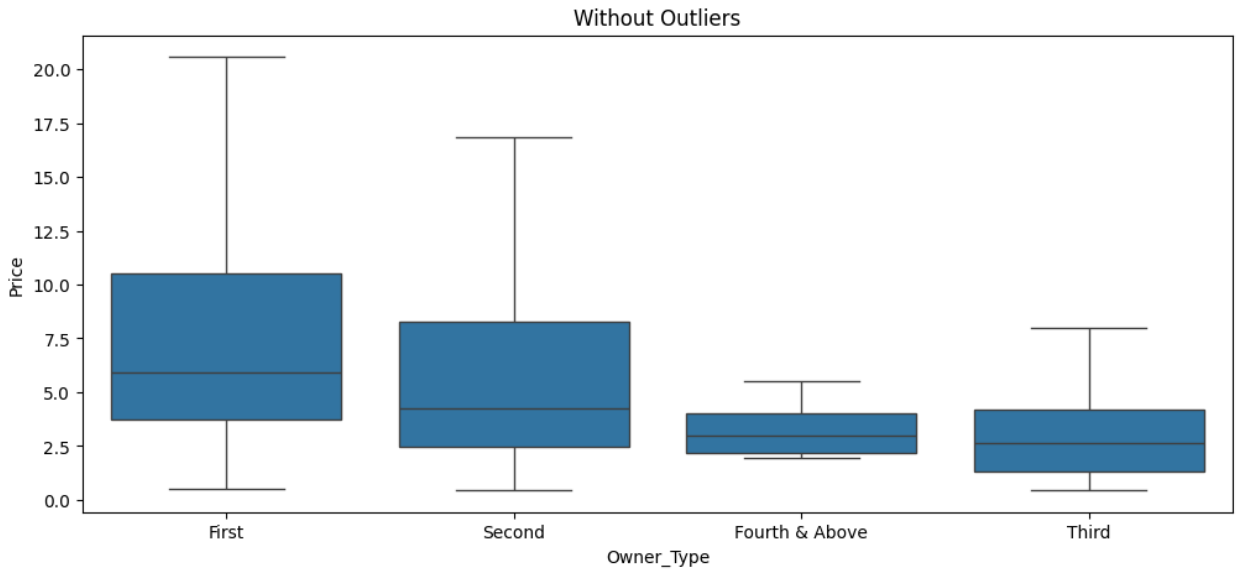
Graph 30. Owner_Type vs. Price Boxplot (With Outliers)



Observations and Implications on the Owner_Type vs. Price Boxplot (With Outliers)

The Owner_Type vs. Price boxplot with outliers indicates the following insights. First, first-owner vehicles are associated with the highest price values. This implies newer used cars are in better condition and that buyers have significant preferences for single-owner histories. Second, second-owner vehicles are associated with mid-priced value, but still show a wide spread and several high-end outliers. This implies that most used cars of preferred models still retain their value after transferring ownership, even after being owned by a single owner for a long time. Third-, fourth-, and higher owners fall into the lower price categories, where there is evidence of tighter distributions and far fewer outliers. This implies that these used cars are older, have considerable wear, and face much less demand. Overall, outliers are most common in the first- and second-owner groups, suggesting that premium or luxury models are more frequently found in former ownership categories.

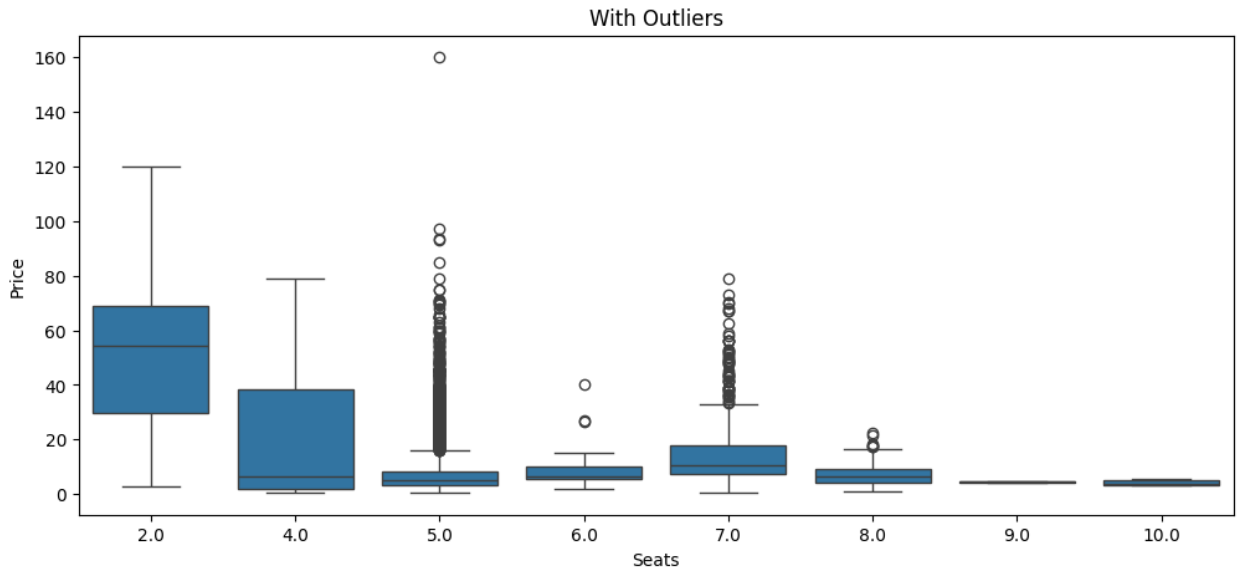
Graph 31. Owner_Type vs. Price Boxplot (Without Outliers)



Observations and Implications on the Owner_Type vs. Price Boxplot (Without Outliers)

The Owner_Type vs. Price boxplot, excluding outliers, indicates the following insights. First, it is much more evident that first-owner used cars comprise the highest-priced group, with the largest box and the widest variability. This implies that these types of used cars are in better condition, have higher retained prices, and command greater buyer trust. Second, it is much more evident that second-hand owner cars comprise a second mid-tier, with slightly lower median values and a narrower range than first-owner cars. This implies that these types of used cars comprise a mix of mid-range models. Third, it is much more evident that third-, fourth-, and higher-owner-type used cars cluster in lower-priced brands, with tighter distributions. Under realistic assumptions, this implies that these used cars are much older and less in demand. Overall, the price gap is now clearer without outliers, revealing a step-down hierarchy from first-owner to second-owner to third-owner to fourth- and above used cars.

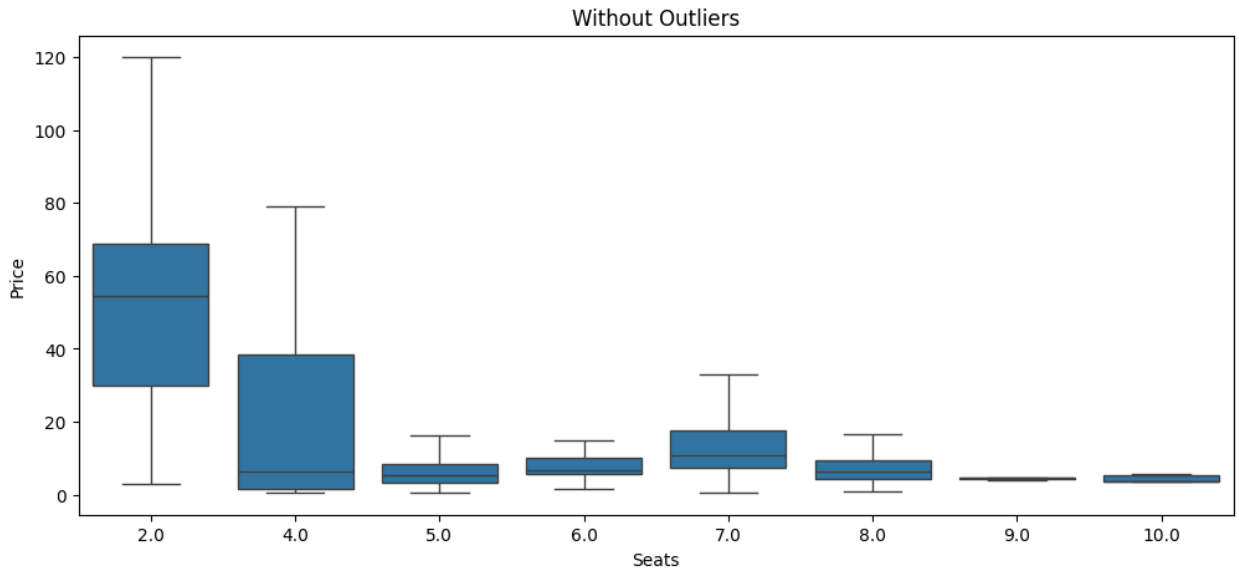
Graph 32. Seats vs. Price Boxplot (With Outliers)



Observations and Implications on the Seats vs. Price Boxplot (With Outliers)

The Seats vs. Price boxplot with outliers indicates the following insights. Foremost, 5-seat used cars constitute the majority of the dataset and exhibit the widest price spread, with many outliers among higher-priced cars. This implies that compact cars and premium SUVs fall into this category. Secondly, 7-seat used cars show substantial price variation, with a notable number of outliers at higher price points. This implies that larger SUVs and MPVs fall into this category. Thirdly, 2-seat and 4-seat used cars lie in the lower to mid-price range. This implies that compact, small coupes are in this category. Fourthly, 8-seat, 9-seat, and 10-seat used cars have moderate price ranges with little variability. This implies that niche used cars, such as vans, are underrepresented in the premium used-car market. Overall, most outliers are found in 5- and 7-seat used cars, suggesting that these are the most costly used cars considered mainstream by buyer preferences.

Graph 33. Seats vs. Price Boxplot (Without Outliers)



Observations and Implications on the Seats vs. Price Boxplot (Without Outliers)

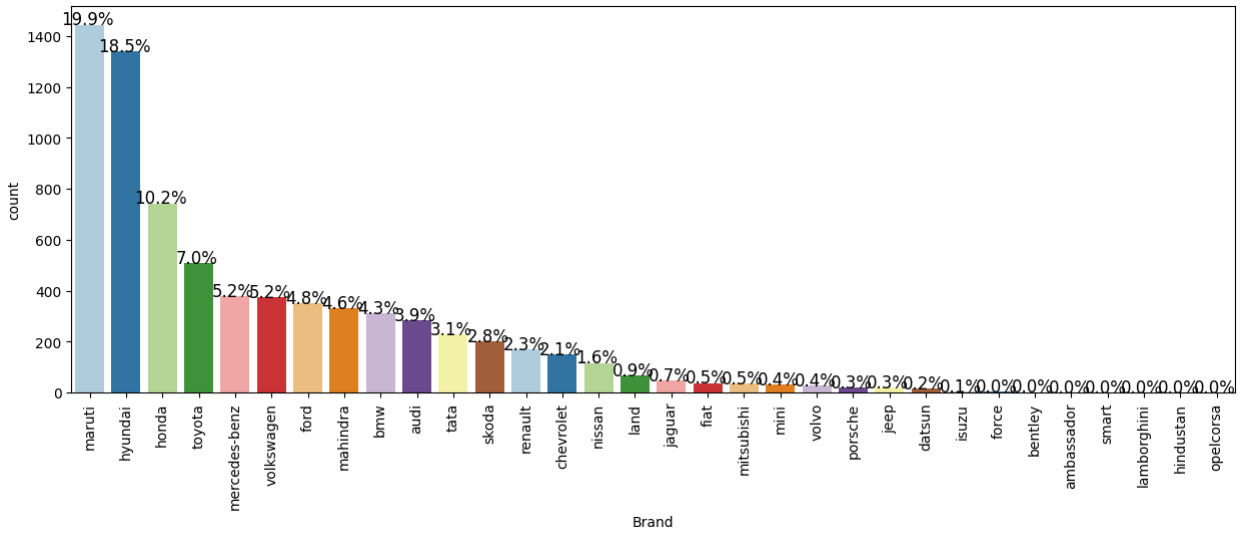
The Seats vs. Price boxplot, excluding outliers, indicates the following insights. First, it is now evident that 2- and 4-seat used cars have higher medians and wider spreads. This implies a mix of compact cars and sporty used cars that still retain their higher value. Second, it is now evident that 5-seat used cars form a stable mid-tier with moderate, balanced price ranges. This implies a dominant presence of 5-seat used cars, which are considered budget-friendly and economically priced. Third, it is now evident that 7-, 8-, 9-, and 10-seat used cars have lower medians and more compact variability. This implies that older or more functional models have defined price variability. Overall, after outliers are removed, there is a clear downward trend in used-car prices with the increasing number of seats.

Feature Engineering

Brand, Model, Car_Age, and Depreciation_Amount are the independent variables considered feature-engineered to enhance the predictive power of the proposed models for predicting used-car prices in the dataset. Brand is extracted from the first word of the original Name categorical variable. Creating a Brand independent variable enhances reputation, authenticity, and market adjustment, with some used-car brands having higher resale values. In this case, Brand is considered a significantly potent categorical mainstay for predicting price. The model is extracted from the second word of the original Name categorical variable. By creating a Model independent variable, more accurate median computations can be made by Brand and

Model for a used car when imputing missing values in several of the aforementioned independent variables, including Seats and New price. Other reasons for using a one-word model name include uniform grouping across the dataset, lower cardinality (i.e., the number of unique values within a specific independent variable), cleaner hierarchical imputation, enhanced alignment with the actual automated structure, and improved downstream modeling. Otherwise, the model's independent variable does not serve other meaningful purposes. Car Age is calculated by subtracting each used car's manufacturing year from the reference year, which in this case is the maximum Year value from the Year independent variable, or 2019. The main rationale for using a fixed reference year in the dataset is to ensure reproducibility and alignment with its temporal context. Another reason is to prevent unrealistic Car Age values, where using the current year as the reference year would make older used cars appear significantly older. Most importantly, using this Car Age calculation makes the proposed models more interpretable by making the distribution more realistic. Depreciation Amount is calculated by subtracting each used car's current used Price (i.e., Price target variable) from its original New price (i.e., New price independent variable). The main rationale for incorporating a variable depreciation amount is to better represent the economic loss associated with the new-to-used price of each used car. The depreciation amount variable best summarizes multiple attributes, including brand strength, model quality, age, and mileage, through price value retention. The Depreciation Amount variable will not be computed until all missing values in the New price and Price target variables have been imputed. Altogether, these engineered independent variables highlight the Indian used-car market behavior, the used-car lifecycle, and value-retention dynamics. At the same time, it also helps comprehend why used cars are valued the way they are. Lastly, additional univariate and bivariate analyses will be conducted on the feature-engineered Brand, Car_Age, and Depreciation Amount variables to better understand their characteristics and impact on the collected dataset, particularly the target Price variable.

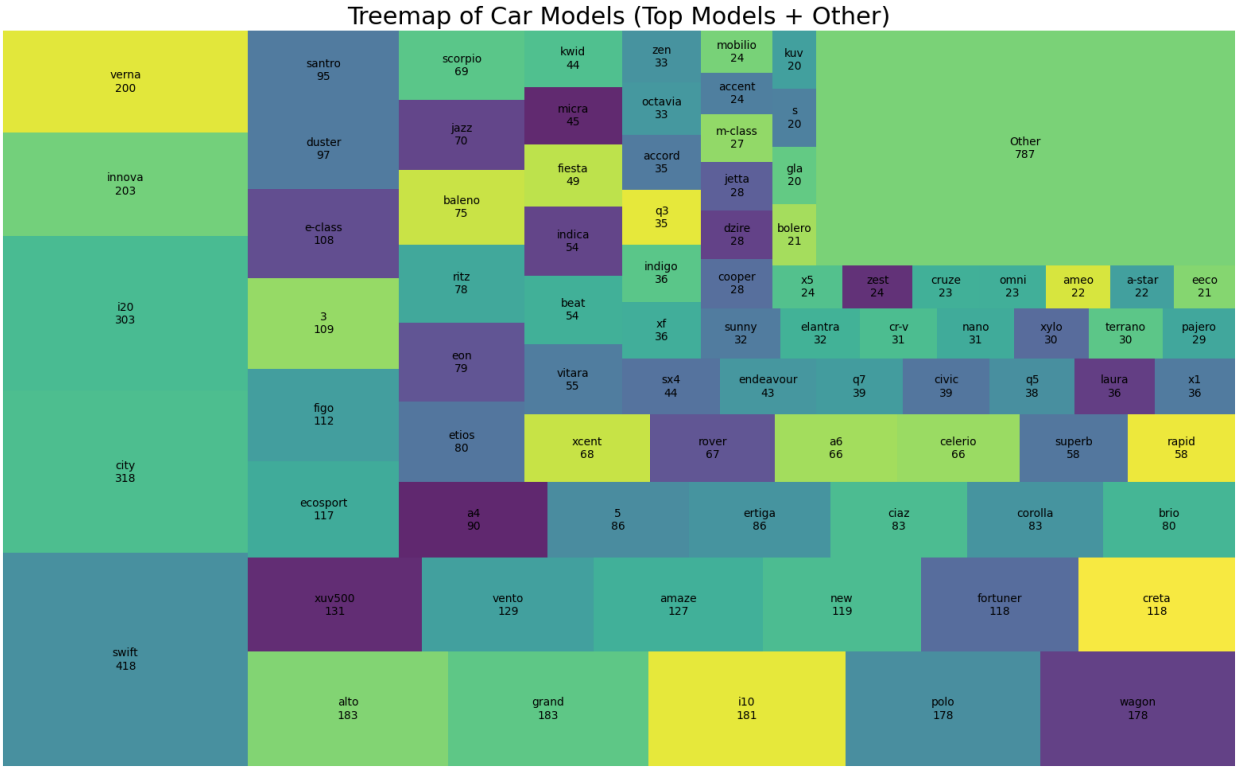
Graph 34. Brand Barplot



Observations and Implications on Brand Barplot

The brand bar plot indicates a highly skewed distribution, dominated by a few mass-market manufacturers, with niche and premium brands occurring much less frequently. Notably, Maruti and Hyundai used cars account for a large share of the dataset,, with 19.9 percent and 18.5 percent representation, respectively. This implies their prominent presence in the Indian used-car market and dominant representation in the used-car ecosystem. It is notable that Honda and Toyota used cars comprise the next tier, representing 10.2 percent and 7.0 percent of the dataset, respectively. Mercedes-Benz, BMW, and Audi luxury brands are represented in much smaller percentages (ranging from 2 to 5 percent). This implies that used cars from luxury brands are present in the dataset but are not predominant. Other, less frequent brands make up less than 2 percent of the dataset, and have frequencies close to zero percent. Overall, it is evident that high-frequency brands that primarily influence price value are considered mainstream, while low-frequency brands trail behind.

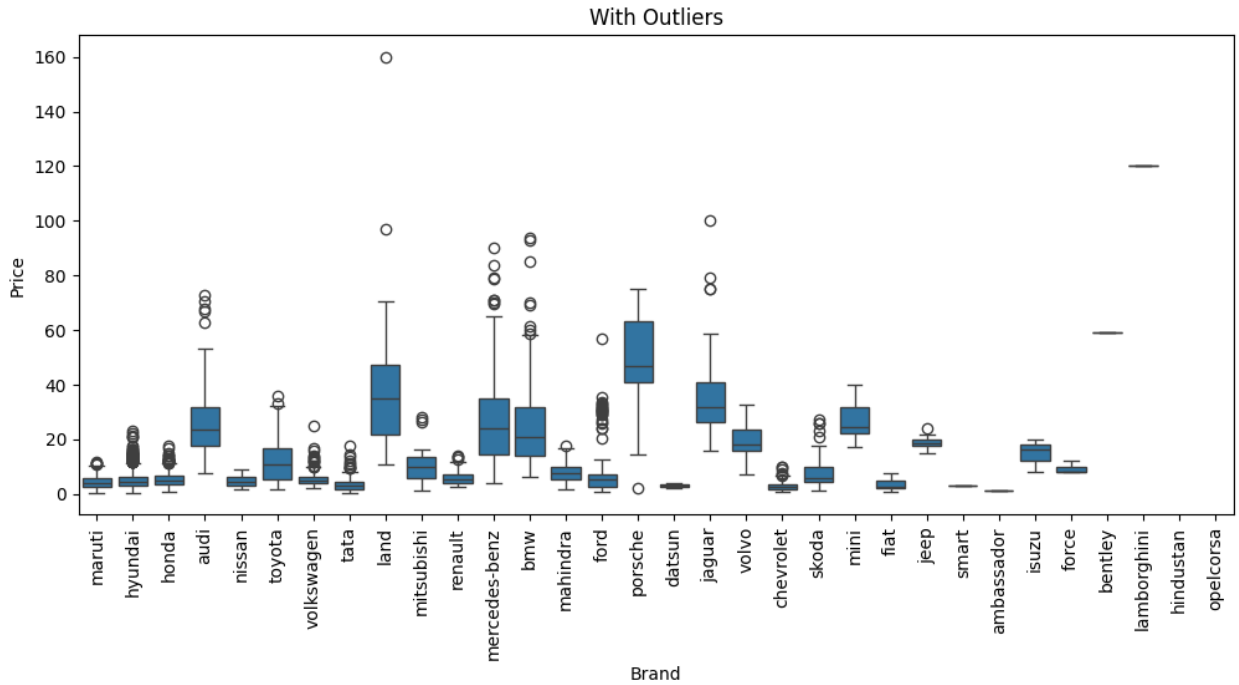
Graph 35. Treemap of the most common Models



Observations and Implications on the most common Models Treemap

The most common models in this treemap reveal there is a highly skewed distribution of car models, in which a small amount of popular models, including Swift (418 models), City (318 models), and i20 (303 models), are the most common models in the dataset. There are also a large number of infrequent models in the Other block (787 models), suggesting they are low-frequency. These less common models may negatively affect further modeling procedures and require considerable feature engineering techniques. The visual partitioning of top models also suggests potential market concentration or consumer preference clustering, which could be invaluable for pricing strategies and segmentation. The overall structure of the treemap illustrates the need to treat uncommon models with caution when they are retained in model processing, as they can introduce noise and instability. Most importantly, due to the overall structure of the treemap, the Models independent variable is decided to be dropped from the initial X feature matrix for reasons discussed later.

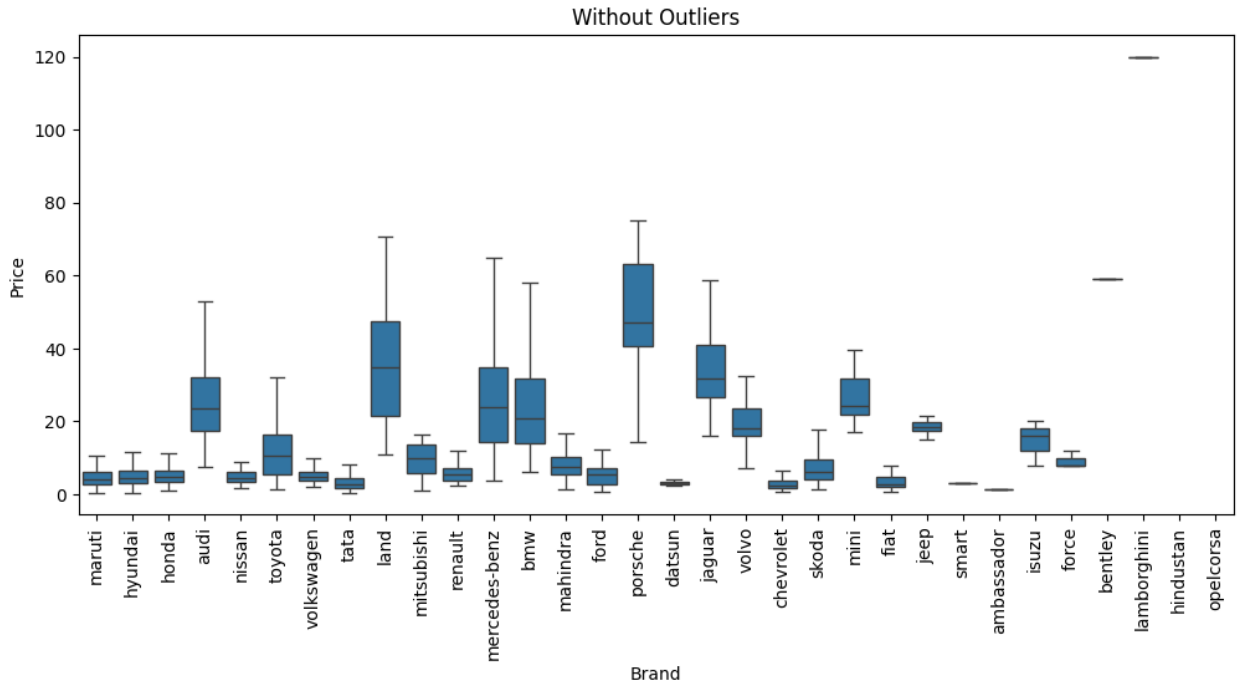
Graph 36. Brand vs. Price Boxplot (With Outliers)



Observations and Implications on Brand vs. Price Boxplot (With Outliers)

The Brand vs. Price with outliers boxplot suggests the following insights. First, luxury brands, including Mercedes-Benz, BMW, Audi, Jaguar, Porsche, and Bentley, have the highest medians, the most variability, and the most extreme outliers. This indicates the presence of premium models with greater variability in price spread. More common mainstream brands, including Maruti, Hyundai, Honda, Toyota, Ford, Mahindra, and Tata, cluster in lower- to mid-range price points, where pricing is more predictable, and there are fewer outliers. Brands, including Land Rover and Mitsubishi, have higher medians and wider spreads, suggesting a mix of rugged SUVs and premium variations. Sparse infrequent brands, including Smart, Ambassador, Isuzu, Force, and OpelCorsa have slim minimal boxplots due to being very low sample size, causing the price patterns to be less trustworthy. In general, outliers are mostly found in luxury and upper-tier brands, which indicate the highest-value used car vehicles.

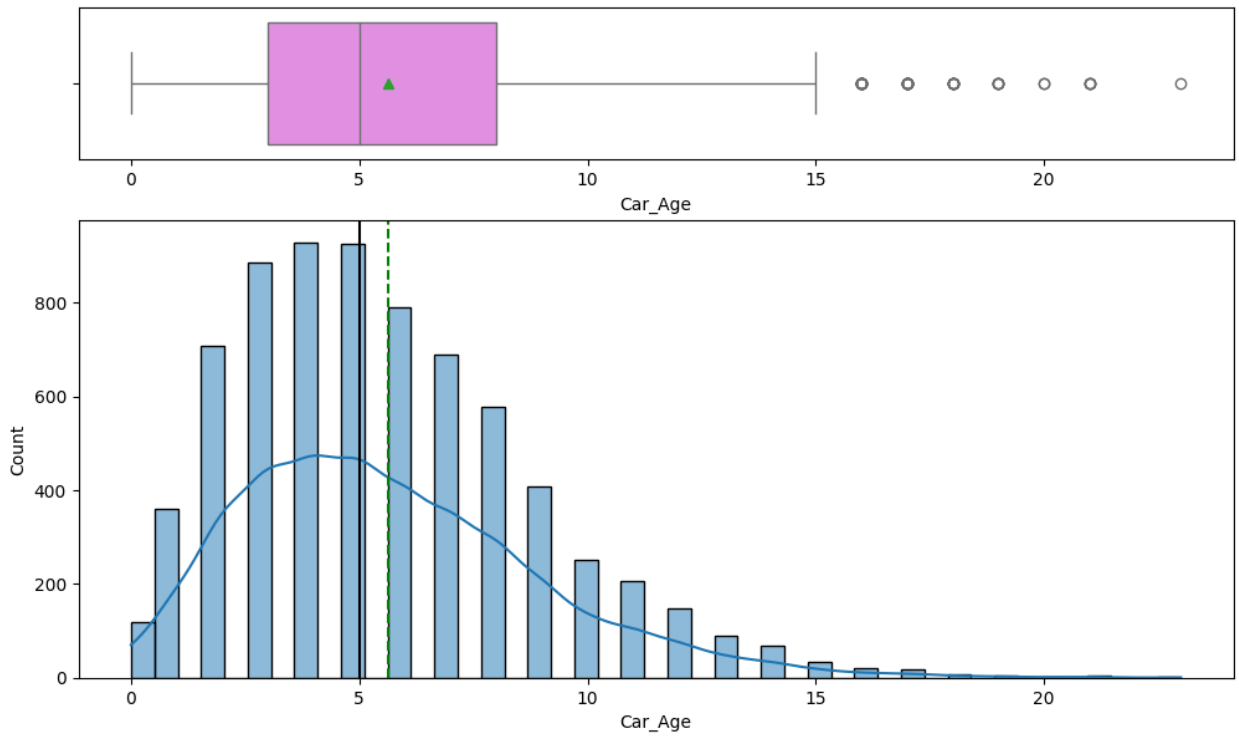
Graph 37. Brand vs. Price Boxplot (Without Outliers)



Observations and Implications on Brand vs. Price Boxplot (Without Outliers)

The Brand vs. Price without outliers boxplot suggests the following insights. First, it is now clearer that luxury and performance brands, including Porsche, Jaguar, Mercedes-Benz, and BMW, still retain substantially elevated median prices once outliers are removed. The wider boxes for these luxury brands indicate diverse premium lineups that command powerful resale prices. Second, it is now clearer that mid-tier brands, including Toyota, Honda, Volkswagen, and Ford, fit in the middle price ranges, indicating a mix of lower-priced and higher-spec models. Third, mass-market brands, including Maruti, Hyundai, Tata, Renault, and Datsun, are grouped in the lower price range with tighter distributions, suggesting more uniform, budget-priced options. Fourth, sparse brands, including Bentley, Lamborghini, Smart, and Ambassador, are shown as single data points or extremely constricted boxes due to little to no representation, resulting in less generalizability. Overall, once outliers are removed, the evidence further reinforces that brand is a stronger categorical predictor of used car prices.

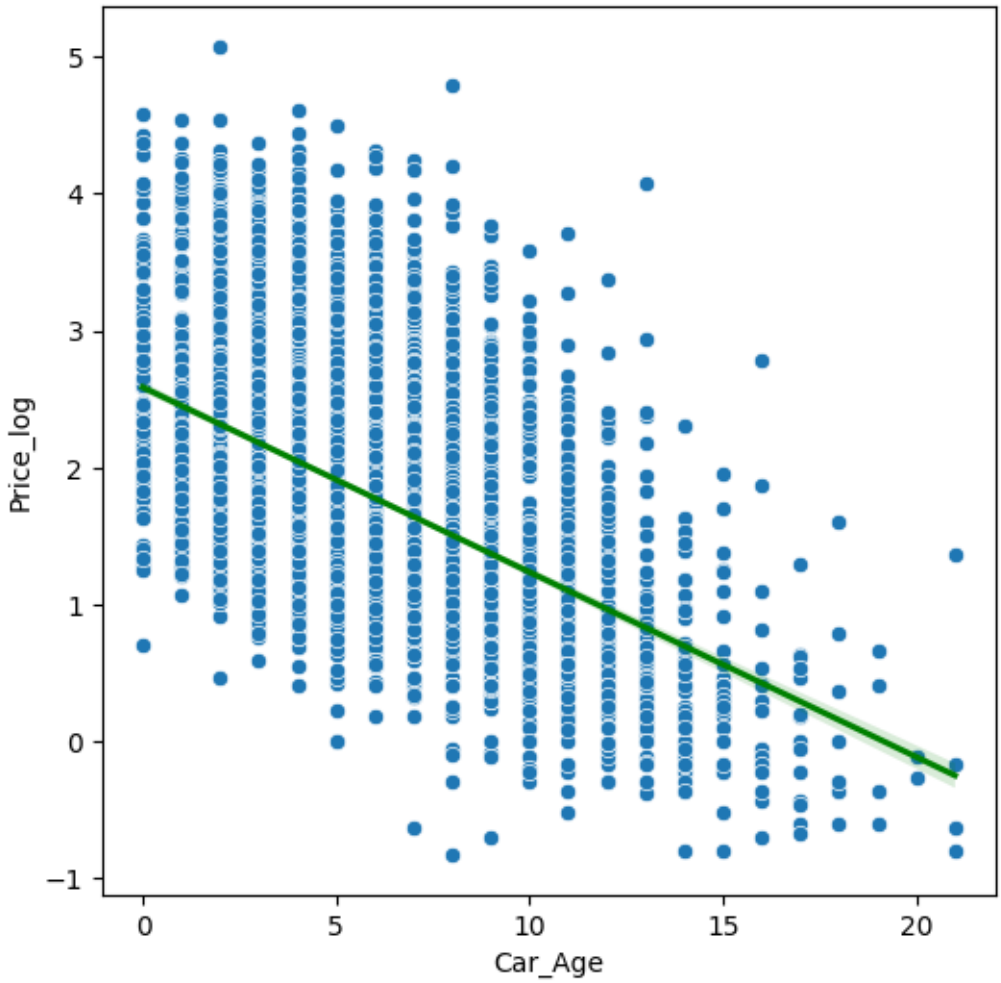
Graph 38. Car Age Boxplot and Histogram



Observations and Implications on Car_Age Boxplot and Histogram

The boxplot and histogram of the feature-engineered Car_Age continuous variable show that the most commonly used cars are between 2 and 8 years old. The distribution is right-skewed. There is a long right tail, indicating a notable number of older used cars, as seen in the outliers. very high-end outlier values. These outliers represent used cars that are mid-teen or older, which are less common. The median (denoted by a solid black line in the histogram) is 5, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 5.64. Overall, Car Age shows a concentrated cluster of young-to-mid-age used cars, with a long right tail of older outliers. This right-skewed pattern best captures the dynamics of depreciation and explains variation in used-car prices.

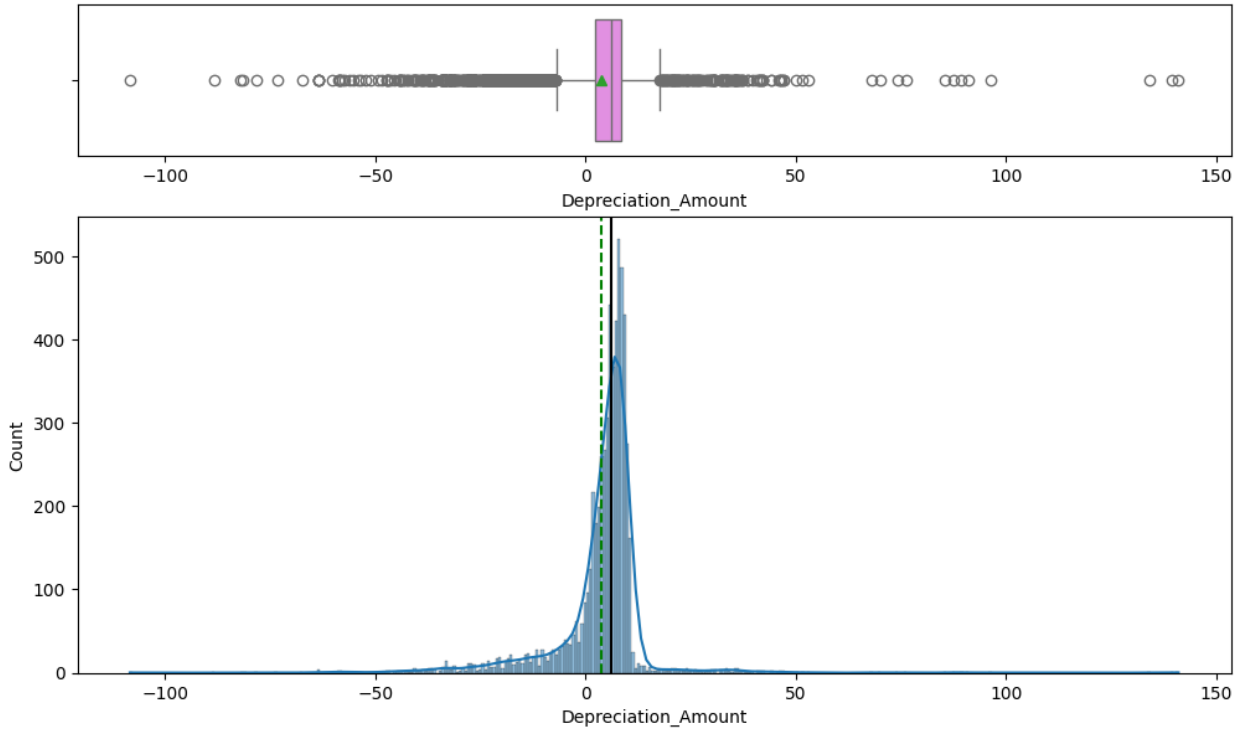
Graph 39. Car_Age and Price_log Scatterplot



Observations and Implications on Car_Age and Price_log Scatterplot

The Car_Age and Price_log scatterplot shows a clear negative linear relationship, with the price of a used car steadily declining as it ages. There is evidence that older cars sell for lower prices, even after applying a log transformation to the target variable. As for younger used cars, most points are clustered closer to the regression line, indicating a rate of early-life depreciation. Older cars indicate greater price spread, likely due to differences in condition and uncommon, infrequent models. Overall, Car_Age is a more robust predictor of used-car prices, with older vehicles introducing more noise into the model.

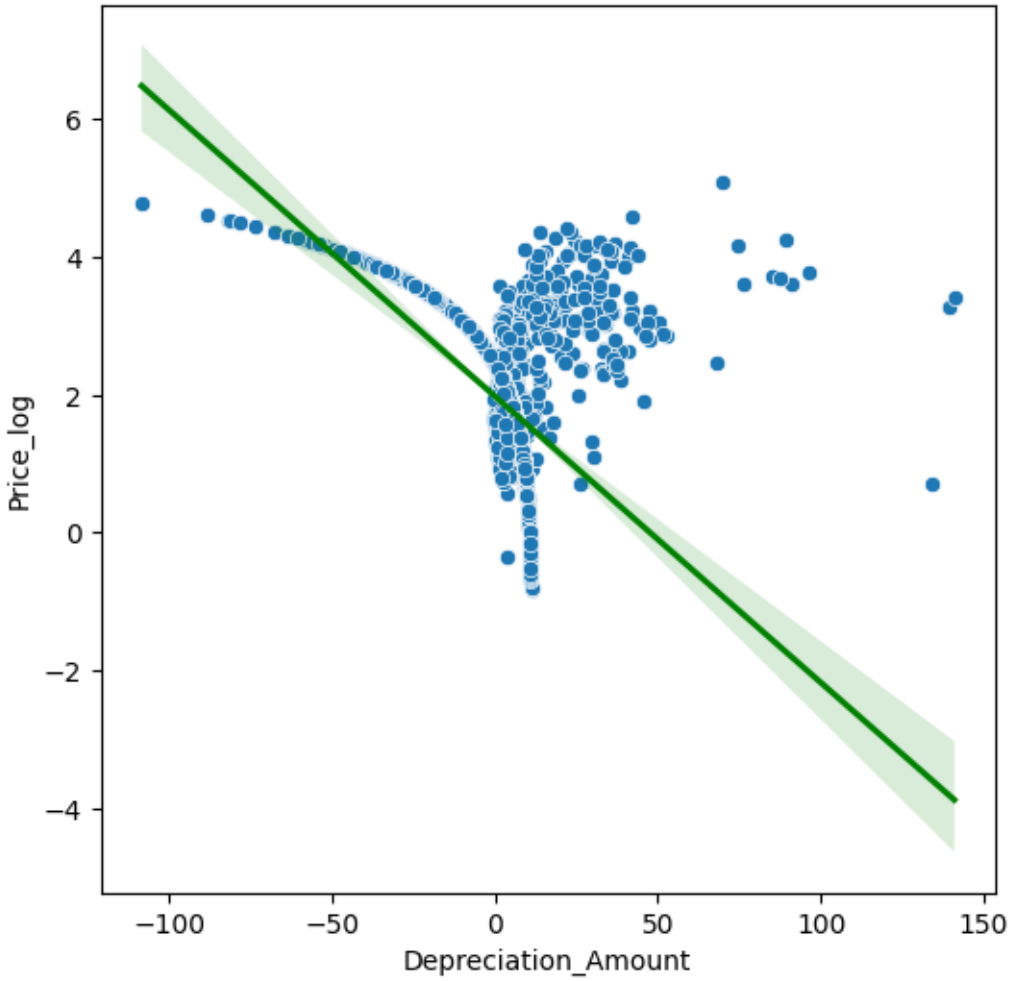
Graph 40. Depreciation_Amount Boxplot and Histogram



Observations and Implications on Depreciation_Amount Boxplot and Histogram

The Depreciation Amount boxplot and histogram show a strongly right-skewed distribution. There is a significant concentration near low depreciation values, with a long tail extending toward large depreciation amounts. This implies a typical Indian used-car market pattern, where mass-market used cars steadily lose value, whereas luxury used cars decline significantly in absolute terms. The median (denoted by a solid black line in the histogram) is 6.04, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 3.68. In particular, the mean is pulled upward, sitting to the right of the median, indicating that a minority of high-depreciation used cars raises the average. The outliers in the visualizations signify realistic economic behavior. The boxplot indicates multiple high-value outliers, which are not errors but rather represent luxury or premium models that incur losses in absolute terms, even when they remain costly. There is also evidence of tight central clustering, with the majority of vehicles within a narrow depreciation band, suggesting consistent value retention across common brands and models. Overall, the depreciation amounts show an evident right-skewed distribution with a balanced central mass and significant outliers.

Graph 41. Depreciation_Amount vs. Price_log Scatterplot

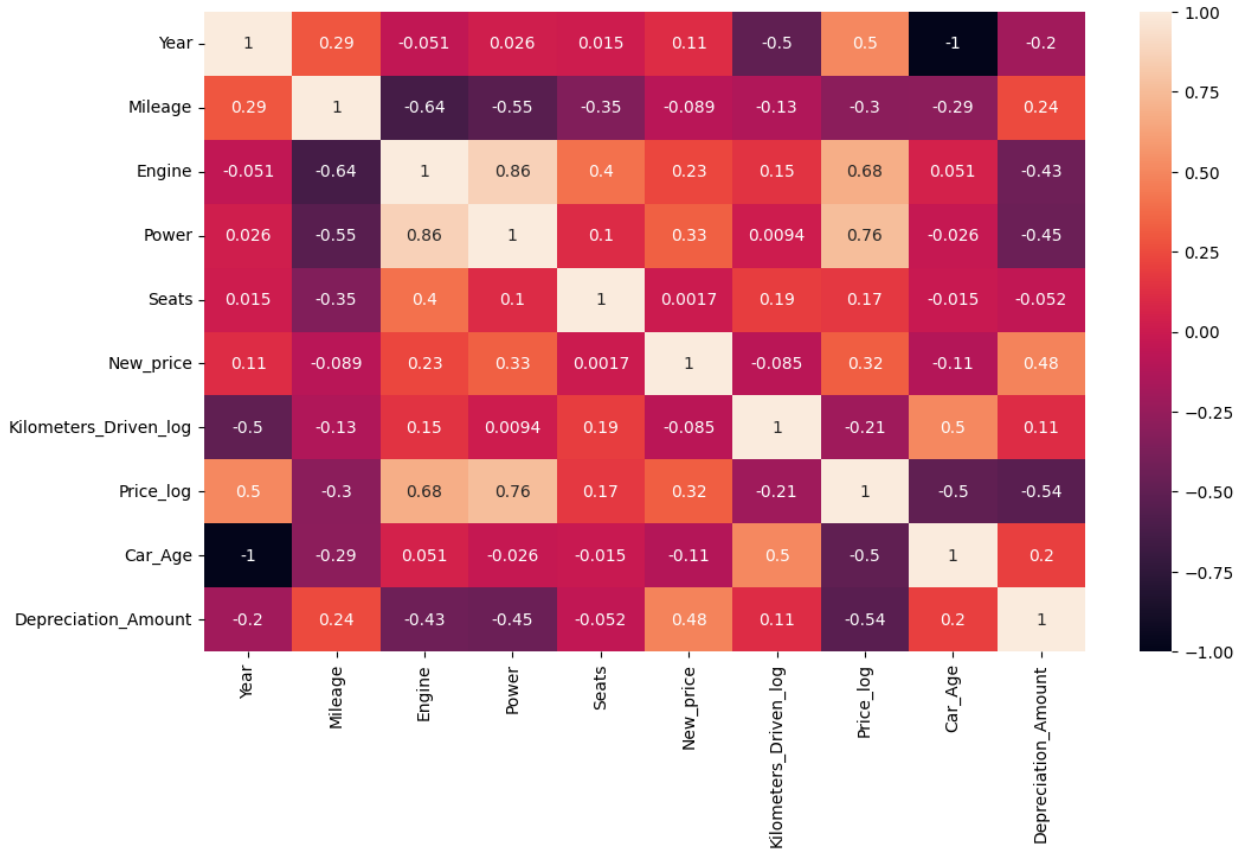


Observations and Implications on Depreciation_Amount vs. Price_log Scatterplot

The Depreciation_Amount and Price_log scatterplot shows a strong negative linear relationship. As the depreciation amount increases, the log-transformed price decreases. The regression line slopes downward, indicating that used cars with greater depreciation usually have lower prices. Using Price_log straightens the relationship between depreciation and price, yielding a more linear, model-friendly visualization. There is evidence that low-depreciation used cars cluster at much higher log prices, suggesting newer or value-retaining models, which are usually represented by strong or premium brands. There is also evidence of high-depreciation vehicles that show greater variability, suggesting realistic variation in how luxury or older used cars fall in value. Overall, the scatterplot validates that Depreciation Amount is a robust explanatory feature, capturing that used cars that lose more value from new to used

consistently lie at lower prices. The log transformation indicates a clear linear relationship.

Graph 42. Updated Heatmap With Car_Age and Depreciation_Amount



Observations and Implications on Updated Heatmap With Car_Age and Depreciation_Amount

The updated Heatmap with Car Age and Depreciation Amount includes the following insights. According to the heatmap, Car Age is a more interpretable transformation of Year, and should replace it in all proposed models. Car Age has a moderate negative correlation with Price_log, with a Pearson correlation coefficient of -0.5. This indicates an expected depreciation pattern in which older cars tend to cost less. Car Age has a moderate positive correlation with Kilometers_Driven_log, with a Pearson correlation coefficient of 0.5. This indicates that older used cars will usually have more kilometers driven. Depreciation Amount has a moderate positive correlation with New_price, with a Pearson correlation coefficient of 0.48. This indicates that higher-priced new used cars lose more value in full terms. Depreciation Amount is also

moderately negatively correlated with Power and Engine, with Pearson correlation coefficients of -0.45 and -0.43, respectively. This implies that used cars with more powerful engines and stronger power-assist systems maintain their value. Overall, the updated heatmap indicates that Car Age and Depreciation Amount augment the essential, non-redundant economic signals for the proposed models, thereby enhancing their predictive power.

Data Preprocessing

Missing Value Treatment

Table 10. Updated list of Missing Values for each independent variable

	0
Name	0
Location	0
Year	0
Kilometers_Driven	0
Fuel_Type	0
Transmission	0
Owner_Type	0
Mileage	2
Engine	46
Power	175
Seats	53
New_price	6246
Price	1234
Kilometers_Driven_log	0
Price_log	1234
Brand	0
Model	0
Car_Age	0

Observations and Implications on Missing Values for each Independent Variable Table

According to this updated table that shows the missing values for each independent variable, there are 2 missing values for Mileage, 46 missing values for Engine, 175 missing values for Power, 53 missing values for Seats, 6,246 missing values for New price, 1,234 missing values for Price, and 1,234 missing values for Price log. Since there are 6,246 missing values for the new price, a hierarchical imputation strategy is implemented to avoid introducing noise into the variable. To clarify, the imputation strategy involves three-level hierarchical steps. The first step involves imputing the median price for the same Brand and Model combination of a used car using a best-guess procedure. The second step involves imputing the median for each brand as a second-best guess. The third final step involves imputing the overall median New price of the entire dataset. Similarly, a hierarchical imputation strategy is used for the Seats variable. To be specific, the imputation strategy uses a two-step hierarchical approach that relies on the typical seating capacity of similar used cars. First, the median for each brand-model combination is calculated and assigned back to the corresponding rows. This step allows most missing entries to be filled in using the seating arrangement for that specific type of used car. The second step involves falling back to the overall median of the Seats column when no brand-model combination is available. This ensures that every missing value is filled in using the most definitive and appropriate information available. As for imputing missing values for Power and Engine, the median of each variable is used to fill them. Unlike other independent variables, the median will be used to fill in Mileage values initially set to 0.0, except for the number of missing values indicated in the table above. Since there are 1,234 missing values in the target Price and transformed target New Price variables, these rows cannot be used for modeling and will be removed. The rationale for dropping rows is to avoid introducing bias into the data.

Combining Infrequent Categories for Particular Independent Variables

As previously stated, some infrequent categories for particular independent variables were combined to stabilize the proposed models and to avoid sparse-categorical issues. To start with, the infrequent CNG, LPG, and Electric categories found in the Fuel Type variable were combined into an “Other_Fuel_Type” variable, since these categories combined roughly represented 0.11 percent of the Fuel Type variable. Secondly, the infrequent Third, Fourth & Above categories in the Owner Type variable were combined into an “Owner_3plus” variable, since these categories roughly accounted for 2.11 percent of the Owner Type variable. Thirdly, the infrequent used car brands (each of these brands are represented less than 1 percent) that

include Jaguar, Fiat, Mitsubishi, Mini, Volvo, Porsche, Jeep, Datsun, Force, Isuzu, Ambassador, Smart, Bentley, and Lamborghini (excluding Hindustan and Opelcorsa– these brands are absent after removing several rows from the Price and Price log variables) were combined into a "Rare_Brands" variable, since these categories combined roughly represented 3.3 percent of the Brand variable. Lastly, the infrequent seat number categories of 2.0, 6.0, 9.0, and 10.0 found in the Seats variable were combined into a "Rare_Seat_Values" variable, since these categories combined roughly represented 0.9 percent of the Seats variable.

Outlier Handling

Based on the above univariate analysis graphs for the numerical independent variables (i.e., features) and the dependent variables Price and Price_log, no outliers will be capped or removed. All outlier values are true, real, and meaningful, reflecting genuine used-car characteristics. Capping or removing outliers would instead distort the distributions of these variables, introduce artificial sanitization bias, reduce model transferability, and remove significant correlation patterns. Specifically, most outliers in the dataset represent actual premium used cars. In addition, when capping the outliers in each numerical independent variable, each proposed model would be weakened in its ability to capture price differences. Moreover, it would reduce variance and degrade predictive performance, and misconstrue important numerical variables, including Depreciation Amount and Price. Otherwise, separate alternative model versions that capped extreme outlier values in the Mileage and Seats variables were run for each proposed regression model. As expected, each model was significantly weaker and less potent when the outliers for the Mileage and Seats variables were capped. To clarify, each model had lower R-squared and adjusted R-squared values and higher Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Definitions of these performance metrics will be provided later in the report.

Important Observations and Insights from the data based on the Exploratory Data Analysis (EDA) and Data Preprocessing

The provided Cars4U initially contained 7,253 used cars with a mix of numerical and categorical features (i.e., independent variables). These features align with the structural patterning found in the Indian used-car market. The EDA indicates that most

cars are of mid-age with manufacturing years between 2011 and 2016, moderately driven, and mostly powered by petrol and diesel engines. Important numerical variables, including Kilometers Driven, Engine, Power, New Price, and Price, all show strong right-skewed distributions. The outliers found in each of these numerical variables represent luxury, high-performance used vehicles. Other significant relationships include newer used cars, larger engines, higher power, and higher original new-car prices, all of which positively correlate with used-car prices. In contrast, higher mileage and older age negatively correlate with used car prices. Categorical variables, including Brand, Fuel Type, Transmission, Owner Type, and Location, reveal significant price segmentation. To be specific, luxury-brand used cars, automatic-transmission vehicles, diesel-powered vehicles, and first-owner used cars command higher prices in the used-car market.

The data preprocessing steps focused on addressing structural issues while preserving market variation. Hierarchical median imputation was conducted for the New price and Seats variables, and simple median imputation procedures were conducted for the Engine, Power, and Mileage (also filled in zero-mile entries with the median) variables. Sparse infrequent categories found in the Fuel Type, Owner Type, Brands, and Seats variables were also combined to stabilize modeling and avoid sparsity. Outliers were deliberately kept, since they represent real market behavior and add important variance for predictive modeling. Feature engineering techniques involved creating Brand, Model, Car Age, and Depreciation Amount independent variables. These variables contributed to economic interpretability and enhanced predictive relationships, most notably the robust negative connection between depreciation and price. Comprehensively, the EDA and preprocessing stages reveal that the provided dataset is noisy, but also informative. Most importantly, the EDA and preprocessing stages reveal clear economic patterns that support robust price prediction modeling.

What it means for Problem formulation

The dataset indicates that Cars4U plans to understand how various used-car attributes (i.e., the independent variables) affect the target prices of used cars in its inventory. As such, the main problem is a supervised regression task focused on predicting the price target. The dataset contains 14 distinct features (i.e., the

independent variables), which include manufacturing year, kilometers driven, engine size, power, fuel type, transmission, ownership history, and new-car price. These features altogether shape the modeling process of the economic depreciation of vehicles. There is a strong presence of right-skewed independent variables, missing values, and illogical entries, which indicate that the problem formulation should also include data cleaning, transformation, and outlier handling to ensure dependable and accurate predictions. In particular, since the Name independent variable comprises over 2,041 unique brand-model combinations, feature engineering is required to extract meaningful Brand and Model predictors (i.e., independent variables) to address the issue of relying solely on raw text. Overall, the problem involves constructing a robust, generalizable model that predicts used car prices using enhanced, structured, vehicle-independent independent variables while mitigating noise, skewness, and high-cardinality categorical variables.

Key patterns in the data

The exploratory data analysis reveals numerous crucial patterns that indicate how the proposed models should be constructed and interpreted. Numerical independent variables, including Kilometers Driven, Engine, Power, New Price, and the target dependent variable Price, each exhibit strongly right-skewed distributions with extreme outliers (possibly due to luxury used cars), necessitating transformation or purging. Mileage contains numerous impossible values of 0.0, demonstrating data quality issues that must be addressed as missing values. Categorical independent variables also have highly imbalanced distributions. For instance, diesel-powered used cars dominate Fuel Type, manual-transmission used cars dominate Transmission, and First-owner-type used cars dominate Ownership histories. The dominant characteristics of these categorical independent variables reveal that Cars4U's inventory reflects a market condensed into a few common configurations. The Name independent variable exhibits extreme cardinality, with a large number of rare models. Thus, this reinforces the need for feature extraction, which involves creating Brand- and Model-independent variables. Overall, these patterns demonstrate that preprocessing, normalization, and cautious encoding are necessary to build stable and interpretable price-prediction regression models.

Building Models

Important Preparation Notes

Table 11. Fully Preprocessed Dataset with Updated Row and Columns

```
<class 'pandas.core.frame.DataFrame'>
Index: 6018 entries, 0 to 6018
Data columns (total 19 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Name                   6018 non-null   object
 1   Location               6018 non-null   object
 2   Year                   6018 non-null   int64
 3   Kilometers_Driven      6018 non-null   int64
 4   Fuel_Type              6018 non-null   object
 5   Transmission           6018 non-null   object
 6   Owner_Type             6018 non-null   object
 7   Mileage                6018 non-null   float64
 8   Engine                 6018 non-null   float64
 9   Power                  6018 non-null   float64
 10  Seats                  6018 non-null   object
 11  New_price              6018 non-null   float64
 12  Price                  6018 non-null   float64
 13  Kilometers_Driven_log  6018 non-null   float64
 14  Price_log              6018 non-null   float64
 15  Brand                  6018 non-null   object
 16  Model                  6018 non-null   object
 17  Car_Age                6018 non-null   int64
 18  Depreciation_Amount    6018 non-null   float64
dtypes: float64(8), int64(3), object(8)
memory usage: 940.3+ KB
```

As the main target variable is Price, the normalized “Price_log” variable will be used primarily for modeling. Hence, after applying a log transformation to the target variable, the skewness has been removed, improving model stability. In particular, there is now greater variance and improved linearity in relationships with other independent variables, making the overall regression modeling more suitable. Before splitting the data, the dataset had 6,018 rows and 19 independent variables after full preprocessing, as shown in the table above.

Table 12. Initial X_train_master and X_test_master Feature Matrix

```
<class 'pandas.core.frame.DataFrame'>
Index: 4212 entries, 2116 to 5158
Data columns (total 43 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Year                       4212 non-null   int64
 1   Mileage                    4212 non-null   float64
 2   Engine                     4212 non-null   float64
 3   Power                      4212 non-null   float64
 4   New_price                  4212 non-null   float64
 5   Kilometers_Driven_log      4212 non-null   float64
 6   Car_Age                    4212 non-null   int64
 7   Depreciation_Amount        4212 non-null   float64
 8   Location_Bangalore         4212 non-null   int64
```

```
 9   Location_Chennai           4212 non-null   int64
 10  Location_Coimbatore        4212 non-null   int64
 11  Location_Delhi             4212 non-null   int64
 12  Location_Hyderabad         4212 non-null   int64
 13  Location_Jaipur            4212 non-null   int64
 14  Location_Kochi             4212 non-null   int64
 15  Location_Kolkata           4212 non-null   int64
 16  Location_Mumbai            4212 non-null   int64
 17  Location_Pune              4212 non-null   int64
 18  Fuel_Type_Other_Fuel_Type  4212 non-null   int64
 19  Fuel_Type_Petrol           4212 non-null   int64
 20  Transmission_Manual        4212 non-null   int64
 21  Owner_Type_Owner_3plus     4212 non-null   int64
 22  Owner_Type_Second          4212 non-null   int64
 23  Seats_5.0                  4212 non-null   int64
 24  Seats_7.0                  4212 non-null   int64
 25  Seats_8.0                  4212 non-null   int64
 26  Seats_Rare_Seat_Values     4212 non-null   int64
 27  Brand_audi                 4212 non-null   int64
 28  Brand_bmw                  4212 non-null   int64
 29  Brand_chevrolet            4212 non-null   int64
 30  Brand_ford                 4212 non-null   int64
 31  Brand_honda                4212 non-null   int64
 32  Brand_hyundai              4212 non-null   int64
 33  Brand_land                 4212 non-null   int64
 34  Brand_mahindra             4212 non-null   int64
 35  Brand_maruti               4212 non-null   int64
 36  Brand_mercedes-benz        4212 non-null   int64
 37  Brand_nissan               4212 non-null   int64
 38  Brand_renault              4212 non-null   int64
 39  Brand_skoda                4212 non-null   int64
 40  Brand_tata                 4212 non-null   int64
 41  Brand_toyota               4212 non-null   int64
 42  Brand_volkswagen           4212 non-null   int64
dtypes: float64(6), int64(37)
memory usage: 1.4 MB
```

Before proceeding with the model-building process, all categorical independent variables (e.g., Location, Fuel Type, Transmission, Owner Type, Brand, and the discrete numerical variable Seats) will be one-hot encoded. This procedure avoids the model assuming meaningful distances between distinct categories, ordinal meaning, or equal spacing between categories within each variable. It would also remove spurious numeric relationships, enable the model to learn independent effects for each accessibility index group found in each variable, and improve overall interpretability. The independent variables (excluding Name, Model, Price, Price_log, and Kilometers_Driven–these will be dropped from the initial X feature matrix) will be grouped into an “X” variable, and the dependent variable (in both the original Price and Log Price transformed variable) will be assigned to a vector variable named “y”. The

reason for dropping Name from the X feature matrix is that keeping it will introduce noise. The reason for dropping Model from the X feature matrix is that it introduces many sparse, unstable dummy variables. Other reasons for dropping the Model independent variable include its multicollinearity (later defined in the report) relationship with Brand, fluctuating text formatting, and rare categories. The reason for dropping Kilometers_Driven from the X feature matrix is that Kilometers_Driven_log has replaced it. The reason for dropping Price and Price_log from the X feature matrix is that they are the target variables and should be included in a separate Y vector. The data will also be split into training and test sets, with 70 percent reserved for training and 30 percent for testing, since the 6,018 rows are suitable for this split. After splitting the data, the initial X_train_master feature matrix contains 4,212 rows and 43 columns, and the X_test_master feature matrix contains 1,806 rows and 43 columns. Each model will start by using the initial features from the X_train_master and X_test_master feature matrices (e.g., 43 independent variables shown in the above table), and the dependent variable y_train will be the "Price_log" target variable. Each proposed model will be trained using the "Price_log" target variable. All X_train_master, X_test_master, y_train, and y_test partitioned variables will also be converted to a float type to ensure all variables are numeric and continuous for each model.

Proposed Regression Models

The proposed models being built are the following: Ordinary Least Squares (OLS) Regression (Multiple Linear Regression), L2-Regularized Linear Regression (Ridge Regression), L1-Regularized Linear Regression (Lasso Regression), Decision Tree Regression (Decision Tree), Random Forest Regression (Random Forest), and k-Nearest Neighbors Regression (KNN Regression). A comprehensive discussion detailing each model's methodology, characteristics, and limitations is provided in separate sections for each proposed model.

Get Model Score (Performance Metrics) Description

The get_model_score function calculates the R-squared, adjusted R-squared, root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) for each proposed model. The R-squared value indicates how

much of the variation in a dependent variable is explained by the independent variables (predictors) in a regression model. The Adjusted R-squared value calculates the proportion of the variance a regression model explains. It also penalizes unnecessary predictors (independent variables), providing a more accurate measure of model fit. Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Difference (MAPE) are evaluation metrics used for regression models. RMSE measures the typical size of a model's prediction errors in the same units as the target variable. It begins by squaring the residuals, averaging them, and taking the square root of the result. MAE measures the average size of a model's prediction errors, where it treats every error equally and maintains the result in the same units as the target variable. It uses absolute residuals, obtained by subtracting the actual values from the predicted values and taking the absolute value of the difference. MAPE measures the average percentage difference between actual and predicted values by taking the residuals and converting them to percentage errors. It divides each error by the actual value, converts the result to a percentage, and then averages those percentages across all observations. The closer the R-squared and Adjusted R-squared values are to 1, the more of the outcome's variance the model captures. It is important to note that it is also ideal for the R-squared and Adjusted R-squared values not to be 1 or too close to 1, as this can indicate overfitting. For optimal model performance, RMSE, MAE, and MAPE evaluation metrics should be as low as possible.

The get_model function operates as follows: First, it generates predictions on the X_train and X_test datasets. Second, it exponentiates predictions and targets to revert them to the original price scale; by default, when the log flag is set to true, this converts the Price_log used in training back into its original Price variable. Third, it computes the R-squared values for the training and test datasets, then rounds them to 2 decimal places. Fourth, it grabs the sample sizes and number of predictors from the y_train and y_test vectors, and the number of columns from the X_train dataset. Fifth, it computes the adjusted R-squared values for the train and test datasets and rounds them to 2 decimal points. Sixth, it computes the error metrics for root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). Seventh, it compiles all the metric results into a dictionary, converts them into a one-row data frame, and then displays the data frame.

Check for Multicollinearity

Before proceeding to discuss and assess the performance of the first Multiple Linear Regression, Ridge Regression, and Lasso Regression models, it is important to note issues related to multicollinearity. Multicollinearity is a statistical phenomenon in which two or more independent variables (i.e., features or predictors) are highly correlated. To put it more simply, when two or more independent variables essentially carry the same information. Due to this phenomenon, multicollinearity can cause the individual effects of each independent variable on the target dependent variable to be isolated. It can lead to unstable coefficients, inflated variance, and inaccurate model results. Likewise, multicollinearity can make it difficult to interpret coefficients, render variables statistically insignificant, and cause numerical issues in the Multiple linear regression and Ridge regression models. Independent variables with high VIFs or infinite VIFs indicate that they do not provide new information, increase standard errors, cause interpretational issues, or are linear combinations of other variables. A multicollinearity function was called multiple times to remove several independent variables with inflated Variance Inflation Factor (VIF) values greater than 10 or 5, or with infinite values in the Multiple Linear Regression model. Checking for multicollinearity is a viable exploratory technique to improve the performance of a multiple linear regression model.

Proposed Model #1 Multiple Linear Regression

Multiple Linear Regression Model Definition, Properties, and Justification

The first proposed model for predicting the price of a used car in 100,000 INR from Cars4U's inventory is an OLS multiple linear regression model. A multiple Linear Regression model describes the relationship between a numeric target and several independent variables (predictors) by fitting a single linear equation that minimizes the sum of squared errors. Thus, a multiple Linear Regression model assumes a linear relationship between the target y variable and the weighted sum of the independent variables (predictors). In this case, the weights or parameters in the linear regression model are the coefficients that indicate the extent to which each predictor contributes to the overall prediction. The main role of a multiple Linear Regression model is to find the optimal coefficients that minimize the difference between the predictions and the true values, using linear algebra. This model isolates each predictor's unique

contribution by holding all other predictors constant, ensuring that each coefficient reflects the relationship with its corresponding predictor. However, a multiple Linear Regression model performs poorly when relationships are nonlinear, independent variables are correlated (multicollinear), outliers and extreme values are present, and when the data is messy. For this proposed model, two versions were generated: one with scaled X_train_ols and X_test_ols features, and the other with unscaled X_train_ols and X_test_ols features. Feature scaling is the process of rescaling all numerical predictors to a common scale, typically by standardizing them to have a mean of 0 and a standard deviation of 1. This prevents variables with larger numeric ranges from dominating the model. In general, feature scaling is not needed for a Multiple Linear Regression Model. However, even after running the model versions, it was observed that scaling the features did not affect the final multiple linear regression model chosen.

A Multiple Linear Regression model was chosen as the first model. The following are reasons why the model is best suited to explore the data and problem at hand. It can provide interpretable relationships between the used car attributes and the target price. The dataset comprises multiple continuous independent variables, including engine, power, and kilometers driven, that consistently fit a linear model groundwork once transformed and preprocessed. It also aligns with the business need for clear pricing insights. After preprocessing stages (e.g., handling outliers, reducing multicollinearity, and applying log transformations), the provided dataset meets the assumptions necessary for a Multiple Linear Regression model to perform accurately. A Multiple Linear Regression model also serves as a robust benchmark against which other follow-up models, such as Ridge Regression, Lasso Regression, and Random Forest, are evaluated. Overall, a Multiple Linear Regression model quantifies how each independent variable contributes to determining the target Price variable, making it useful for both prediction and critical decision-making.

Multiple Linear Regression Model Building Process

Addressing Multicollinearity (Technique #1)

A multicollinearity function was called multiple times to remove several independent variables from the X_train (X_train_ols) and X_test (X_test_ols) datasets with inflated Variance Inflation Factor (VIF) values, including infinite values, values

greater than 10 or 5, or values in the Multiple Linear Regression model. Between each removal of these independent variables, the model was trained on an updated X_train_ols dataset and y_train (Price_log) vector, and an output Multiple Linear Regression model summary was produced. The first independent variable (feature) removed from the X_train_ols and X_test_ols datasets was "Year" due to its INF VIF score. Even though "Car Age" also had an INF VIF score, it was kept because it is more interpretable, generalizes better across the training and test datasets, and captures depreciation better. The second independent variable (feature) removed was "Engine," due to having a VIF score (approximately 10.62) above 10. The third independent variable removed was "Seats_5.0" because its VIF (approximately 9.35) exceeded 5. The final independent variable that was removed because of having a VIF score greater than 5 or 10 was "Brand_maruti," due to having a VIF score (approximately 7.89) above 5. Up to this point, four consecutive multiple linear regression models have been trained.

Table 13. Updated Final VIF Diagnostics

```
=== VIF Diagnostics ===

Features with VIF > 5:
  feature         VIF
0   const  585.217762

Features with VIF > 10:
  feature         VIF
0   const  585.217762

Features with infinite VIF:
Empty DataFrame
Columns: [feature, VIF]
Index: []

Features with NaN VIF:
Empty DataFrame
Columns: [feature, VIF]
Index: []
```

Observations and Implications on Updated Final VIF Diagnostics Table

As seen in the updated final VIF Diagnostics table, after removing the "Year," "Engine," "Seats_5.0," and "Brand_maruti" variables, severe multicollinearity is now absent. It is now evident that after removing these predictors (features), all remaining features have VIF scores below 5, indicating acceptable multicollinearity. By removing these independent variables, the Multiple Linear Regression model has become more stable.

Addressing statistically insignificant independent variables (Technique #2)

Following the removal of features with severe multicollinearity, features with p-values greater than 0.05 were removed, since they are considered statistically insignificant. Again, between each removal of these independent variables, the model was trained on an updated X_train_ols dataset and y_train (Price_log) vector, and an output Multiple Linear Regression model summary was produced. Keeping count of the previously removed independent variables with severe VIF scores, the fifth independent variable removed was "Location_Chennai," with a p-value of 0.883. The sixth independent variable removed was "Brand_hyndai," because it had a p-value of 0.528. The seventh independent variable removed was "Location_Mumbai," with a p-value of 0.408. The eighth independent variable removed was "Location_Kochi," with a p-value of 0.376. The ninth independent variable removed was "Brand_volkswagen," with a p-value of 0.317. The tenth independent variable removed was "Brand_nissan," with a p-value of 0.301. The eleventh and final independent variable removed was "Seats_8.0," with a p-value of 0.174. Up to this point, seven consecutive multiple linear regression models have been trained. Checking p-values is a second viable exploratory technique for improving a multiple linear regression model's performance.

Table 14. Updated Final Multiple Linear Regression Summary Results

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              Price_log   R-squared:                       0.927
Model:                            OLS   Adj. R-squared:                  0.927
Method:                 Least Squares   F-statistic:                     1661.
Date:                Mon, 23 Mar 2026   Prob (F-statistic):               0.00
Time:                        15:46:39   Log-Likelihood:                 119.37
No. Observations:                4212   AIC:                            -172.7
Df Residuals:                    4179   BIC:                             36.67
Df Model:                          32
Covariance Type:            nonrobust
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                          2.7584      0.083     33.141      0.000       2.595       2.922
Mileage                       -0.0217      0.001    -15.061      0.000      -0.025      -0.019
Power                          0.0042      0.000     29.835      0.000       0.004       0.004
New_price                      0.0181      0.001     25.756      0.000       0.017       0.020
Kilometers_Driven_log         -0.0344      0.007     -5.056      0.000      -0.048      -0.021
Car_Age                       -0.1031      0.002    -62.734      0.000      -0.106      -0.100
Depreciation_Amount           -0.0210      0.001    -30.918      0.000      -0.022      -0.020
Location_Bangalore             0.1323      0.017      8.013      0.000       0.100       0.165
Location_Coimbatore            0.0695      0.013      5.361      0.000       0.044       0.095
Location_Delhi                -0.0416      0.014     -3.040      0.002      -0.068      -0.015
Location_Hyderabad             0.0881      0.012      7.116      0.000       0.064       0.112
Location_Jaipur               -0.0823      0.016     -5.288      0.000      -0.113      -0.052
Location_Kolkata              -0.1947      0.014    -13.886      0.000      -0.222      -0.167
Location_Pune                 -0.0382      0.013     -2.886      0.004      -0.064      -0.012
Fuel_Type_Other_Fuel_Type     -0.2339      0.036     -6.554      0.000      -0.304      -0.164
Fuel_Type_Petrol              -0.3334      0.010    -32.340      0.000      -0.354      -0.313
Transmission_Manual           -0.1471      0.012    -12.441      0.000      -0.170      -0.124
Owner_Type_Owner_3plus        -0.1347      0.027     -5.080      0.000      -0.187      -0.083
Owner_Type_Second             -0.0460      0.011     -4.317      0.000      -0.067      -0.025
Seats_7.0                      0.0450      0.015      2.966      0.003       0.015       0.075
Seats_Rare_Seat_Values         0.0843      0.040      2.105      0.035       0.006       0.163
Brand_audi                     0.3487      0.023     15.137      0.000       0.303       0.394
Brand_bmw                      0.3560      0.022     15.897      0.000       0.312       0.400
Brand_chevrolet               -0.2939      0.026    -11.216      0.000      -0.345      -0.243
Brand_ford                    -0.0719      0.017     -4.126      0.000      -0.106      -0.038
Brand_honda                    0.0853      0.013      6.526      0.000       0.060       0.111
Brand_land                     0.4842      0.042     11.414      0.000       0.401       0.567
Brand_mahindra                -0.0544      0.022     -2.495      0.013      -0.097      -0.012
Brand_mercedes-benz            0.3670      0.021     17.464      0.000       0.326       0.408
Brand_renault                 -0.0582      0.024     -2.425      0.015      -0.105      -0.011
Brand_skoda                    0.0694      0.022      3.117      0.002       0.026       0.113
Brand_tata                    -0.4939      0.021    -23.207      0.000      -0.536      -0.452
Brand_toyota                   0.2432      0.017     13.935      0.000       0.209       0.277
==============================================================================
Omnibus:                      505.497   Durbin-Watson:                   1.992
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             3679.086
Skew:                          -0.327   Prob(JB):                         0.00
Kurtosis:                       7.532   Cond. No.                     2.94e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.94e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Observations and Implications on Updated Final Multiple Linear Regression Summary Results

As seen in the updated final Multiple Linear Regression summary results table, all independent variables now have p-values less than 0.05, indicating they are statistically significant. The model's R-squared value of 0.927 indicates strong predictive performance. Most independent variables, such as Mileage, Power, New Price, Car Age, and Depreciation Amount, are considered to be highly significant and operate in economically intuitive directions. A considerable number of categorical variables, such as Location, Fuel type, transmission, Owner Type, Seats, and Brand, also show strong, meaningful effects on the target variable, Price log. There are also a few predictors, such as Fuel_Type_Other_Fuel_Type, that may not meaningfully contribute to the model. Overall, the model remains stable, given its robust

significance and persistent coefficient patterns. This model is also chosen as the ideal, represented by a multiple linear regression model.

Table 15. Final Multiple Linear Regression model coefficients

	Feature	Coefs
0	const	2.76
1	Mileage	-0.02
2	Power	0.00
3	New_price	0.02
4	Kilometers_Driven_log	-0.03
5	Car_Age	-0.10
6	Depreciation_Amount	-0.02
7	Location_Bangalore	0.13
8	Location_Coimbatore	0.07
9	Location_Delhi	-0.04
10	Location_Hyderabad	0.09
11	Location_Jaipur	-0.08
12	Location_Kolkata	-0.19
13	Location_Pune	-0.04
14	Fuel_Type_Other_Fuel_Type	-0.23
15	Fuel_Type_Petrol	-0.33
16	Transmission_Manual	-0.15
17	Owner_Type_Owner_3plus	-0.13
18	Owner_Type_Second	-0.05
19	Seats_7.0	0.04
20	Seats_Rare_Seat_Values	0.08
21	Brand_audi	0.35
22	Brand_bmw	0.36
23	Brand_chevrolet	-0.29
24	Brand_ford	-0.07
25	Brand_honda	0.09
26	Brand_land	0.48
27	Brand_mahindra	-0.05
28	Brand_mercedes-benz	0.37
29	Brand_renault	-0.06

	Feature	Coefs
30	Brand_skoda	0.07
31	Brand_tata	-0.49
32	Brand_toyota	0.24

Observations and Implications on the Final Multiple Linear Regression Model Coefficients

The final chosen Multiple Linear Regression model shows that key used-condition variables (e.g., Mileage, Kilometers Driven_log, Car Age, and Depreciation Amount) have negative coefficients. This validates that wear, usage, and aging all consistently lower used car prices. Since Power and New Price have positive coefficients, this suggests that performance and original market value remain strong predictors of higher used-car prices. Location effects vary widely, with Indian cities such as Bangalore and Hyderabad commanding premium used-car prices. On the other hand, Indian cities such as Kolkata and Jaipur show significant discounts, suggesting differences in regional demand. Fuel type, transmission, and ownership type also matter, with petrol-used cars, manual-transmission used cars, and higher-order owners all lowering used-car prices relative to their corresponding baselines. Specific used-car brands are also significant, with luxury brands such as Land Rover, Mercedes-Benz, BMW, and Audi showing strong positive premiums. Conversely, budget used-car brands such as Tata, Renault, and Chevrolet are associated with substantial negative impacts on used-car resale values.

Checking for the four multiple linear regression assumptions is an important technique to ensure the reliability and validity of the model

After building the final multiple linear regression model, it is critical to ensure that it meets four assumptions, each of which safeguards the model's validity. First, the residuals in the model should have a mean of 0. Second, the residuals should be homoscedastic, meaning they are symmetrically distributed around the regression line. A Goldfeld-Quandt Test will be used to assess whether the residuals are homoscedastic (i.e., not heteroscedastic) in the model by verifying that the p-value exceeds 0.05, and that the F-statistic score is close to 1. In other words, there is a failure in the rejection of the null hypothesis. Third, the independent variables (features) in the model must be linearly related to the target Price log variable. Under

this assumption, the residuals are plotted and do not exhibit a strong pattern, being randomly and uniformly scattered along the x-axis. Fourth, the residuals in the model must be normally distributed, or, in other words, the error terms are normally distributed. To test this assumption, a Q-Q plot and a histogram of the residuals are used.

Mean of residuals should be 0

Code Output 3. Mean of residuals

np.float64(4.133023985309225e-15)

Observations and Implications on the Mean of residuals should be 0 output

The model's mean residuals are very close to 0. The number in decimal form is 0.000000000000004133023985309225. This means that this first assumption is met. This further implies that the model is not over- or under-predicting. Also, this implies that the residual diagnostics are functioning normally.

Check for Homoscedasticity

Code Output 4. Check for Homoscedasticity

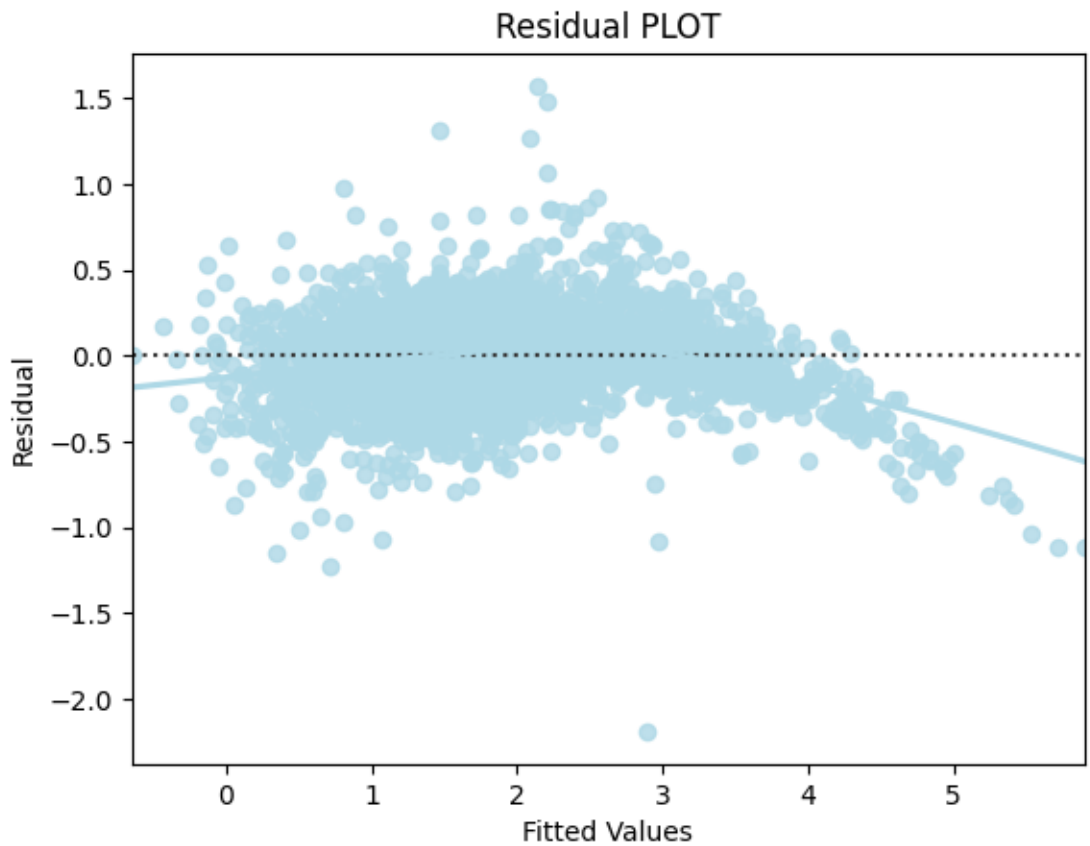
```
[('F statistic', np.float64(1.039400664107524)),
 ('p-value', np.float64(0.18953605799920994))]
```

Observations and Implications on Checking for Homoscedasticity

Here, the p-value based on the code output is 0.18953605799920994, and the F-statistic value is 1.039400664107524. The p-value is greater than 0.05, indicating that the null hypothesis cannot be rejected. The F-statistic is close to 1, indicating that the residual variances are very similar. The interpretation of these results is that the model residuals are homoscedastic. These results imply that the p-values and F-statistics are reliable and, when interpreted together, provide no statistical evidence of heteroscedasticity in the residuals.

Linearity of variables

Graph 43. Residual Plot

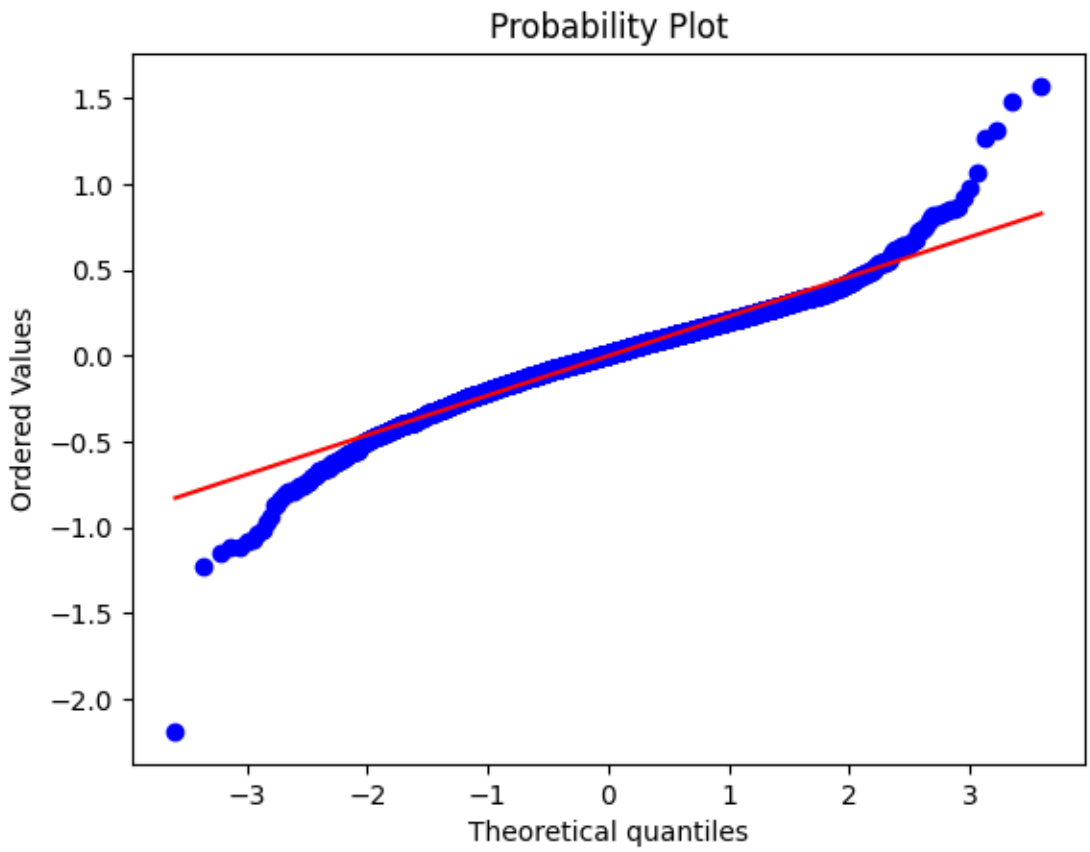


Observations and Implications on Residual Plot

In this residual plot, the residuals and the fitted values do not form a noticeable strong pattern. They are randomly and uniformly scattered on the x-axis. Thus, the condition for the linearity of the independent variables in the model is satisfied. Other implications include constant variance (so that standard errors and p-values are trustworthy), independence of errors, and no strong indication of omitted nonlinear terms.

Normality of error terms

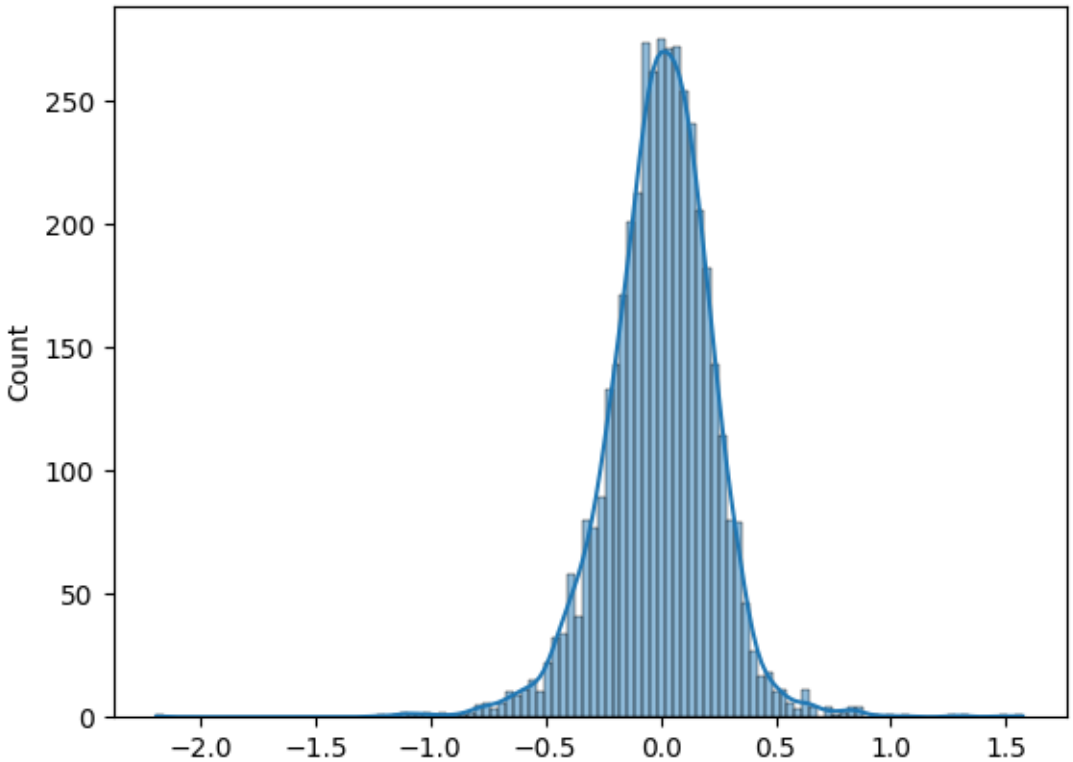
Graph 44. Residuals Q-Q plot



Observations and Implications on the Residuals Q-Q Plot

In this Q-Q residuals plot, the theoretical red straight reference line captures most of the ordered residuals (represented as blue) in the plot. In other words, the points stay close to the theoretical red line throughout the majority of the distribution, and do not deviate much away, except at the tails of the blue graphed line. These plots indicate that the residuals follow a normal distribution, with no significant anomalies, such as skewness or outliers.

Graph 45. Residual Histogram



Observations and Implications on Residuals Histogram

In this residuals histogram plot, the residuals follow a strict bell-shaped curve. The bell-shaped curve peaks near zero, resulting in a symmetric relationship on both sides of zero. Most residuals are near zero, indicating that the model's predictions are overall accurate. There is no evidence of asymmetry or extreme outliers, suggesting no signs of non-normality, such as tails or skewness. These findings are aligned with the Q-Q plot.

Multiple Linear Regression Final Model Performance Results

Table 16. Final chosen Multiple Linear Regression model's performance metrics

	train_r2	test_r2	train_adj_r2	test_adj_r2	train_rmse	test_rmse	train_mae	test_mae	train_mape	test_mape
0	0.45	0.20	0.45	0.19	8.29	9.97	2.06	2.11	0.18	0.18

Observations, Insight, Implications, and Improvements on the final chosen Multiple Linear Regression model's performance metrics

Based on the performance metrics table for the final chosen Multiple Linear Regression model, it has modest explanatory power. It has an R-squared of 0.45 and a significantly lower value of 0.20, suggesting it cannot generalize to unseen data. The noticeable difference between the train and test R-squared values indicates that the model is underfitting. This means that it is not effectively capturing enough of the basic relationships in the data. Specifically, there is a noticeable decline in the R-squared value from the training dataset to the test dataset, from 0.45 to 0.20. This suggests some evidence of overfitting, causing the model to learn patterns from the training dataset that are not transferable to the test dataset. The RMSE values of 8.29 and 9.97 for the train and test datasets, respectively, indicate high prediction errors in both. This reinforces that the model is having difficulty generating precise price estimates. A MAPE value of 18 percent suggests that the predictions are commonly within a fairly wide range of actual used car prices, but still provides evidence of possible improvement. The RSME, MAE, and MAPE values are more similar across the train and test datasets, indicating more stable, but not highly accurate, error behavior. Overall, the performance metrics indicate that although the model is consistent, it lacks predictive durability and would benefit from additional feature engineering, transformations, or more flexible modeling approaches.

The following improvements can be made to the final chosen Multiple Linear Regression model. First, it is important to introduce nonlinear transformations, including interaction terms, polynomial features, or spline terms. This would allow the model to capture curvature, which linear terms cannot illustrate. Further feature engineering techniques, such as those relating to depreciation patterns, brand-location interactions, and power-to-engine ratios, could enhance explanatory power. Even though multicollinearity and insignificant variables were removed, exploring ancillary encodings for categorical variables may reduce noise and improve generalization. Lastly, applying cross-validated model selection instead of sequential manual removal could generate a more stable and statistically defendable final independent variable set.

Proposed Model #2 Ridge Regression

Ridge Regression Model Definition, Properties, and Justification

The second proposed model for predicting the price of a used car in 100,000 INR from Cars4U's inventory is Ridge Regression. A Ridge Regression model extends the already structured framework of a Multiple Linear Regression model by incorporating regularization penalties. In simpler terms, a regularization penalty encourages the model to generalize better by penalizing overly complex weights during training. In this case, the weights are better interpreted as learned coefficients that indicate each feature's (independent variable or predictor) importance to the model. Ridge Regression uses an L2 penalty, which increases with the square of the weights and encourages them to remain small but not reach zero. Also, Ridge Regression penalizes the weights, favoring small, smooth weights and preventing any feature (independent variable or predictor) from becoming dominant. In simpler terms, Ridge Regression attempts to reduce the weights (like reducing the overall volume and tone control on an audio device). The L2 penalty term in Ridge Regression pushes the weights toward more interpretable values. Unlike a Multiple Linear Regression model, a Ridge Regression model can further prevent overfitting and make the linear regression model much simpler and more stable. However, like a Multiple Linear Regression model, a Ridge regression model cannot capture nonlinear patterns or complex interactions in the data. In particular, a Ridge Regression model maintains all independent features, preventing it from performing accurate feature selection. Likewise, the performance of a Ridge Regression model deteriorates when a linear model does not well approximate the model's generating process. Instead, scaling the independent variables (features) is important for a Ridge Regression model because its regularization penalty is sensitive to feature magnitudes. Similar to a Multiple Linear Regression model, a Ridge Regression model also assumes smooth, additive relationships that are most robust when the underlying structure is relatively linear, and the predictors are not highly correlated.

A Ridge Regression model was chosen as a second model. The following are reasons why the model is best suited to explore the data and problem at hand. It can primarily handle multicollinearity in the provided used car dataset better than a Multiple Linear Regression model. It can stabilize coefficient estimates when independent variables are correlated. L2 regularization reduces overfitting and enhances generalization. This is necessary as the used car prices in the dataset naturally have noisy, high-variance values. A Ridge Regression model retains all independent variables rather than removing them, making it reliable when each

variable contributes some predictive power. Furthermore, after scaling and log-transforming the data, a Ridge Regression model can deliver more reliable performance than an unregularized model.

Ridge Regression Model Building Process

Two distinct Ridge Regression models were created: one trained using the initial X feature matrix, and a second trained using the same X feature matrix without the "Year" independent variable. This was because the multicollinearity function that was previously called to evaluate the VIF score of the Year independent variable revealed that the "Year" independent variable (feature) had an infinite VIF score. It is important to note this as an exploratory technique to improve the performance of a Ridge Regression model, since Ridge Regression cannot easily handle independent variables with INF VIF scores, or, more clearly, with extremely high multicollinearity. Hence, the Ridge Regression model computes the optimal coefficients by minimizing the penalized loss function, thereby shrinking the coefficients to reduce variance and better assess multicollinearity, in addition to removing the "Year" independent variable. The trained Ridge Regression model also stores the learned coefficients and intercepts internally, where the coefficient values are more stable than those found in a multiple linear regression model. Based on both models' performance, the second model was chosen as the final Ridge Regression model because it had a slightly higher adjusted R-squared on the training dataset.

Ridge Regression Final Model Performance Results

Table 17. Final chosen Ridge Regression model's performance metrics

	train_r2	test_r2	train_adj_r2	test_adj_r2	train_rmse	test_rmse	train_mae	test_mae	train_mape	test_mape
0	0.51	0.31	0.51	0.29	7.78	9.26	1.99	2.01	0.18	0.18

Observations, Insights, Implications, and Improvements on the final chosen Ridge Regression model's performance metrics

The final Ridge Regression model also shows moderate explanatory capability. It has an R-squared of 0.51 on the training dataset and a definite improvement to 0.31 on the testing dataset. Compared to the multiple linear regression model, it generalizes better. The small difference between the R-squared values for the training

and testing datasets indicates that the model is not overfitting and is capturing more important structure in the data. The RMSE values of 7.78 and 9.26 on the training and testing datasets, respectively, indicate that the prediction errors remain substantial but are smaller than those of the multiple linear regression model, which suggests enhanced stability. The MAE and MAPE scores are nearly identical across the train and test datasets, indicating consistent error behavior and more trustworthy performance across both. The consistent values of RMSE, MAE, and MAPE across the train and test datasets indicate that the model is stable and not overfitting. A MAPE of about 18 percent indicates that the model achieves reasonable accuracy, as observed in a commonly used car dataset. Overall, the Ridge regression model suggests a more powerful and more generalizable solution than the multiple linear regression model, mainly due to its ability to stabilize coefficients in the presence of correlated independent variables. Despite this enhanced performance, further feature engineering or nonlinear modeling can still improve predictive accuracy.

The following improvements can be made to the final chosen Ridge Regression model. First, it is recommended to extend the hyperparameter tuning search to much wider alpha values, specifically by using a cross-validated grid or randomized search. The main rationale behind this technique is that the current regularization strength in both models provides evidence that it may not be optimal. Thus, this would assist in identifying a penalty that better equilibrizes bias and variance. Second, it would be beneficial to add engineered nonlinear independent variables prior to regularization as it would allow it to better capture more complicated pricing patterns in the dependent target price variable. Ensuring that all independent variables are standardized consistently can enhance the L2 penalty's effect and amplify coefficient shrinkage behavior. Lastly, the Ridge Regression model can also benefit from removing or amalgamating insignificant categorical dummy variables that contribute to noise but cannot be set to a zero value by L2 regularization.

Proposed Model #3 Lasso Regression

Lasso Regression Model Definition, Properties, and Justification

The third proposed model for predicting the price of a used car in 100,000 INR from Cars4U's inventory is Lasso Regression. Lasso Regression extends the already structured framework of a Multiple Linear Regression model by also incorporating

regularization penalties. In simpler terms, a regularization penalty encourages the model to generalize better by penalizing overly complex weights during training. Again, weights are better interpreted as learned coefficients that indicate each feature's (independent variable or predictor) importance to the model. Lasso Regression uses an L1 penalty, which penalizes nonzero weights and encourages the model to drop unnecessary independent variables entirely. Also, Lasso Regression sets some weights to exactly zero, which results in the removal of some features (independent variables or predictors). In simpler terms, Lasso Regression attempts to shrink the weights but may also set them to zero (like disabling or stopping the device's function). The L1 penalty term in Lasso Regression pushes the weights toward more interpretable values. Unlike a Multiple Linear Regression model, Lasso Regression can further prevent overfitting and make the model much simpler and more stable. However, like multiple linear regression and Ridge regression, Lasso regression cannot capture nonlinear patterns or complex interactions in the data. In particular, a Lasso Regression model cannot handle multicollinearity among independent variables and may remove one of them. Likewise, the performance of a Lasso Regression model deteriorates when a linear model does not well approximate the model's generating process. As with Ridge Regression, scaling the independent variables (features) is important for a Lasso Regression model because its regularization penalty is sensitive to feature magnitudes. Similar to Multiple Linear Regression and Ridge Regression, Lasso Regression assumes smooth, additive relationships. It is most robust when the underlying structure is relatively linear, and the predictors are not highly correlated.

A Lasso Regression model was chosen as a third model. The following are reasons why the model is best suited to explore the data and problem at hand. It can primarily perform regularization and feature selection at the same time. Its abilities are beneficial for the provided used car dataset, since it can filter out independent variables that are highly correlated or weakly correlated. Applying L1 penalties can reduce some coefficients to zero, simplifying the model and enhancing interpretability without losing predictive power. As a result, a Lasso Regression model refrains from overfitting to a vast, noisy set of independent variables, even when the target price variable can differ among used cars that share similar attributes. Overall, a Lasso Regression model emphasizes the most important features, resulting in a more stable and favorable final model.

Lasso Regression Model Building Process

Three distinct Lasso Regression models were created: one trained on the initial X feature matrix, a second trained on the same X feature matrix with a hyper-tuned alpha of 0.1, and a third trained on the same X feature matrix with a hyper-tuned alpha of 0.2. The second and third Lasso Regression models were hyperparameter-tuned because the initial model had negative R-squared and adjusted R-squared values. Negative R-squared values indicate that a model fits the data much worse than simply predicting the average of the target variable. This suggests that the model is either incorrectly specified or performing poorly on the dataset. Analogously, negative adjusted R-squared values indicate the same facts, however, with an added penalty for non-essential independent variables. This indicates that the included independent variables cannot explain variance and instead cause the model to perform worse. It is important to note that this process is better described as hyperparameter tuning for a Lasso Regression model, where alpha values of 0.1 and 0.2 are used as an exploratory technique to improve its performance. In particular, when hyperparameter-tuning a Lasso Regression model, it seeks the best alpha values (in this example, 0.1 and 0.2) that control how strongly the model penalizes large coefficients. Thus, in this setting, this procedure provides the Lasso Regression model with a suitable balance between model simplicity and predictive performance, ensuring it avoids overfitting and underfitting. In this case, a Lasso Regression model has a fixed regularization strength that controls how aggressively it shrinks coefficients and performs feature selection. Moreover, the alpha value enables the Lasso Regression model to fit the prepared training features and the target Price log dependent variable, allowing it to learn coefficients as noted under L1 regularization.

Based on the performance of both the second and third hyperparameter-tuned models, the second model was chosen as the final Lasso Regression model. The second model's performance shows a higher, more robust R-squared and lower test RMSE, MAE, and MAPE, suggesting better predictive accuracy on the test dataset. This model further captures the price patterns for used cars better than the third model, as evidenced by lower RSME, MAE, and MAPE on the test dataset. Even though the model shows evidence that Lasso's stronger regularization oversimplifies it, leading to underfitting on the training dataset, it still maintains a suitable balance between training and testing performance.

Lasso Regression Final Model Performance Results

Table 18. Final chosen Lasso Regression model's performance metrics

	train_r2	test_r2	train_adj_r2	test_adj_r2	train_rmse	test_rmse	train_mae	test_mae	train_mape	test_mape
0	0.52	0.65	0.52	0.64	7.72	6.61	2.94	2.91	0.29	0.28

Observations, Insights, Implications, and Improvements on the final chosen Lasso Regression model's performance metrics

The final chosen Lasso Regression model shows solid explanatory power, with R-squared and adjusted R-squared values in the test datasets higher than in the training dataset. Specifically, the R-squared value on the test dataset is 0.65, and the adjusted R-squared is 0.64. This indicates that the model can capture more meaningful structure in the data and performs better on unseen data. RMSE, MAE, and MAPE are much lower on the test dataset, indicating that the model is stable and is not overfitting on the training dataset. Also, the model's lower test RMSE than its train RMSE indicates that L1 regularization helped the model identify the most informative independent variables and filter out noise. Since the MAE and MAPE values are consistent across the train and test datasets, this indicates stable predictive behavior. The close alignment between the R-squared values and adjusted R-squared values affirms that the model's predictors contribute significantly without unessential complications. These patterns indicate that the chosen level of regularization is favorably balancing simplicity and accurate predictions. This allows Lasso Regression to retain the most informative independent variables while shrinking the coefficients of weaker ones. Overall, the model demonstrates improved generalization, trustworthy error behavior, and adequate regularization, making it an effective and reliable Lasso Regression model. Its feature-selection process improves performance by simplifying the model and highlighting the strongest independent variables.

The following improvements can be made to the final chosen Lasso Regression model. First, it is recommended that a more extensive alpha hyperparameter tuning process be conducted, specifically through cross-validation, to help avoid overly shrinking important independent variables. Thus, this would prevent the Lasso Regression model from removing moderately informative independent variables when the regularization penalty process is too strong. Second, adding interaction terms or

polynomial features before applying L1 regularization would allow the Lasso Regression model to select significant nonlinear relationships rather than forcing it to rely strictly on linear terms. Third, applying correlation clustering or Principal Component Analysis (PCA) based preprocessing could possibly enhance coefficient reliability. Lastly, the Lasso Regression model could also benefit from re-evaluating the removed independent variables to ensure that business-relevant variables were not excluded due to the penalty.

Proposed Model #4 Decision Tree Regression

Decision Tree (Regression) Model Definition, Properties, and Justification

The fourth proposed model for predicting the price of a used car in 100,000 INR from Cars4U's inventory is a Decision Tree Regression model. Decision Tree Regression takes an alternative approach by recursively splitting the data into specific regions based on feature thresholds. A Decision Tree Regression model can also capture nonlinear patterns and interactions without requiring transformations. Thus, the Decision Tree Regression model can optimally split the data and minimize the splitting of features (independent variables) into groups that are as comparable as possible. In simpler terms, a Decision Tree Regression model works by asking yes-or-no questions about the features. To start with, a Decision Tree regression model searches for the best optimal split based on a chosen feature and its corresponding threshold. It then divides the data into two groups based on that rule, and later reiterates the process in each group, splitting again iteratively. A Decision Tree regression model searches for the split that optimally reduces variance by creating groups in which the target values are as similar as possible. Soon enough, the Decision Tree regression model stops splitting, and the final region is known as a leaf. Within each leaf, the prediction is the average of the target values in that particular region. Unlike a single global equation used in previous linear regression models, Decision Trees can model curved, stepped, and erratic shapes, as well as feature interactions. However, Decision Trees are susceptible to overfitting and can result in significant jumps in predictions. An important note about feature importance should be stated for a Decision Tree Regression model. Feature importance in a Decision Tree Regression model is determined by measuring how much each feature reduces impurity (e.g., variance) across all the splits the model uses. These reductions are then summed and

normalized so that the importance values sum to 1. It is also better known as Gini Importance.

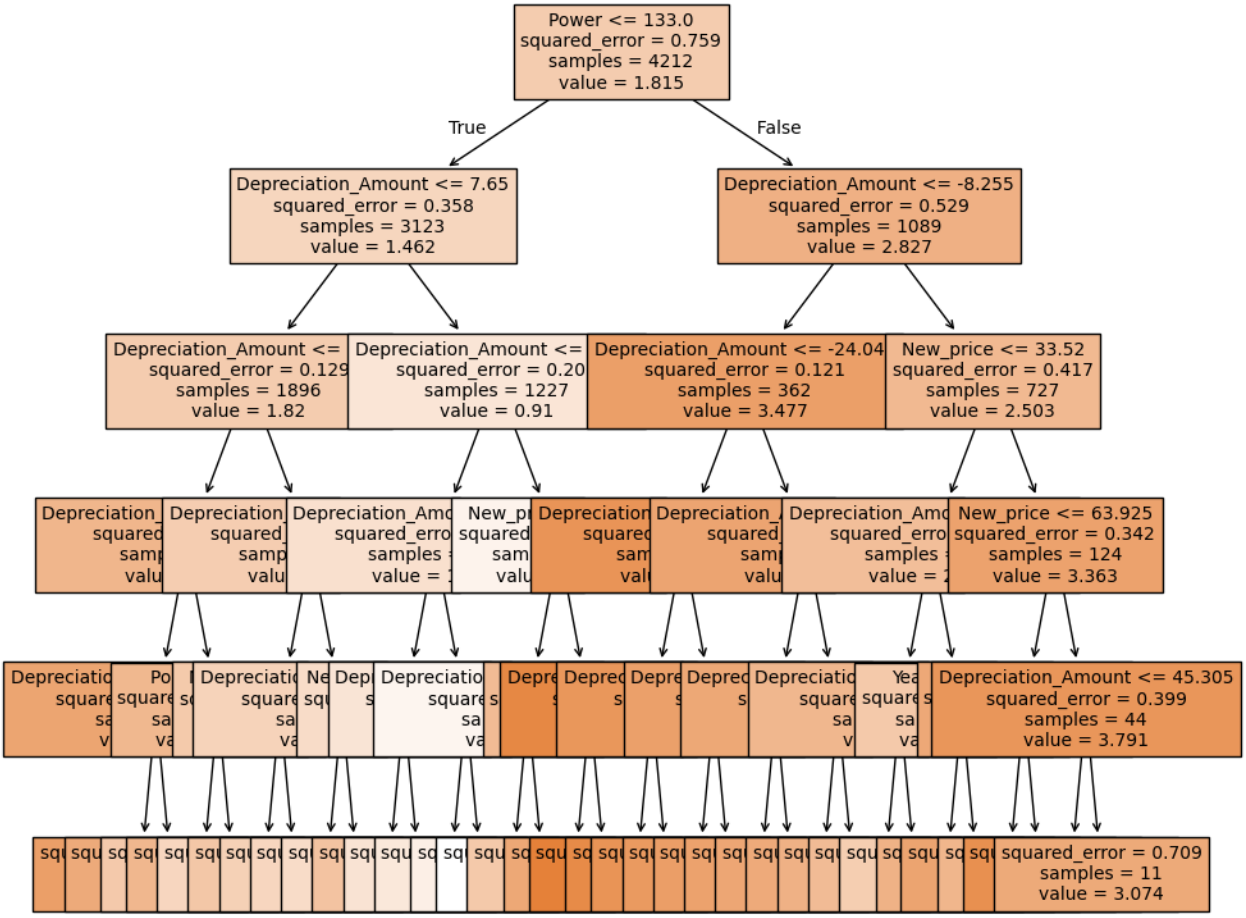
A Decision Tree Regression model was chosen as a fourth model. The following are reasons why the model is best suited to explore the data and problem at hand. It can primarily capture nonlinear relationships and complex interactions among independent variables (used-car features) that linear models may miss. Its abilities include handling mixed data types without needing scaling. Therefore, it can be suitable for a dataset that comprises both continuous variables and one-hot-encoded categorical independent variables. Also, a Decision Tree Regression model is not sensitive to outliers and can capture sharp differences between independent variables such as brands, fuel types, or different locations. Hyperparameter tuning was conducted on the Decision Tree Regression model primarily because it is highly sensitive to parameter changes, and tuning significantly improves its performance. Overall, a Decision Tree Regression model can provide transparent, visual explainability through tree structures, with the best explanatory power for detailing how certain independent variables affect the determinability of the target variable, used car price.

Decision Tree Regression Model Building Process

Four distinct Decision Tree Regression models were created: one trained on the initial X feature matrix using a random state of 1, and default parameters, a second trained on the same X feature matrix with a max depth of 5, a random state of 1, and other default parameters, a hyperparameter tuned Decision Tree model with a max depth of 10, a minimum samples leaf amount of 5, and a random state of 1, and a fourth hyperparameter tuned Decision Tree model with a max depth of 5, a minimum samples leaf amount of 5, and a random state of 1. When hyperparameter tuning the third and fourth Decision Tree models, RandomizedSearchCV was used instead of GridSearchCV because it is computationally less expensive, even though it explores a smaller set of combinations in a more suggested manner. The hyperparameter tuning process for a Decision Tree model, entails seeking the best combination of different parameters (includes the most possible depth levels a Decision Tree is permitted to grow from the top root node to the deepest leaf, the least possible number of samples needed for a node to be qualified for splitting into two nodes, and the least possible

number of samples allowed in each leaf node) that commands how the Decision Tree is built. The most important independent variables in the first Decision Tree Regression model, ranked by Gini Importance from highest to lowest, are Power, Depreciation Amount, New Price, Year, and Engine. In general, this first Decision Tree Regression model is overfitting the training dataset and is unlikely to generalize well without pruning or tuning. In general, this second Decision Tree Regression model generalizes better while still capturing considerable structure in the overall dataset. Similarly, the most important independent variables in the second Decision Tree Regression model, ranked by Gini Importance from highest to lowest, are the same as in the first Decision Tree Regression model. The third hyperparameter-tuned Decision Tree had the following tuned parameters: a max depth of 10, a minimum of 5 samples per leaf node for the split to be permitted, and a random state of 1. Likewise, the most important independent variables in the third hyperparameter-tuned Decision Tree Regression model, ranked by Gini Importance from highest to lowest, remain the same as in the first and second Decision Tree Regression models. This model was also not selected as the final Decision Tree model because it is not very interpretable when plotted, given its max depth of 10.

Graph 46. Final Decision Tree Model Plot

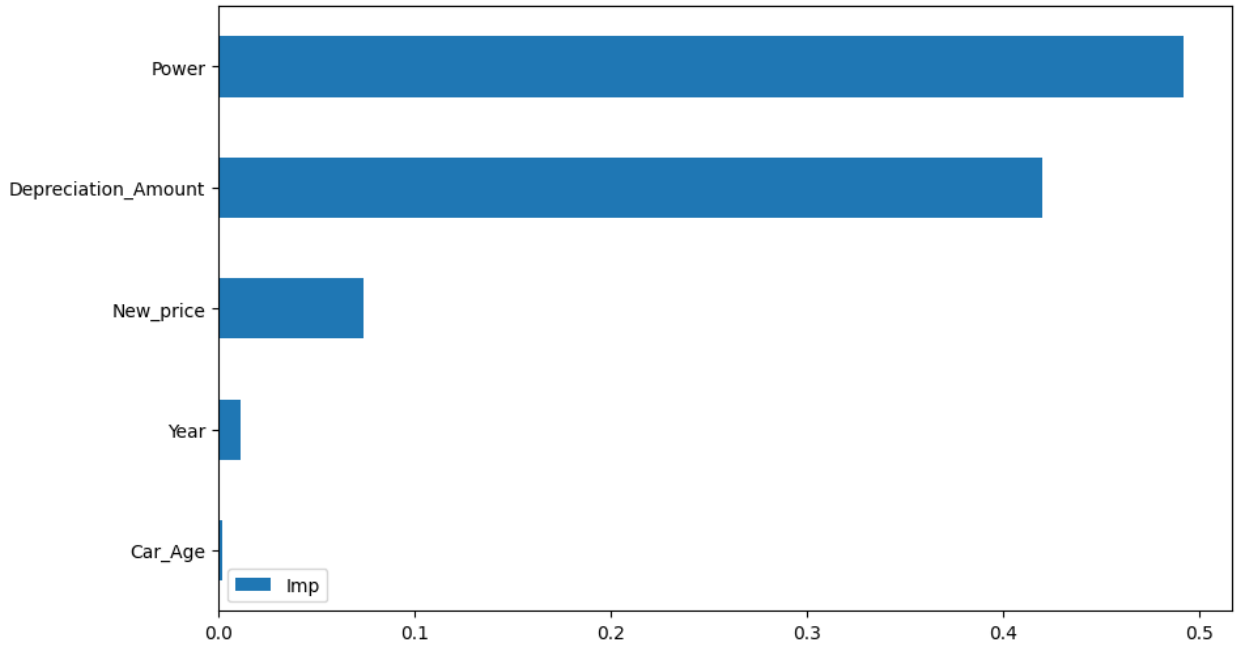


Observations, Insights, and Implications on the final chosen hyperparameter-tuned Decision Tree model

The fourth final chosen hyperparameter-tuned Decision Tree Regression model indicates a clear hierarchy of predictive power. It shows Power as the dominant primary split, indicating that it accounts for the largest share of the variance in the target Price variable. The most influential independent variable is Depreciation Amount, which frequently appears in deeper splits and helps divide used cars into distinct groups with noticeable value patterns. The branches with elevated predicted values usually follow paths where Power is high, and Depreciation Amount is low. This suggests a robust interaction effect between performance and maintained value. The slightly low squared errors in many leaf nodes suggest that the model shapes tight, comparable groups, indicating that it captures significant data structure. Overall, the final, hyperparameter-tuned Decision Tree Regression model focuses on a small

set of features with the strongest explanatory power, capturing most of the variation, and reinforces the model's explainability and the importance of key independent variables.

Graph 47. Final Decision Tree Hyperparameter-tuned Model Feature Importance Barplot



Observations, Insights, and Implications on the final chosen hyperparameter-tuned Decision Tree Feature Importance Barplot

The Feature importance bar plot for the final, hyperparameter-tuned Decision Tree Regression model clearly shows that Power and Depreciation Amount dominate the model's overall predictive capability. New Price also makes a meaningful contribution, but in smaller quantities, indicating that it plays a secondary role, following Price and Depreciation Amount. Year and Car Age show minimal importance, suggesting that time-aging effects still add only minuscule informative value beyond what Depreciation Amount captures. The abrupt drop-off after the top two independent variables suggests that the model relies heavily on a small set of independent variables. Overall, these patterns reinforce the conclusion that the Decision Tree Regression model's behavior is determined mostly by used-car performance and value-loss properties, with other features playing minimal roles.

Decision Tree Regression Final Model Performance Results

Table 19. Final chosen Decision Tree Regression model's performance metrics

	train_r2	test_r2	train_adj_r2	test_adj_r2	train_rmse	test_rmse	train_mae	test_mae	train_mape	test_mape
0	0.96	0.86	0.96	0.86	2.29	4.2	1.05	1.21	0.12	0.13

Observations, Insights, Implications, and Improvements on the Final chosen Decision Tree model's performance metrics

The final Decision Tree Regression model shows strong performance with some overfitting, as indicated by a high R-squared value of 0.96 on the training dataset. There is also a solid R-squared value of 0.86 in the test dataset. The increase in error from the training to the test dataset suggests some overfitting, but it is not sufficient to be classified as weakened generalization. MAE and MAPE remain low and very similar across the train and test datasets, indicating that the model's predictions are stable and trustworthy. The consistency between R-squared and adjusted R-squared further validates that the model's complexity is appropriate for the number of independent variables used. Overall, the Decision Tree model achieves robust predictive accuracy with minimal overfitting and remains well-balanced. Most importantly, this model was chosen as the final Decision Tree Regression model because it is more interpretable when plotted and does not overly overfit the training dataset.

The following improvements can be made to the final chosen Decision Tree Regression model. First, it would be ideal to increase regularization by using more rigorous limits on max depth, min samples leaf (e.g., the minimal number of training samples that must appear in any leaf node), or min samples split (e.g., controls when a node in a Decision Tree is allowed to split) parameters to reduce variance and improve generalization. These preliminary procedures are required to ensure that the model does not memorize training patterns. Second, it would be ideal to remove low-importance dummy variables or merge sparse categories to help stabilize splits. Third, implementing more robust hyperparameter tuning, such as including cost-complexity pruning, would prevent the tree from growing unnecessarily deep. Lastly, it would be ideal to feature-engineer independent variables that better capture depreciation, power-to-weight relationships, or brand-specific pricing patterns, thereby reducing reliance on raw splits.

Proposed Model #5 Random Forest Regression

Random Forest (Regression) Model Definition, Properties, and Justification

The fifth proposed model for predicting the price of a used car in 100,000 INR from Cars4U's inventory is a Random Forest Regression model. Within the same scope, a Random Forest Regression model builds on the Decision Tree Regression process by averaging predictions from multiple randomized Decision Trees, thereby reducing overfitting and improving stability. Specifically, a Random Forest Regression model generates many Decision Trees, each trained on slightly different versions of the data. The Decision Trees generated by a Random Forest Regression model are randomized, independent, and individually imperfect. Each Decision Tree in a Random Forest Regression model is trained on a random sample of the data drawn with replacement, a process known as bootstrapping. In simpler terms, some rows might pop up multiple times in a tree's training dataset, and be left out in their entirety, or every Decision Tree accesses a slightly modified dataset. Furthermore, in the Random Forest Regression model, each Decision Tree considers a random subset of features (independent variables), which prevents all Decision Trees from creating the same splits and allows each Decision Tree to learn alternative patterns. Put together, each Decision Tree can make its own prediction, which allows the overall Random Forest Regression model to formulate a full averaging of all Decision Tree predictions initially generated from the Random Forest model. However, a Random Forest regression model is less interpretable, struggles with high-dimensional data, and cannot extrapolate data well (i.e., cannot predict beyond the range of the training targets). As with a Decision Tree Regression model, an important note about feature importance should be stated in a Random Forest Regression model. In a Random Forest Regression model, feature importance is calculated in the same way as in a Decision Tree Regression model. Still, it is instead averaged across all trees in the model. This makes the calculations more stable and less sensitive to noise.

A Random Forest Regression model was chosen as the fifth model primarily because it can significantly reduce overfitting by averaging numerous decision trees. The following are reasons why the model is best suited to explore the data and problem at hand. It can generate more durable and trustworthy predictions for the provided used car dataset. It can capture more complex nonlinear relationships and

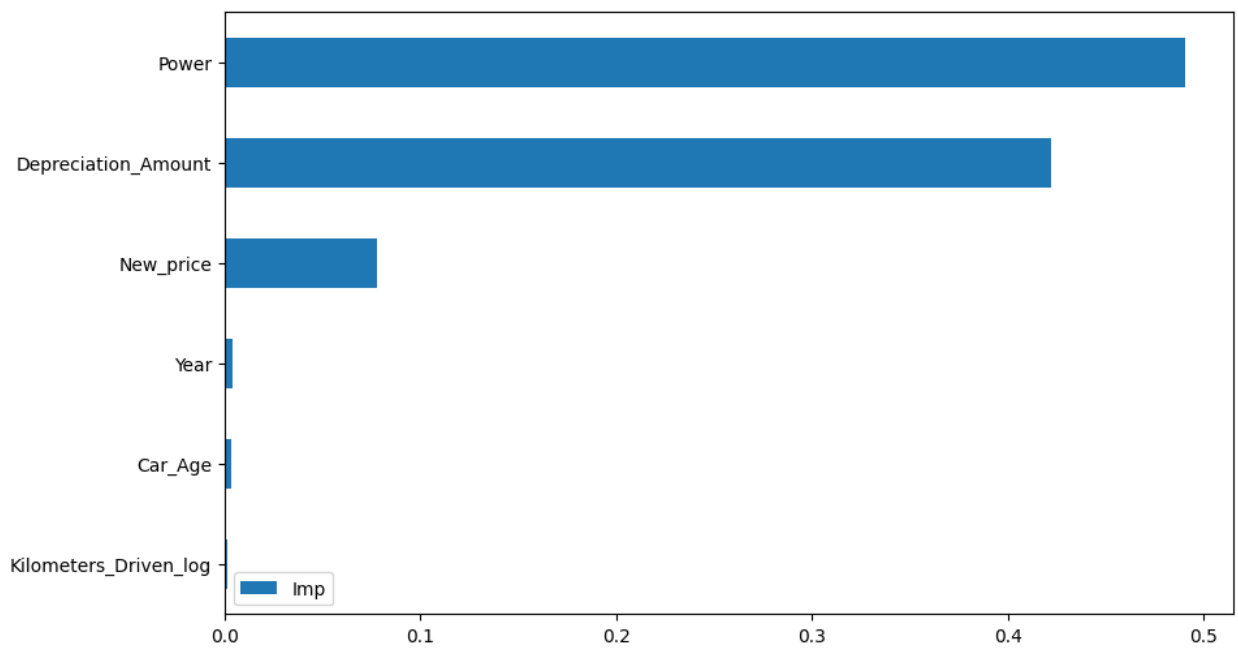
interactions among independent variables (used-car features) that linear models cannot adequately represent. The algorithm behind a Random Forest Regression model handles high-dimensional data and one-hot-encoded categorical variables with dexterity. Thus, these properties make the Random Forest Regression model best suited for the provided used car dataset. Additionally, a Random Forest Regression model is not sensitive to outliers or price variance, as noted under the used car prices target variable. Hyperparameter tuning was also conducted on the Random Forest Regression model, primarily because it is also highly sensitive to parameter changes, and tuning significantly improves its performance. Overall, a Random Forest Regression model usually delivers strong predictive performance, making it a strong benchmark compared to other regression models.

Random Forest Regression Model Building Process and Results

Four distinct Random Forest Regression models were created: one trained on the initial X feature matrix using a random of state of 1, and default parameters, a second trained on the same X feature matrix with a max depth of 5, and a random state of 1, and other default parameters, a hyperparameter tuned Random Forest Regression model with a max depth of 10, a criterion of Friedman MSE (method used in regression tree to measure the quality of a split at each node of the tree), and a random state of 1, and a fourth hyperparameter tuned Random Forest Regression model with a max depth of 5, and a random state of 1. When hyperparameter-tuning the third and fourth Random Forest Regression models, RandomizedSearchCV was also used instead of GridSearchCV for the same reasons. The hyperparameter tuning process for a Random Forest Regression model, entails seeking the best combination of different parameters (includes the number of Decision Trees in the model, most possible depth levels a Decision Tree in the model is permitted to grow from the top root node to the deepest leaf, the least possible number of samples needed for a node to be qualified for splitting into two nodes, and the least possible number of samples allowed in each leaf node) that commands how the overall Random Forest Regression model is built. The first Random Forest Regression model serves as a robust baseline due to its high accuracy, stable generalization, and reliable predictive performance. There is also alignment between the adjusted and unadjusted R-squared values, suggesting that the independent variables do not artificially inflate the first model's performance. The second Random Forest Regression model's performance shows

robust predictive accuracy. The second Random Forest model has a good balance between accuracy and generalization. The third hyperparameter-tuned Random Forest Regression model performs well, with robust generalization and low error. However, due to a perfect R-squared value of 1.0 on the training dataset, it is severely overfitting and is not chosen as the final Random Forest Regression model.

Graph 48. Final Random Forest Regression Hyperparameter-tuned Model Feature Importance Barplot



Observations, Insights, and Implications on the final chosen hyperparameter-tuned Random Forest Feature Importance Barplot

The Feature importance bar plot for the final, hyperparameter-tuned Random Forest Regression model clearly shows a concentrated importance structure that is similar to that of the final chosen hyperparameter-tuned Decision Tree Regression model. To clarify, Power and Depreciation Amount dominate the model's overall predictive signal. New Price also contributes purposefully, but is still far behind Power and Depreciation Amount. In clearer terms, New Price construes predictions rather than driving both Power and Depreciation Amount. The Year, Car Age, and Kilometers Driven Log have little importance. Overall, this suggests that temporal and usage-

based effects contribute little to feature importance after Power and Depreciation Amount. The steep drop-off after the major independent variables indicates that the final hyperparameter-tuned Random Forest Regression model relies on a compact set of independent variables, thereby improving interpretability. Overall, the final hyperparameter-tuned Random Forest Regression model is driven primarily by performance and value-loss attributes of used cars, with only a small number of other independent features playing a marginal role.

Table 20. Final chosen Random Forest Regression model's performance metrics

	train_r2	test_r2	train_adj_r2	test_adj_r2	train_rmse	test_rmse	train_mae	test_mae	train_mape	test_mape
0	0.97	0.9	0.97	0.9	2.07	3.59	0.79	0.88	0.09	0.09

Observations, Insights, Implications, and Improvements on the final chosen Random Forest model's performance metrics

The final Random Forest Regression model shows strong performance, as indicated by a high R-squared of 0.97 on the training dataset and 0.90 on the test dataset. The performance results show that it is generalizing quite well. The slight increase in RMSE from 2.07 in the training dataset to 3.59 in the test dataset suggests some overfitting, although the gap is modest and normal for an ensemble machine learning model. MAE is low on both the training and test datasets, and the identical train and test MAPE values of 0.09 suggest exceptionally stable percentage-based accuracy in the test dataset. The close values between adjusted and unadjusted R-squared values suggest that non-essential independent variables do not elevate the model's performance. Overall, this Random Forest model has a healthy balance between accuracy and generalization. This makes the Random Forest a dependable choice for prediction. Most importantly, this Random Forest Regression model is chosen as the preferred model due to its greater interpretability and its tendency not to overfit the training dataset.

The following improvements can be made to the final chosen Random Forest Regression model. First, it would be ideal to increase the number of trees and tune max_features (e.g., the number of independent variables each split is allowed to consider) to reduce correlation among trees and improve generalization. Second, it would be ideal to increase the minimum sample leaf (i.e., the minimum number of samples required in any leaf node), which would flatten predictions and avoid overly

specific leaf nodes that capture noise. Third, it would be ideal to implement bootstrap settings, max samples (e.g., controls how many samples each tree draws from the training set), or out-of-bag evaluation (e.g., a built-in, unbiased validation method that allows the model to estimate its generalization error without requiring a separate validation set or cross-validation), which would aid in diagnosing variance and enhance stability. Lastly, it would be ideal to lower the dimensionality of sparse dummy variables, in which case trees would split on low-value categories that do not generalize well.

Proposed Model #6 KNN Regression

KNN Regression Model Definition, Properties, and Justification

The sixth proposed model for predicting the price of a used car in 100,000 INR from Cars4U's inventory is a KNN Regression model. A KNN Regression model is a non-parametric, instance-based method that predicts a value by averaging the target values of the k most similar observations. Its overall methodology makes it flexible but sensitive to scaling and local density. Thus, a KNN Regression model functions entirely based on the structure of the training data. This model begins with choosing several neighbors to consider. It then calculates the distances between the query point and all training points. Afterward, it selects the K points in the dataset that are closest according to a specified distance metric. Lastly, it accumulates their target values to generate a prediction. Other key characteristics of a KNN Regression model include its reliance on the local neighborhood around each query point, which causes the model to adapt to different patterns across regions of the feature space shaped by the data's structure. Furthermore, a KNN Regression model is shaped by its distance metric, the scaling of all independent variables (features), and the choice of K. Moreover, the smoothness of a KNN Regression model is determined by K; a small K value yields high adaptability with low bias and high variance. In contrast, a larger K value leads to smoother predictions with higher bias and lower variance. However, a KNN Regression model has limitations: it is computationally expensive (slow for large datasets), sensitive to feature scaling, computationally challenging in high-dimensional spaces, susceptible to extraneous or noisy features, and lacks interpretability because it is solely geometric.

A KNN Regression model was chosen as the sixth and final model primarily because it can make predictions based on the prices of the most comparable used cars. The following are reasons why the model is best suited to explore the data and problem at hand. This modeling approach can work quite well when there are potential local patterns in the provided dataset. Its characteristics as a fully non-parametric model allow it to capture nonlinear relationships without making assumptions about the independent variables or the underlying distributions. A KNN Regression model can also manage complicated independent variables efficiently after they are scaled. This model approach makes it applicable to datasets with diverse numerical properties. Moreover, the model can be seen as an additional innate benchmark for assessing how well other, more practical algorithmic models generalize beyond local neighborhoods.

KNN Regression Model Building Process and Results

Two distinct KNN Regression models were created: one trained using the initial X feature matrix with default parameters, and a second trained using the same X feature matrix with a specified 10-neighbor parameter. The default parameters in a KNN Regression model are 5 neighbors (using the closest training samples for prediction), uniform weights (each neighbor contributes the same to the prediction), and an auto algorithm (allowing the model to choose the best neighbor-searching algorithm based on the provided X feature matrix dataset). It is important to note that the KNN Regression model also requires scaling the X feature matrix before training. This ensures that the training and test datasets are properly scaled by computing the mean and standard deviation for each independent variable, which are then used to transform the datasets. This further ensures that there is no data leakage and maintains a fair evaluation. Once the scaled independent variables are used to train the KNN Regression model with the target Price log dependent variable, the model stores the training points in a standardized space and can make predictions. Based on both models' performance results, the first model was chosen as the final KNN Regression model due to its slightly higher R-squared value on the test dataset.

Table 21. Final chosen KNN Regression model's performance metrics

	train_r2	test_r2	train_adj_r2	test_adj_r2	train_rmse	test_rmse	train_mae	test_mae	train_mape	test_mape
0	0.89	0.80	0.89	0.80	3.66	5.04	1.79	2.29	0.20	0.25

Observations, Insights, Implications, and Improvements on the final chosen KNN Regression model's performance metrics

The final KNN Regression model shows robust yet moderate performance, with R-squared values of 0.89 and 0.80. This suggests that the model captures most of the data's structure while still generalizing well. The increase in RMSE from 3.66 in the training dataset to 5.04 in the test dataset suggests that the KNN Regression model is sensitive to noise. The MAE and MAPE values both show some increase on the test dataset, even though they remain within an acceptable range, indicating that prediction errors are still adequately contained. The nearby alignment between R-squared and adjusted R-squared suggests that non-essential independent variables do not inflate the model's performance. Overall, this KNN Regression model still provides a healthy balance between generalization and stabilized error behavior.

The following improvements can be made to the final chosen KNN Regression model. First, it would be ideal to reduce the dimensionality through Principal Component Analysis (PCA), feature selection, or combining sparse categories to markedly improve neighborhood quality. These primary methods are ideal to implement because the KNN regression model is adversely affected by high dimensionality, particularly with over 40 dummy variables, which leads to unstable distance calculations and poor test performance. Second, it would be ideal to tune the number of neighbors (the main k parameter) and experiment with various distance metrics, such as Manhattan or Minkowski, to possibly compute more stable predictions. It is also important to note that all independent variables must be standardized to enable meaningful distance comparisons. Lastly, it is ideal to remove noisy or low-important independent variables that deform distance calculations and yield unstable predictions.

Models/Techniques Comparison

Key Measures of Success

The key measures of success across all models used to predict used-car prices, as provided by the Cars4U dataset, include the performance metrics generated by the initial get model score function: R-squared, which calculates the amount of variance in price the model explains, Adjusted R-squared, which penalizes non-essential predictors and provides a more reliable measure of model fit, Root Mean Squared Error (RMSE), which calculates the typical size of prediction errors in the original target used car prices, Mean Absolute Error (MAE), which calculates the average absolute error, and equally addresses all errors, and Mean Absolute Percentage Error (MAPE), which is advantageously used for interpretations across the target dependent variable. These performance metrics are fundamental because they assess explanatory power, generalizability, and predictive accuracy. As such, these are the most important properties for a regression model's success. Along the same lines, it is important to note that these performance metrics for each model's test dataset are considered more important than those for the corresponding training dataset.

Important Metrics to Consider

The previously stated key measures of success are also important metrics, as each has its own purpose and adaptability for assessing each model's performance. Both R-squared and Adjusted R-squared calculate how well a regression model captures the basic structure of the data. RMSE penalizes large errors more heavily, making it preferred for evaluating practical prediction quality. MAE offers a more understandable, stronger measure of average error. MAPE calculates error according to percentage, making it simpler to report to stakeholders. It is critical to ensure that each model has higher optimal R-squared and Adjusted R-squared values, and lower RMSE, MAE, and MAPE values. The overall multi-metric approach across these five distinct performance metrics ensures that each model is statistically robust and applicable in a business setting, which is needed to effectively determine the price of a used car in INR 100,000.

Table 22. Performance Metrics of Each Final Chosen Regression Model

	train_r2	test_r2	train_adj_r2	test_adj_r2	train_rmse	test_rmse	train_mae	test_mae	train_mape	test_mape
Linear Regression	0.45	0.20	0.45	0.19	8.29	9.97	2.06	2.11	0.18	0.18
Ridge Regression	0.51	0.31	0.50	0.29	7.78	9.25	1.99	2.01	0.18	0.18
Lasso Regression	0.52	0.65	0.52	0.64	7.72	6.61	2.94	2.91	0.29	0.28
Decision Tree	0.96	0.86	0.96	0.86	2.29	4.20	1.05	1.21	0.12	0.13
Random Forest	0.97	0.90	0.97	0.90	2.07	3.59	0.79	0.88	0.09	0.09
KNN Regression	0.89	0.80	0.89	0.80	3.66	5.04	1.79	2.29	0.20	0.25

Performance of Different Techniques/Models (Observations, Insights, and Implications)

As there are six distinct models used to address the business problem of predicting the price of a used car in INR 100,000, as requested by the Cars4U startup, each model yields unique performance metrics.

To begin with, the final chosen Multiple Linear Regression model has the following highlighted performance metrics. It has an R-squared of 0.45 on the training dataset and 0.20 on the test dataset, with high RMSE values ranging from 8.29 to 9.97. These highlighted performance results suggest that a Multiple Linear Regression model underfits and poorly generalizes unseen data. Consequently, a Multiple Linear Regression model is not ideal for capturing the overall structure of the dataset. Furthermore, Ridge and Lasso Regression models, similar in structure to Multiple Linear Regression, show some stability improvements over Multiple Linear Regression but remain limited under linear assumptions. Regularization found in both Ridge and

Lasso Regression mildly improves model performance, but is incapable of capturing nonlinear relationships in the target variable, the used car price. The overall performance of both the Ridge and Lasso Regression models shows slight improvements, but tree-based models still outperform them.

In addition to linear regression models, a Decision Tree Regression and Random Forest Regression model were chosen as alternative tree-based models, and a KNN regression model was chosen as a non-parametric geometric-based model. The final tuned Decision Tree Regression model has an R-squared of 0.96 on the training dataset and 0.86 on the test dataset. The Decision Tree Regression model also has low MAE and MAPE values, with some overfitting but robust generalization. It can be viewed as a preliminary baseline tree-based model because it efficiently captures nonlinear interactions. The final chosen Random Forest Regression-tuned model has an R-squared of 0.97 on the training dataset and 0.90 on the test dataset. The Random Forest Regression model also has very low RMSE, MAE, and MAPE. Compared to the Decision Tree Regression model, the Random Forest Regression model offers high accuracy, stable generalization, and dependable predictive performance. The final KNN Regression model has an R-squared of 0.89 on the training dataset and 0.80 on the test dataset. The KNN Regression model has modest RMSE and MAE values, is sensitive to local noise, and performs fairly well but is not as effective as the final chosen Random Forest Regression model.

Best Performing Technique/Model and Reasons for Performance

Among the models, the Random Forest Regression model outperforms all others. It has the highest R-squared (0.97) and adjusted R-squared (0.97) values in the training dataset among the models. It also has the highest R-squared (0.90) and highest adjusted R-squared (0.90) values in the test datasets. With respect to error metrics, it has the lowest RMSE (2.07), MAE (0.79), and MAPE (0.09) in the training dataset. Likewise, it has the lowest RMSE (3.59), MAE (0.88), and MAPE (0.09) on the test dataset. Reasons why Random Forest Regression is the best-performing include the following: First, it can capture nonlinear relationships that linear regression models (e.g., Multiple Linear Regression, Ridge Regression, and Lasso Regression) cannot. Second, a Random Forest Regression model can reduce overfitting by bootstrap

aggregation. Bootstrap aggregation is an ensemble machine learning algorithm that enhances model stability and accuracy by reducing variance and avoiding overfitting. Third, a Random Forest Regression model can efficiently handle high-dimensional, amalgamated data of various types. Fourth, it can generate durable, low-error predictions across train and test datasets. Lastly, it can avoid the instability of single decision trees while maintaining interpretability through feature importance.

Scope to improve the performance of each technique/model further

The following are scope improvements that can further enhance each linear regression technique (model). To improve the Multiple Linear Regression model, the following scope improvements should be made. First, it would be beneficial to introduce nonlinear transformations into some independent variables. This would include using polynomial terms to better capture curvature in relationships that a linear model cannot. Second, it would be beneficial to include interaction terms, since price depends on combinations of the independent variables. Third, adding feature-engineered independent variables can enhance linear separability. To improve the Ridge Regression model, the following scope improvements should be made. First, it would be beneficial to tune the regularization parameter alpha more broadly. Second, it would be beneficial to add polynomial and interaction features before training Ridge regression. Third, it would be beneficial to standardize all independent variables consistently. To improve the Lasso Regression model, the following scope improvements should be made. First, it would be beneficial to tune alpha more attentively. Second, it would be beneficial to run an Elastic Net Regression model rather than relying solely on Lasso Regression. Third, it would be beneficial to group independent variables, such as categorical brand and location variables, into a single independent variable category.

The following are scope improvements that can further enhance each tree regression technique (model) (e.g., Decision Tree Regression model and Random Forest Regression model) and the KNN Regression model. To improve the Decision Tree Regression model, the following scope improvements should be made. First, it would be beneficial to increase the regularization parameters, such as the maximum depth, the minimum number of samples required to split a node, the minimum number

of samples required in a leaf node, and the maximum number of leaves. Second, it would be beneficial to utilize pruning to simplify the Decision Tree Regression model. Pruning is the method of reducing the size of overfitting trees by removing nonessential, redundant branches. Third, it would be beneficial to engineer smoother independent variables, such as binning mileage or Car Age, to enhance model stability. To improve the Random Forest Regression model, the following scope improvements should be made. First, it would be beneficial to further hyperparameter-tune the model (e.g., max depth, max features, minimum samples required in a leaf node, number of Decision Trees, and bootstrap parameters). Second, it would be beneficial to reduce noise in categorical dummy independent variables by grouping uncommon categories. Third, it would be beneficial to use feature importance to remove weak independent variables.

To improve the KNN Regression geometric-based model, the following scope improvements should be made. First, it would be beneficial to extensively tune the main K variable more broadly. Second, it would be beneficial to reduce the dimensionality of the X feature matrix either by applying Principal Component Analysis (PCA) or by removing weak dummy variables. Principal Component Analysis is a dimensionality reduction method that reduces complex, high-dimensional datasets to a smaller set of uncorrelated variables (principal components) while preserving the original information. Third, it would be beneficial to experiment with other distance metrics, such as the Manhattan and Euclidean distances, in higher-dimensional spaces.

Taken together, the following scope improvements apply across all models. Foremost, it would be beneficial to incorporate more advanced feature engineering, such as interaction terms, nonlinear transformations (e.g., a log transformation of the New Price independent variable), and domain-specific independent variables. Secondly, it would be beneficial to handle categorical variables more effectively by grouping uncommon categories. Thirdly, it would be beneficial to incorporate outlier treatment methods, such as clipping extreme values in variables, most notably Mileage and New price. Finally, it would be beneficial to hyperparameter-tune each model using cross-validation, most notably the Multiple Linear Regression, Ridge Regression, and KNN Regression models, since these were not hyperparameter-tuned.

Final Solution Design

Model to be adopted

Of the six proposed models, the Random Forest Regression model is selected for the fully preprocessed dataset of 6,018 used cars, trained on the same 43-feature X_train matrix and the log-transformed Price target variable. After thorough consideration, the Random Forest Regression model achieves the highest R-squared and adjusted R-squared values in the test dataset, and the lowest overall RMSE, MAE, and MAPE. Among all the models, the Random Forest Regression model has the most robust generalization performance compared to the linear, single-tree, and geometry-based models. Due to these high-achieving characteristics, the Random Forest Regression model is the best candidate for the final solution.

Reasons for the best solution (Expected Benefits)

The following are reasons why the Random Forest Regression model is considered the best solution for the business problem. First, the final chosen Random Forest Regression model achieves a notable R-squared of 0.90 on the test dataset, with significantly lower RMSE and MAE, and the lowest MAPE of approximately 0.10. This implies that the Random Forest Regression model captures much more of the variance in used car prices and incurs much less prediction error on unseen data than the other linear-based regression models (e.g., Multiple Linear Regression, Ridge Regression, and Lasso Regression), the single Decision Tree Regression model, and the geometric KNN Regression model. Second, the final chosen Random Forest Regression model has optimal robustness to nonlinear relationships and interactions. To better clarify, the initially provided used car dataset is highly nonlinear and interaction-heavy. Some of these interactions include Brand with Car Age, Power and Fuel Type, and Location with New Price. Random Forest Regression models can typically handle nonlinear relationships and higher-order interactions without requiring feature engineering of polynomial or interaction terms. Third, the final chosen Random Forest Regression model handles bias and variance better than the single Decision Tree Regression model. In fact, averaging numerous decorrelated Decision Trees preserves high accuracy, stability, and robustness to noise. Lastly, the chosen Random Forest Regression model meets the practical business fit criteria by delivering high accuracy, strong generalization, and flexibility for the convoluted data structure, making it the most dependable engine for generating used-car prices.

Costs and Tradeoffs of the chosen model

The following are possible costs or drawbacks for the chosen Random Forest Regression model. First, the Random Forest Regression model has a higher computational cost due to longer training times, higher memory usage, and more costly hyperparameter tuning. Second, the Random Forest Regression model has lower interpretability than linear regression models (e.g., Multiple Linear Regression, Ridge Regression, and Lasso Regression). Third, the Random Forest Regression model is more complex to deploy and monitor. Fourth, the Random Forest Regression model is sensitive to irrelevant or noisy features. Lastly, the Random Forest Regression model is considerably larger, which can be an issue on some resource-limited systems or devices.

Reasons why the model best solves the business problem

The following are the reasons and benefits for why the chosen Random Forest Regression model best solves the business problem. First, the Random Forest Regression model can learn nonlinear relationships between the target variable, the log-transformed Price, and independent variables such as Car Age, Depreciation Amount, Power, New price, and categorical dummy variables for Location and Brand. Second, the Random Forest Regression model can handle mixed independent variables more smoothly, such as one-hot-encoded categorical variables and continuous numerical variables. In particular, the Random Forest Regression model can leverage detailed numeric signals in the independent variables, such as Mileage and Power, as well as the categorical structure, such as luxury versus economy brands and high-demand versus low-demand Indian cities. Third, the Random Forest Regression model provides stable, low-error predictions, meaning that the Cars4U startup can rely on the model to produce price estimates close to actual market values, while avoiding mispricing risks and enhancing revenue and competitiveness against other companies. Lastly, the Random Forest Regression model supports operational decision-making because it is initially trained on the target log-transformed Price variable, and predictions are exponentiated back to the original price scale; the outputs remain applicable as price recommendations in typical business flows.

Reasons for the model to be production-ready and interpretable

The following are reasons why the chosen Random Forest Regression model is production-ready and interpretable. First, given its high R-squared of 0.90 on the test dataset, low RMSE and MAE, and a substantially low, stable MAPE, the Random Forest Regression model is viable for production deployment. It outperforms every other model while maintaining precise, flexible approaches. Second, the Random Forest Regression model offers sufficient interpretability for business use. To clarify, the Random Forest Regression model quantifies global feature importance by identifying which independent variables (e.g., Power, Depreciation Amount, New price, Year, Car Age, and Kilometers Driven log) contribute the most to price predictions, which align with already common economic interpretations as seen in previous linear regression models. Similarly, the Random Forest Regression model best captures and refines the patterns in which independent variables related to a used car's wear, usage, and age consistently lead to lower used-car prices. At the same time, performance and original market value remain robust predictors of elevated used-car prices, as do luxury used-car brands and specific locations. Third, the Random Forest Regression model has the best operational considerations. To be specific, the Random Forest Regression model is the quickest for real-time used-car pricing, the most robust to moderate data drift, and the most adaptable to standardized versioning, monitoring, and retraining in machine learning operations.

Altogether, the Random Forest Regression model has the optimal combination of significant accuracy, enhanced generalization, robustness to nonlinearity, and constructive interpretability. It also best solves the problem by using reliable, low-error price predictions based on used-car attributes and market characteristics, and is the superior production-ready model.

Conclusions and Recommendations

Most Refined Key Findings and Insights

The project's analysis confirms that India's used car market offers high-growth prospects and poses a configurational pricing challenge for Cars4U. The dataset depicts a marketplace detailed by unreliable valuations, regional discrepancies, and information gaps that negatively affect buyers and sellers. These conditions establish manual or intuition-driven pricing that is not sustainable as Cars4U scales, which later increases the probability of incorrectly priced used cars, slower turnover, and revenue

loss. The key findings further underscore the necessity of a machine-learning pricing model not only for external clarity but also for Cars4U's core operational stability and its long-term competitive edge in the Indian used-car market. By using consistent, data-driven pricing, Cars4U's business model is fundamentally validated.

The exploratory data analysis phase of the project shows strong predictive signals from several independent variables. However, there is evidence of considerable data-quality issues that necessitate careful treatment. Numerical independent variables, including Engine, Power, and New Price, indicated heavy right-skew due to luxury used car vehicles. Categorical independent variables, including Fuel Type and Owner Type, aligned with common Indian market characteristics, as diesel-fueled and first-owner used cars comprised most of the dataset. Key variables had a significant number of missing values, especially New Price, with over 6,247 missing values, necessitating robust preprocessing and imputation methods. Indian cities, such as Mumbai and Hyderabad, indicate higher listing frequencies. Outlier analysis revealed that numerous extreme values should be retained to maintain model consistency, as the outliers represent luxury used-car segments rather than errors. These insights highlight the importance of thorough feature engineering, as independent variables such as Car Age and Depreciation Amount enhance the dataset's predictive power.

The modeling results from the six models indicated that nonlinear machine-learning models are better suited to the complexity of used-car pricing than linear machine-learning and geometry-based machine-learning models. Even though linear and regularized linear machine learning models are interpretable, they fail to capture nonlinear interactions among independent variables such as brand, power, mileage, and depreciation. Instead, tree-based machine learning models, most notably Random Forest Regression, outperformed all other models, achieving the highest R-squared and adjusted R-squared values and the lowest RMSE, MAE, and MAPE. The Random Forest Regression ensemble-based methodology allowed it to manage mixed independent variable types, outliers, and nonlinear relationships with stability and consistency. The Random Forest Regression model's feature importance analysis confirmed that Power, Depreciation Amount, New price, Year, Car Age, and Kilometers Driven (log-transformed) are the most powerful predictors of price, aligning with actual used-car valuation logic.

The Random Forest Regression model's predictive accuracy best supports Cars4U's operational goals by lowering the risk of mispricing used cars, enhancing inventory turnover, and enabling more accurate procurement decision-making. Insights from depreciation and brand-specific valuations can steer Cars4U toward selecting high-margin used cars while avoiding low-yield inventory. The Random Forest Regression model also improves clarity for customers by providing insight into which independent variables drive prices, lowering negotiation friction, and bolstering customer trust. From a strategic viewpoint, the Random Forest Regression model helps position Cars4U to better compete against other platforms that also leverage algorithmic or model-based pricing, helping ensure that Cars4U does not fall behind in a data-driven marketplace. As time passes, Cars4U can keep retraining the Random Forest Regression model and use data enrichment tools to help it become a sustained competitive asset.

Key Takeaways

Cars4U operates in a rapidly expanding yet highly fragmented Indian used-car marketplace where pricing can vary widely across cities, brands, and ownership histories. The project validates that without a systematic, data-driven valuation system, Cars4U will default to mispricing used vehicles, incur revenue loss, and erode customer trust. A machine learning pricing engine model is then a critical business requirement for scaling operations, improving clarity, and competing with platforms that already use algorithmic pricing.

The provided dataset originally included over 7,253 used cars, which provides robust predictive value but still has considerable missing data and skewed distributions in variables such as New Price, Power, and Engine. Despite these challenges, the dataset still exhibits meaningful market patterns. These include diesel fuel used cars, which make up most of the dataset; first-owner cars still have the highest value; and luxury-segment used cars are identified as sensible outliers. Feature engineering methods, such as creating Car Age, Depreciation Amount, and Brand extraction, substantially enhance the dataset's composition and predictive robustness.

Across the proposed models, nonlinear models consistently outperform linear models due to the complex interactions present in used-car pricing. The Random Forest Regression model is the most accurate and trustworthy, as it delivers the best

performance across R-squared, Adjusted R-squared, RMSE, MAE, and MAPE. Its ability to handle mixed data types, nonlinear relationships, and outliers makes it the most robust candidate model for Cars4U's production pricing engine. The feature importance analysis from the Random Forest Regression model also indicates that Power, Depreciation Amount, New price, Year, Car Age, and Kilometers Driven (log-transformed) are the most important predictors of used car price.

The final Random Forest Regression model chosen supports Cars4U's expansive goals by enabling dynamic pricing, inventory optimization, and fraud detection. Having trustworthy price predictions helps Cars4U by pinpointing undervalued used cars worth acquiring, uncovering overpriced listings that could be stifling, and forecasting depreciation trends across used-car brands and segments. The model's interpretability comes from Cars4U's ability to validate price valuations for customers, reduce negotiation friction, and build trust. By implementing continuous data enrichment and regular model retraining, the Random Forest Regression model can become a competitive asset that compounds value for Cars4U.

Business Recommendations

Cars4U should begin with primarily improving the consistency and completeness of new incoming data as its number one priority. Due to the provided dataset initially having a large number of missing values in critical independent variables, such as New Price (over 6,247 missing values), Power, Engine, and Seats, these missing values negatively affect model accuracy and necessitate preprocessing methods. Cars4U would also benefit from incorporating standardized data-collection procedures across seller entities, mandating required fields, and validating data entries upon addition to Cars4U's inventory. Authenticating automated irregularity detections can also help highlight impossible mileage, engine displacement, or price entries, thereby reducing noise and safeguarding model performance. Over time, a tidier, more structured data pipeline would substantially improve the pricing model's predictive performance.

The Random Forest Regression model should also be ingrained in Cars4U's operational processes rather than used solely as an analytical tool. Integrating model outputs into seller enlistments, listing formation, and buyer-facing price displays will ensure price valuations are consistent across Cars4U's platform. Cars4U should also

use predicted prices to guide attainment decisions, identify undervalued used cars worth acquiring, and uncover overpriced listings that may stagnate. Also, incorporating explanatory features, such as indicating the top attributes that affect each price valuation, would help build trustworthy relationships between buyer and seller entities. Therefore, this would allow Cars4U to operate as a transparent, data-driven platform, where pricing opaqueness is the norm.

Given the evolving used-car market, Cars4U should also adopt a periodic model retraining schedule supported by continuous data enrichment. As new used-car brands, models, and regional trends emerge, ongoing retraining of the Random Forest Regression model will ensure the pricing engine remains accurate and relevant. Also, by monitoring model drift, tracking error metrics over time, and maintaining a feedback loop with Cars4U's operations team, we can determine when retraining or feature updates are required. Moreover, expanding the dataset with further attributes, such as service history, accident records, or seller reputation, could improve predictive robustness. A prolonged model-governance structure will also ensure the pricing engine remains a stable, competitive asset.

The Random Forest Regression model's generated insights can also inform broader strategies beyond valuation pricing. Cars4U can leverage depreciation patterns and brand-specific value retention to enhance the inventory mix and centralize on high-margin or fast-turnover segments. Likewise, regional price trends across different Indian cities can better inform targeted marketing campaigning or city-based strategic acquisitions. The Random Forest Regression model can also support fraud detection by flagging listings with prices that are abnormally low or high relative to the model's predictions. By treating pricing intelligence as a strategic resource, Cars4U can enhance its position in the Indian used car marketplace and set itself apart from competitors.

It is important to note that clear communication of how used car prices are conveyed to buyers and sellers will mostly help reduce negotiation friction and increase user confidence. Cars4U should provide buyers and sellers with streamlined explanations of the key factors that affect the valuation of each used car. Visual breakdowns or confidence ranges should back these streamlined explanations. Furthermore, educating users about how the Random Forest Regression model's

pricing engine works can better underpin Cars4U's commitment to fairness and accuracy. This clarity will be invaluable in a used car market where pricing is inconsistent and has historically been fraught with trust issues. The ideal next step would be first to decide which operational domain (i.e., seller onboarding, procurement, or buyer-facing pricing) should receive the primary integration of the Random Forest Regression model engine to maximize early impact.

Conclusions

The project shows that Cars4U can effectively change its pricing operations by using a well-designed machine-learning engine based on a Random Forest Regression model. The combination of strong predictive performance, explainability, and scalability makes the model best suited for real-world deployment across India's diverse used-car market ecosystem. The key findings collectively show that Cars4U's success will depend on sustaining data quality, incorporating model insights into business techniques, and treating pricing intelligence as a strategic pillar rather than a supplementary tech add-on. As Cars4U continues to grow, the pricing engine will standardize used-car pricing, enable more intelligible procurement, build customer trust, and enhance long-term operational efficacy.